

BOX 11.1

Experimental
Methods

Amino Acid Sequencing

Protein-sequencing techniques have been known since 1953, when F. Sanger worked out the complete sequence of the protein hormone insulin. The basic strategy is to purify the protein and then sequence it, beginning at one end. However, since most proteins contain too many amino acids to do this successfully, proteins are first broken into small peptides in several different ways. These peptides are sequenced, and the whole protein sequence can be determined by the overlap pattern of the sequenced subunits.

A protein can be broken into peptide fragments by many different methods, including acid and alkaline hydrolysis. For the most part, proteolytic enzymes (proteases) that hydrolyze the peptides at specific

points are used. *Pepsin*, for example, preferentially hydrolyzes peptide bonds involving aromatic amino acids, methionine, and leucine; *chymotrypsin* hydrolyzes peptide bonds involving carboxyl groups of aromatic amino acids; and *trypsin* hydrolyzes bonds involving the carboxyl groups of arginine and lysine.

The proteolytic digest is usually separated into a *peptide map*, or *peptide fingerprint*, by using a two-dimensional combination of paper

chromatography, electrophoresis, or column chromatography. In two-dimensional chromatography, a sample is put onto a piece of paper that is then placed in a solvent system. After an allotted time, the paper is dried, turned 90 degrees, and placed in a second solvent system for another allotted time (fig. 1). In each solvent, different peptides travel through the paper at different rates. The spots are then developed using ninhydrin, which reacts with the N-terminal amino acid and produces a colored product when heated.

The spots, which represent small peptides, can be cut out of a second, identical chromatogram that has not been sprayed with ninhydrin. These spots can then be sequenced by, for example, the Edman method, whereby the peptide is sequentially degraded

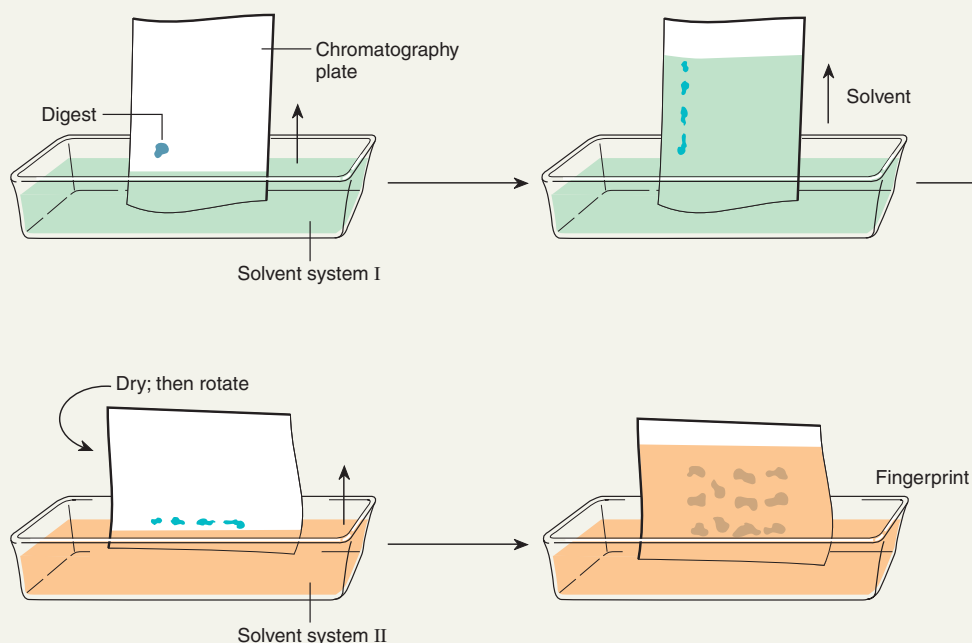


Figure 1 Two-dimensional paper chromatography of a protease digest. Chromatography is done first in one solvent system. The paper is then dried, rotated, and placed into a second solvent system. The pattern on the resulting plate is called a peptide fingerprint.

from the N-terminal end. Phenylisothiocyanate (PITC) reacts with the amino end of the peptide. When acid is added, the N-terminal amino acid is removed as a PITC derivative and can be identified. The process is then re-

peated until the whole peptide has been sequenced (fig. 2).

If the fingerprint pattern is worked out for two different digests of the same polypeptide, the unique sequence of the original polypeptide

can be determined by overlap. In figure 3, the letters A–J represent the ten amino acids in a polypeptide. A is known to be the first (N-terminal) amino acid since the Edman method sequences peptides from this end. We can thus summarize the methodology as follows:

1. A protein is purified. If it is made up of several subunits, these subunits are separated and purified. (If disulfide bridges exist within a peptide, they must be reduced. The bridges are later determined by digestion, keeping the bridges intact, and then resequencing.)
2. Different proteolytic enzymes are used on separate subsamples so that the protein is broken into different sets of peptide fragments.
3. Two-dimensional chromatography, electrophoresis, or column chromatography can be used to isolate the peptides.
4. The Edman method of sequentially removing amino acids from the N-terminal end is used to sequence each peptide.
5. The amino acid sequence from the N- to C-terminal ends of the protein is deduced from the overlap of sequences in peptide digests generated with different proteolytic enzymes.

Today, a machine known as an amino acid sequencer (*sequenator*) can automatically sequence protein. Taking about two hours per amino acid residue, sequenators can carry out Edman degradation on polypeptides up to about fifty amino acids long.

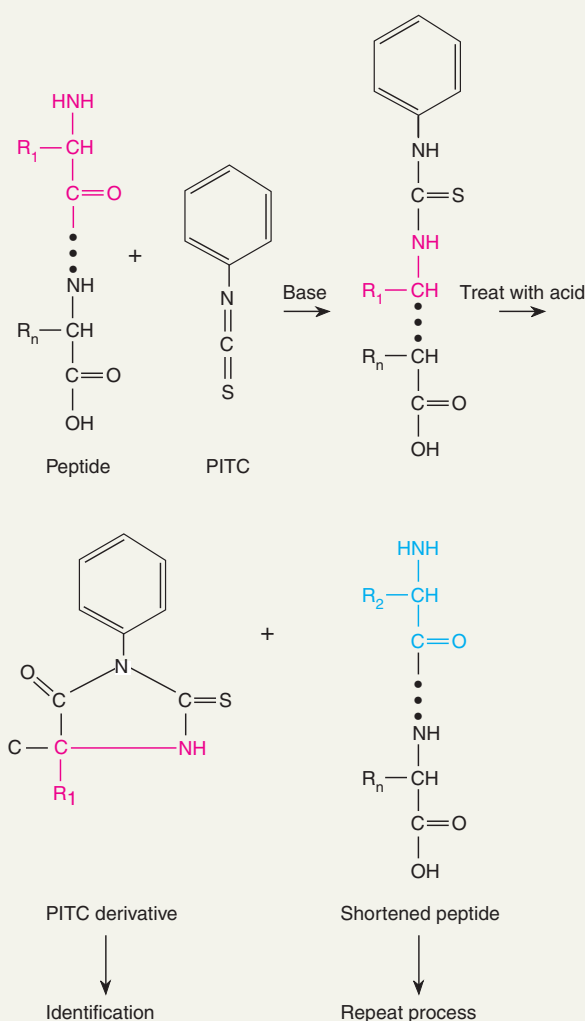


Figure 2 Isolation of amino acids from a peptide for sequencing purposes. First, the peptide reacts with PITC (phenylisothiocyanate) at the amino end. Acid treatment produces a PITC derivative of the amino-terminal amino acid and a peptide one amino acid shorter than the original. The PITC derivative can be identified. These steps are then repeated, isolating one amino acid at a time.

continued

BOX 11.1 CONTINUED

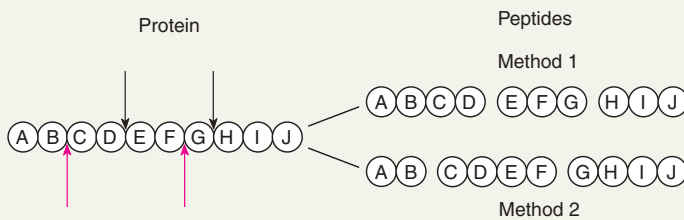


Figure 3 The overlap of peptides digested two different ways provides the sequence of the original peptide.

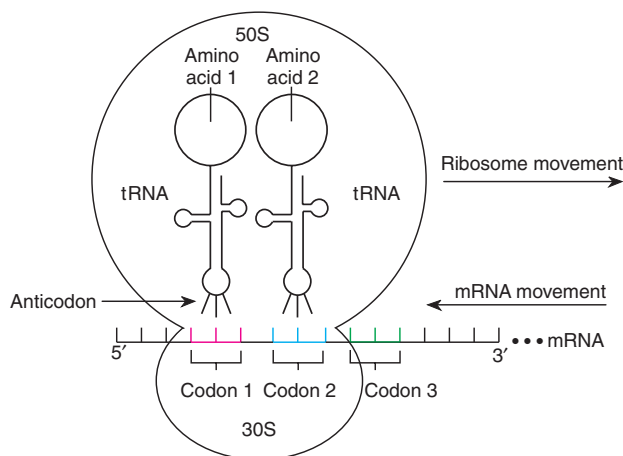


Figure 11.5 The initiation of the translation process at the ribosome. Note the two charged transfer RNAs and the messenger RNA. They are in position to form the first peptide bond between the two amino acids attached to the transfer RNAs.

code for that amino acid. In eukaryotes, there are separate sets of twenty cytoplasmic and twenty mitochondrial synthetases, all coded in the nucleus.

Aminoacyl-tRNA synthetases are a heterogeneous group of enzymes. In *E. coli*, they vary from monomeric proteins (one subunit) to tetrameric proteins, made up of two copies each of two subunits. The enzymes fall into two categories based on sequence similarity, structural features, and whether the amino acid is attached at the 2'-OH (in class I enzymes) or 3'-OH (in class II enzymes) of the 3'-terminal adenosine of the transfer RNA.

To add its appropriate amino acid to the appropriate transfer RNA, a synthetase recognizes many parts of the transfer RNA. This can be shown by experiments that introduce specific changes in transfer RNAs by site-directed mutagenesis (see chapter 13). In seventeen of

the twenty *E. coli* synthetases, recognition involves part of the anticodon itself. This makes sense since the anticodon is the defining element of a transfer RNA in protein synthesis.

A synthetase can initially make errors and attach the “wrong” amino acid to a tRNA. For example, isoleucyl-tRNA synthetase will attach valine about once in 225 times. This type of error occurs because a similar, but smaller, amino acid can sometimes occupy the active site of the enzyme (compare isoleucine and valine in fig. 11.1). However, because of a proofreading step, only 1 in 270 to 1 in 800 of the errors are released intact from the enzyme. The amino acids on the rest of the incorrectly charged transfer RNAs are hydrolyzed before the transfer RNAs are released. The overall error rate is the product of the two steps; this means only about one incorrectly charged transfer RNA occurs per 60,000 to 80,000 formed.

In several cases, the number of amino acyl-tRNA synthetases in a particular organism is below twenty. For example, in some archaea, there is no cysteinyl-tRNA synthetase. However, the prolyl-tRNA synthetase activates the tRNAs for both cysteine and proline with their appropriate amino acids. Similarly, in some eubacteria, there is no glutamyl-tRNA synthetase; the glutamyl tRNA is charged with glutamic acid, rather than glutamine. An amino transferase enzyme then converts the glutamic acid to glutamine (see fig. 11.1).

There are sixty-four possible codons in the genetic code (four nucleotide bases in groups of three = $4 \times 4 \times 4 = 64$). Three of these codons are used to terminate translation. Thus, sixty-one transfer RNAs are needed because there are sixty-one different nonterminator codons. About fifty transfer RNAs are known in *E. coli*. The number fifty can be explained by the wobble phenomenon, which occurs in the third position of the codon. We examine this phenomenon in the section on the genetic code. The transfer RNAs for each amino acid are designated by the convention tRNA^{Leu} (for leucine), tRNA^{His} (for histidine), and so on.

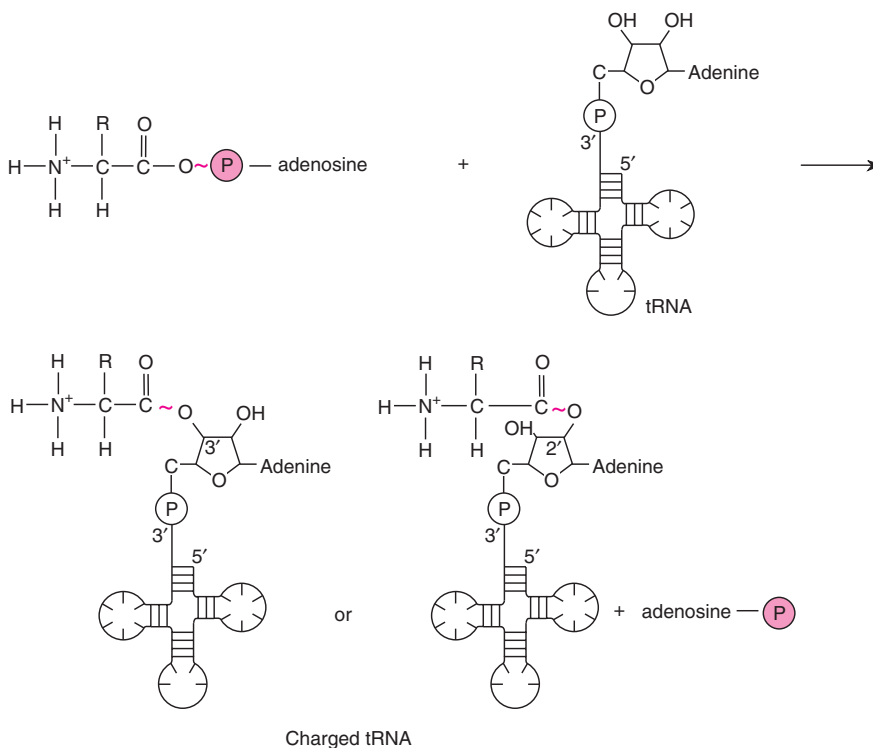
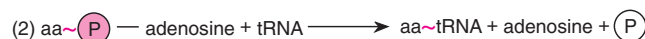
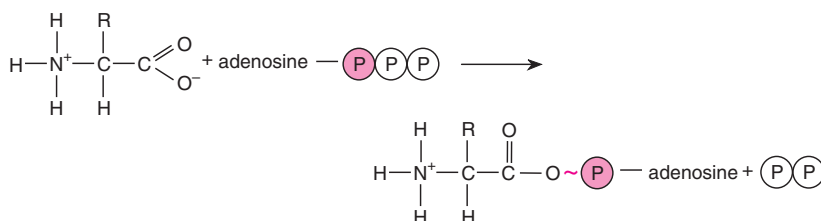


Figure 11.6 It takes a two-step process to attach a specific amino acid to its transfer RNA by an aminoacyl synthetase. High-energy bonds are indicated by ~. In the first step, an amino acid is attached to AMP with a high-energy bond. In the second step, the high-energy bond is transferred to the tRNA, which is then referred to as “charged.” Depending on which class of aminoacyl-tRNA synthetase is involved, the amino acid will be attached to either the 2' or 3' carbon of the sugar of the 3' terminal adenosine.

Recognition of the Aminoacyl-tRNA During Protein Synthesis

Although amino acids enter the protein-synthesizing process attached to transfer RNAs, it was theoretically possible that the ribosome recognized the amino acid itself during translation. A simple experiment was done to determine whether the amino acid or the transfer RNA was recognized.

In 1962, F. Chapeville and colleagues isolated transfer RNA with cysteine attached. They chemically con-

verted the cysteine to alanine by using Raney nickel, a catalytic form of nickel that removes the SH group of cysteine (fig. 11.7). When these transfer RNAs were used in protein synthesis, alanine was incorporated where cysteine should have been, demonstrating that the transfer RNA, not the amino acid, was recognized during protein synthesis. The synthetase puts a specific amino acid on a specific transfer RNA; then, during protein synthesis, the anticodon on the transfer RNA—not the amino acid itself—determines which amino acid is incorporated.

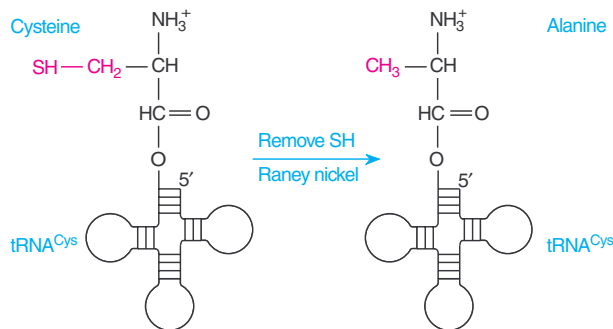


Figure 11.7 Cysteine-tRNA^{Cys} treated with Raney nickel becomes alanine-tRNA^{Cys} by the removal of the SH group of cysteine. During protein synthesis, alanine is incorporated in place of cysteine in proteins, indicating that the specificity of amino acid incorporation into proteins resides with the tRNA.

Initiation Complex



Translation can be divided into three stages: initiation, elongation, and termination. Elongation is the repetitive process of adding amino acids to a growing peptide chain. However, added complexity enters the picture in the initiation and termination of protein synthesis.

It is especially important that the translation process start precisely. Remember that the genetic code is translated in groups of three nucleotides (codons). If the reading of the messenger RNA begins one base too early or too late, the reading frame is shifted so that an entirely different set of codons is read (fig. 11.8). The protein produced, if any, will probably bear no structural or functional resemblance to the protein the gene is coded for.

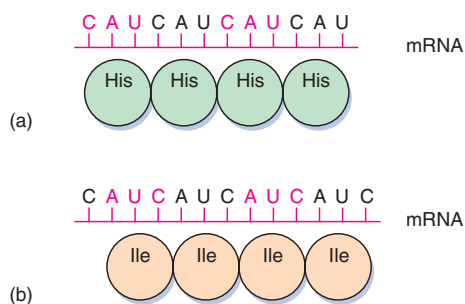


Figure 11.8 (a) In the normal reading of the messenger RNA, these codons are read as repeats of CAU, coding for histidine. (b) A shift in the reading frame of the messenger RNA causes the codons to be read as AUC repeats coding for isoleucine.

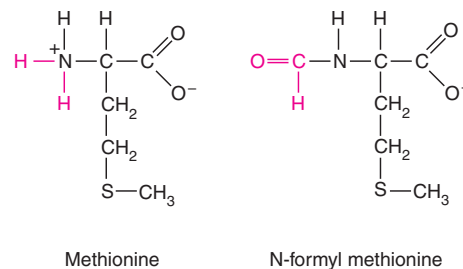


Figure 11.9 The structures of the amino acids methionine and N-formyl methionine.

Role of N-Formyl Methionine

The synthesis of every protein in *Escherichia coli* begins with the modified amino acid N-formyl methionine (fig. 11.9). However, none of the completed proteins in *E. coli* contains N-formyl methionine. Many of these proteins do not even have methionine as their first amino acid. Obviously, before a protein becomes functional, the initial amino acid is modified or removed. In eukaryotes the initial amino acid, also methionine, does not have an N-formyl group.

Methionine, with a codon of 5'-AUG-3', known as the **initiation codon**, has two transfer RNAs with the same complementary anticodon (3'-UAC-5') but with different structures (fig. 11.10). One of these transfer RNAs (tRNA^{Met}) serves as a part of the initiation complex. Before the initiation of translation, this transfer RNA will have its methionine chemically modified to N-formyl methionine (fMet). The other transfer RNA will not have its methionine modified (tRNA^{Met}). The translation machinery will use it to insert methionine into proteins, where called for, in all but the first position. The cell thus has a mechanism to make use of methionine in the normal way as well as to use a modified form of it to initiate protein synthesis. Because of the structure of the prokaryotic initiation transfer RNA, it can recognize AUG, GUG, and, rarely, UUG as initiation codons. In eukaryotes, CUG as well as AUG can serve as an initiation codon. Since the initiation methionine is not formylated in eukaryotes, the eukaryotic transfer RNA is designated tRNA^{Met}; there is a separate internal methionine transfer RNA, termed tRNA^{Met}, in eukaryotes, as in prokaryotes.

Translation Initiation

The subunits of the ribosome (30S and 50S) usually dissociate from each other when not involved in translation. To begin translation, an **initiation complex** forms, consisting of the following components in prokaryotes: the 30S subunit of the ribosome, a messenger RNA, the charged N-formyl methionine tRNA (fMet-tRNA^{Met}), and

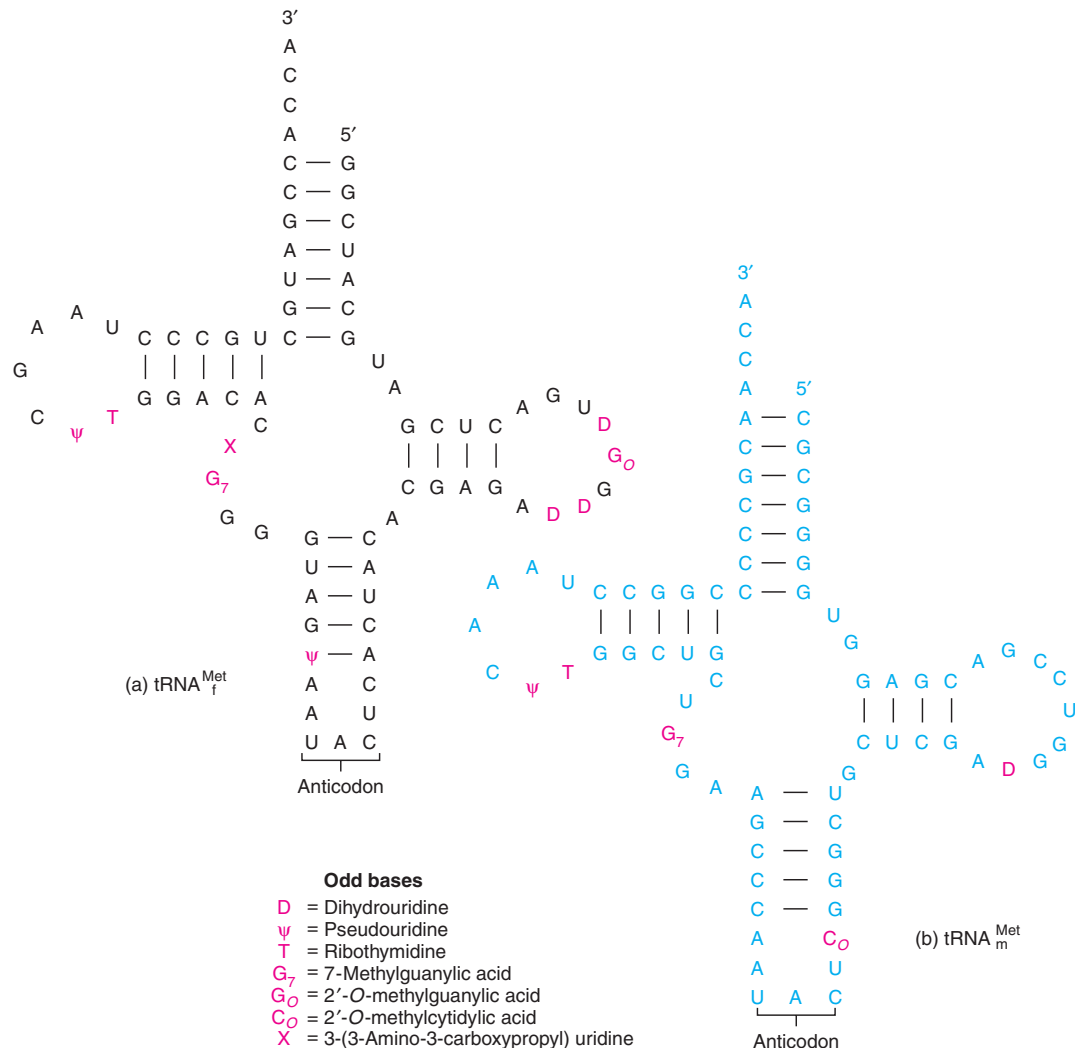


Figure 11.10 The two tRNAs for methionine in *E. coli*. (a) The initiator tRNA. (b) The interior tRNA.

three **initiation factors (IF1, IF2, IF3)**. Initiation factors (as well as elongation and termination factors) are proteins loosely associated with the ribosome. They were discovered when ribosomes were isolated and then washed, losing the ability to perform protein synthesis.

The components that form the initiation complex interact in a series of steps. It is known that IF3 binds to the 30S ribosomal subunit, allowing the 30S subunit to bind to messenger RNA (fig. 11.11, *step 1*). Meanwhile, a complex forms with IF2, the charged N-formyl methionine tRNA (fMET-tRNA_f^{Met}) and GTP (guanosine triphosphate; fig. 11.11, *step 2*). It is IF2 that brings the initiator transfer RNA to the ribosome. IF2 binds only to the charged initiator transfer RNA, and, without IF2, the initiator transfer RNA cannot bind to the ribosome. The final step in

initiation-complex formation is bringing together the first two components (fig. 11.11, *step 3*).

The hydrolysis of GTP to GDP + P_i (inorganic phosphate, PO₄³⁻—see fig. 9.8) produces conformational changes; these changes allow the initiation complex to join the 50S ribosomal subunit to form the complete ribosome and then allow the initiation factors and GDP to be released. Frequently, the hydrolysis of a nucleoside triphosphate (e.g., ATP, GTP) in a cell occurs to release the energy in the phosphate bonds for use in a metabolic process. However, in the process of translation, the hydrolysis apparently changes the shape of the GTP so that it and the initiation factors can be released from the ribosome after the 70S particle has been formed. Thus, hydrolysis of GTP in translation is for conformational

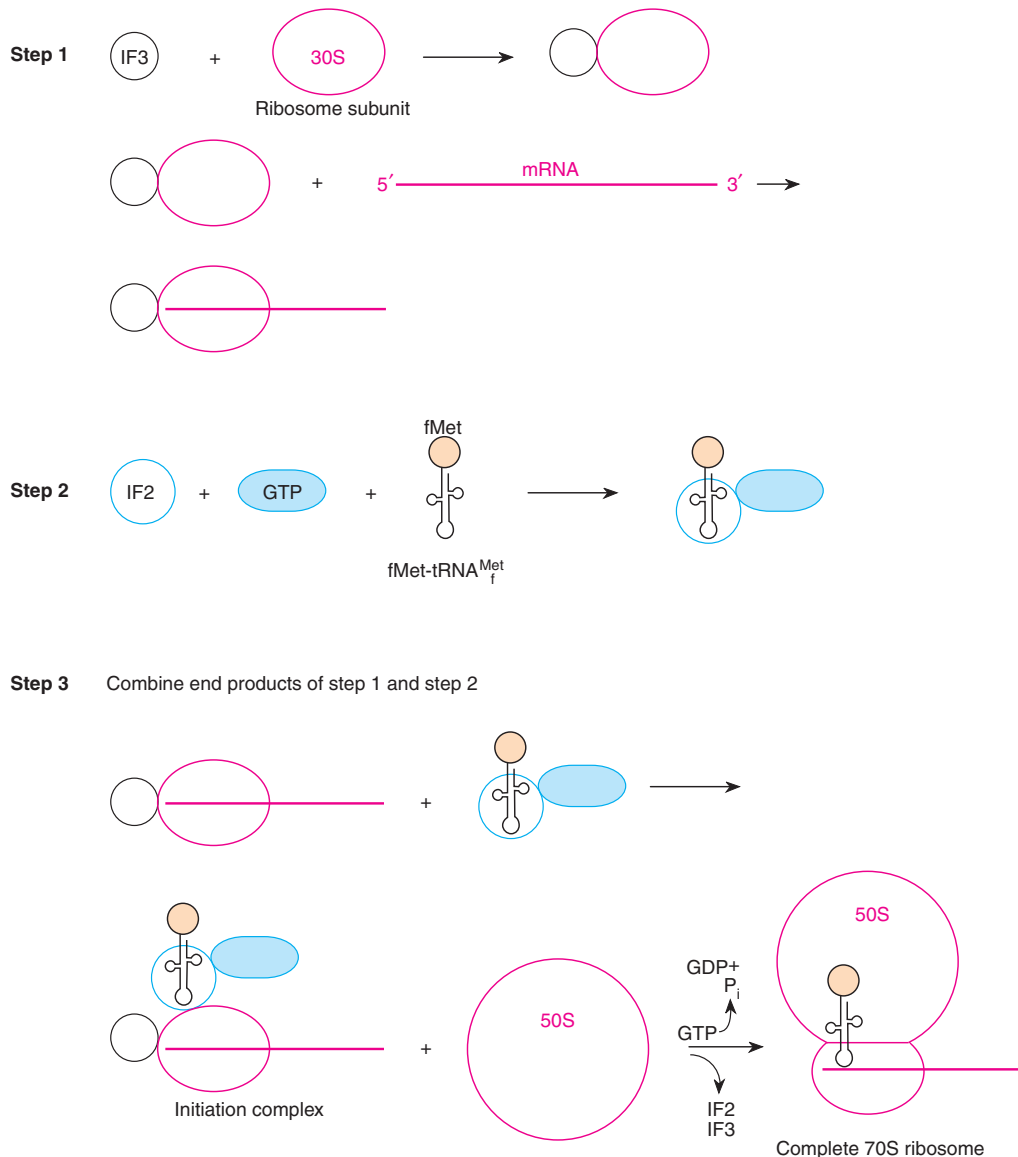


Figure 11.11 The prokaryotic 70S ribosome forms in a three-step process. In the first step, the 30S ribosomal subunit and the mRNA combine. In the second step, the initiator tRNA combines with IF2. In the final step, the components from steps 1 and 2 combine to form the initiation complex, followed by the formation of the 70S ribosome.

change rather than covalent bond formation. IF1 helps the other two initiation factors bind to the 30S ribosomal subunit or stabilizes the 30S initiation complex.

The process in eukaryotes is generally similar, but more complex. The eukaryotic initiation factor abbreviations are preceded by an “e” to denote that they are eukaryotic (eIF1, eIF2, etc.). At least eleven initiation factors are involved, including a specific cap-binding protein, eIF4E.

The ribosome apparently recognizes the prokaryotic messenger RNA through complementarity of a region at the 3' end of the 16S ribosomal RNA and a region slightly upstream from the initiation sequence (AUG) on the messenger RNA. This idea, the **Shine-Dalgarno hypothesis** (fig. 11.12), is named after the people who first suggested it. The sequence (AGGAGGU) of complementarity between the messenger RNA and the 16S ribosomal RNA is referred to as the Shine-Dalgarno sequence. Although

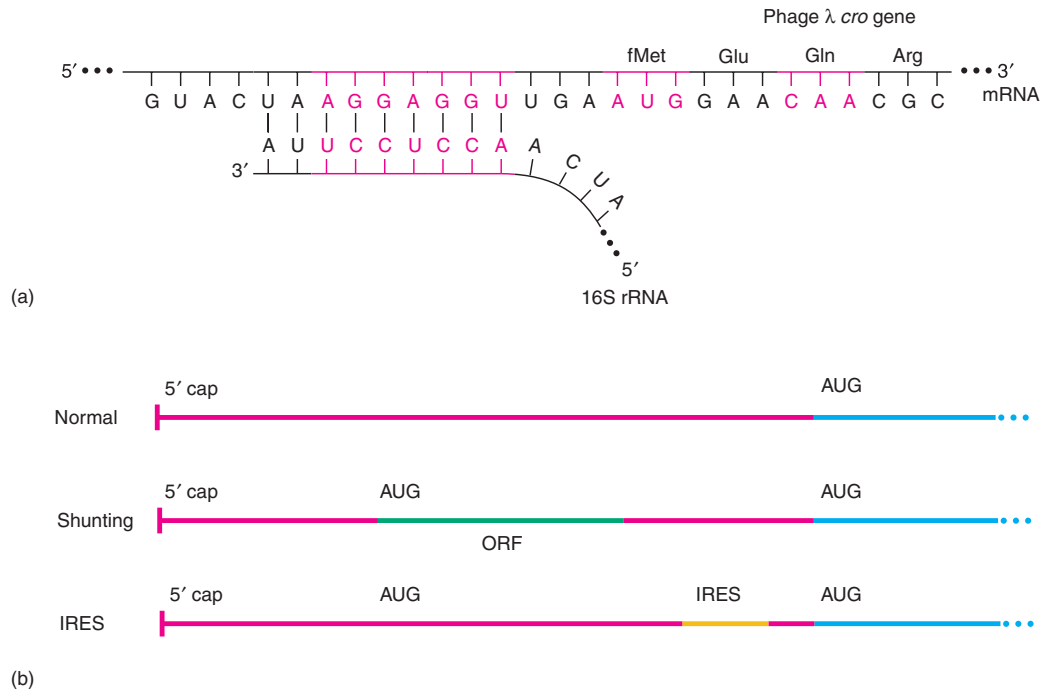


Figure 11.12 Translation initiation signals. (a) The Shine-Dalgarno hypothesis for prokaryotic translation. The Shine-Dalgarno sequence (AGGAGGU) is on the prokaryotic messenger RNA just upstream from the initiation codon AUG. Complementarity exists between this sequence and a complementary sequence (UCCUCCA) on the 3' end of the 16S ribosomal RNA. (b) Scanning, shunting, and internal ribosome entry in eukaryotic messenger RNA. The 5' untranslated region of a eukaryotic gene is shown in red; the beginning of the gene in blue. Normally, in the scanning model, the initiation codon of the gene is the first AUG encountered. In shunting, an open reading frame (ORF, green) may or may not be present to provide secondary structure in the messenger RNA to shunt scanning to the main gene. If the open reading frame is translated, reinitiation of translation at the same ribosome may occur at the main gene. Finally, an internal ribosome entry site (IRES, yellow) allows translation to begin within the messenger RNA without scanning.

there is a good deal of homology between prokaryotic and eukaryotic small ribosomal RNAs, the Shine-Dalgarno region is absent in eukaryotes. The actual mechanism for recognizing the 5' end of eukaryotic messenger RNA appears to be based on recognition of the 5' cap of the messenger RNA by the cap-binding protein with recruitment of other initiation factors and the small subunit of the ribosome. This is followed by the small subunit's movement down the messenger RNA. The ribosome scans the messenger RNA until it recognizes the initiation codon. This model is referred to as the **scanning hypothesis**.

In several known cases in eukaryotes, a process called **shunting** occurs, in which the first AUG does not serve as the initiation codon; rather, scanning begins, but it bypasses a region of the messenger RNA upstream of the initiation codon, called the *leader* or **5' untranslated region (5' UTR)**, in favor of an AUG further down the messenger RNA. The cause of shunting seems to be secondary structure in the messenger RNA, upstream from

the AUG codon that actually serves as the initiation codon. In some cases, very small genes, called **open reading frames (ORFs)**, are present in this region of the messenger RNA and play some role in shunting. It seems also that some ORFs are translated, and then the main gene is translated by the same ribosome in a process called **reinitiation** (fig. 11.12b). We have seen this in the genes of some plant and animal viruses; it is a topic under current study.

Under some circumstances, eukaryotic ribosomes can initiate protein synthesis within the messenger RNA if that messenger RNA contains a sequence called an **internal ribosome entry site**. These sequences were discovered in the poliovirus RNA and in several cellular messenger RNAs. They are at least four hundred nucleotides long. Thus, although scanning accounts for the initiation of most eukaryotic messenger RNAs at their 5' ends, some initiation can take place internally in messenger RNAs that have internal ribosome entry sites.

Aminoacyl and Peptidyl Sites in the Ribosome

When the initiator transfer RNA joins the 30S subunit of the prokaryotic ribosome with its messenger RNA attached, it fits into one of three sites in the ribosome. These sites, or cavities in the ribosome, are referred to as the aminoacyl site (**A site**), the peptidyl site (**P site**), and the exit site (**E site**, fig. 11.13). Here, we concentrate on the A and P sites, each of which contains a transfer RNA just before forming a peptide bond: the P site contains the transfer RNA with the growing peptide chain (peptidyl-tRNA); the A site contains a new transfer RNA with its single amino acid (aminoacyl-tRNA). The exit site helps eject depleted transfer RNAs after a peptide bond forms. When the complete 70S ribosome of figure 11.11 has formed, the initiation fMET-tRNA^{Met} is placed directly into the P site (fig. 11.13), the only charged transfer RNA that can be placed directly there. The association of transfer RNA and ribosome is aided by a G-C base pairing between the 3'-CCA terminus of all transfer RNAs and a guanine in the 23S ribosomal RNA.

Elongation

Positioning a Second Transfer RNA

The next step in prokaryotic translation is to position the second transfer RNA, which is specified by the codon at the A site. The second transfer RNA is positioned in the A site of the ribosome so it is able to form hydrogen bonds between its anticodon and the second codon on the messenger RNA. This step requires the correct transfer RNA, another GTP, and two proteins called

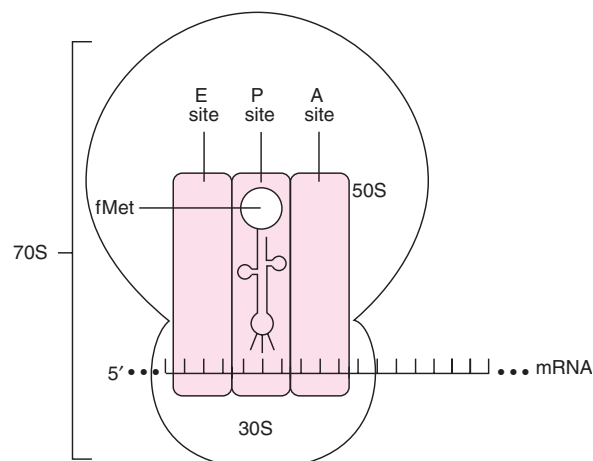


Figure 11.13 The 70S ribosome contains an A site, a P site, and an E site that can receive tRNAs. The messenger RNA runs through the bottom of the sites.

elongation factors (EF-Ts and EF-Tu). EF-Tu, bound to GTP, is required to position a transfer RNA into the A site of the ribosome (fig. 11.14). After the transfer RNA is positioned, the GTP is hydrolyzed to GDP + P_i. Upon hydrolysis of the GTP, the EF-Tu/GDP complex is released from the ribosome. EF-Ts is required to regenerate an EF-Tu/GTP complex. EF-Ts displaces the GDP on EF-Tu. Then a new GTP displaces EF-Ts, and now the EF-Tu/GTP complex can bind another transfer RNA. Here again, the hydrolysis of GTP changes the shape of the GTP so that the EF-Tu/GDP complex can depart from the ribosome after the transfer RNA is in place in the A site (fig. 11.15). Figure 11.16 shows the ribosome at the end of this step. EF-Tu does not bind fMet-tRNA^{Met}, so this blocked (formylated) methionine cannot be inserted into a growing peptide chain.

It takes several milliseconds for the GTP to be hydrolyzed, and another few milliseconds for the EF-Tu/GDP to actually leave the ribosome. During those two intervals of time, the codon-anticodon fit of the transfer RNA is scrutinized. If the correct transfer RNA is in place, a peptide bond forms. If not, the charged transfer RNA is released and a new cycle of EF-Tu/GTP-mediated testing of transfer RNAs begins. The error rate is only about one mistake in ten thousand amino acids incorporated into protein. The speed of amino acid incorporation is about fifteen amino acids per second in prokaryotes and about two to five per second in eukaryotes.

Peptide Bond Formation

The two amino acids on the two transfer RNAs are now in position to form a peptide bond between them; both amino acids are juxtaposed to an enzymatic center, **peptidyl transferase**, in the 50S subunit. This enzymatic center, an integral part of the 50S subunit, was originally believed to be composed of parts of several of the 50S proteins. Now, however, it is believed to have ribozymic activity, enzymatic activity of the ribosomal RNA of the ribosome. The enzymatic activity involves a bond transfer from the carboxyl end of N-formyl methionine to the amino end of the second amino acid (phenylalanine in fig. 11.16). Every subsequent peptide bond is identical, regardless of the amino acids involved. The energy used is contained in the high-energy ester bond between the transfer RNA in the P site and its amino acid (fig. 11.17). Immediately after the formation of the peptide bond, the transfer RNA with the dipeptide is in the A site, and a depleted transfer RNA is in the P site (box 11.2).

Translocation

The next stage in elongation is translocation of the ribosome in relation to the transfer RNAs and the messenger

RNA. Elongation factor EF-G, earlier called translocase, catalyzes the translocation process. The ribosome must be converted from the *pretranslocational state* to the *posttranslocational state* by the action of EF-G, which physically moves the messenger RNA and its associated transfer RNAs (fig. 11.18). This movement is accomplished by the hydrolysis of a GTP to GDP after EF-G enters the ribosome at the A site. After the first posttranslocational state is reached, the depleted transfer RNA in the E site is ejected, leaving the ribosome ready to accept a new charged transfer RNA in the A site. A computer-generated diagram of a ribosome with all three transfer RNA sites occupied is shown in figure 11.18*b*. In eukary-

otes, three elongation factors perform the same tasks that EF-Tu, EF-Ts, and EF-G perform in prokaryotes. The factor eEF1 α replaces EF-Tu, eEF1 $\beta\gamma$ replaces EF-Ts, and eEF2 replaces EF-G.

When translocation is complete, the situation is again as diagrammed in figure 11.13, except that instead of fMet-tRNA^{Met}, the P site contains the second transfer RNA (tRNA^{Phe}) with a dipeptide attached to it. The process of elongation is then repeated, with a third transfer RNA coming into the A site. The process repeats from here to the end (fig. 11.19), synthesizing a peptide starting from the amino (N-terminal) end and proceeding to the carboxyl (C-terminal) end. During the repetitive aspect of

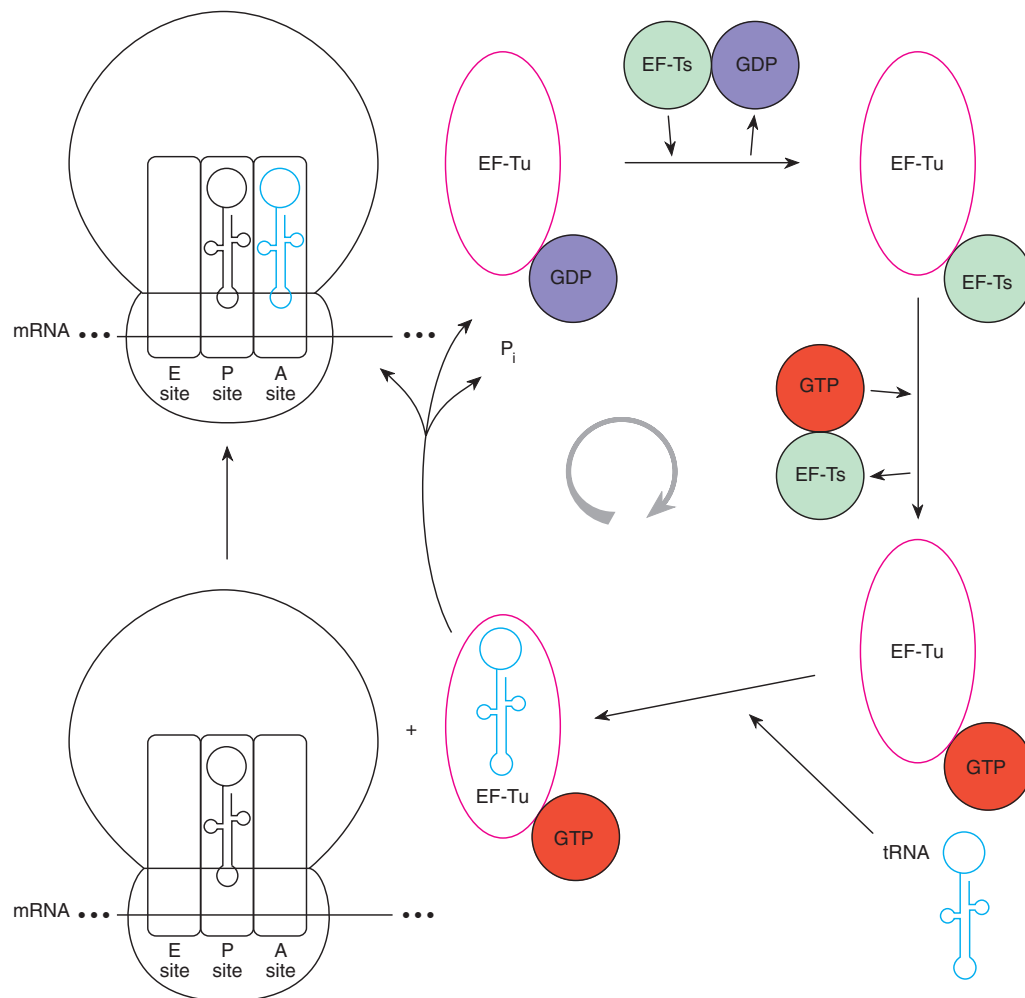


Figure 11.14 The EF-Ts/EF-Tu cycle. EF-Ts and EF-Tu are required for a transfer RNA to attach to the A site of the ribosome. At *top center*, we have EF-Tu attached to a GDP. The GDP is then displaced by EF-Ts, which in turn is displaced by GTP. A transfer RNA attaches and is brought to the ribosome. If the codon-anticodon fit is correct, the transfer RNA attaches at the A site with the help of the hydrolysis of GTP to GDP + P_i, allowing the EF-Tu to release. The EF-Tu is now back where we started. Since EF-Tu has a strong affinity for GDP, the role of EF-Ts is to displace the GDP, and later to be replaced by GTP.

BOX 11.2

Antibiotics, substances living organisms produce that are toxic to other living organisms, are of interest to us for two reasons: They have been extremely important in fighting the diseases that strike human beings and farm animals, and many are useful tools for analyzing protein synthesis. Some antibiotics impede the process of protein synthesis in a variety of ways, often poisoning bacteria selectively; the effectiveness of antibiotics normally derives from the metabolic differences between prokaryotes and eukaryotes. For example, an antibiotic that blocks a 70S bacterial ribosome without affecting an 80S human ribosome could be an excellent antibiotic. About 160 antibiotics are known.

PUROMYCIN

Puromycin resembles the 3' end of an aminoacyl-tRNA (fig. 1). It is bound to the A site of the bacterial ribosome, where peptidyl transferase creates a bond from the nascent peptide attached to the transfer RNA in the P site to puromycin. Elongation can then no longer occur. The peptide chain is released prematurely, and protein synthesis at the ribosome terminates.

Experiments with puromycin helped demonstrate the existence of the A and P sites of the ribosome. It was found that puromycin could not bind to the ribosome if translocation factor EF-G were absent. With EF-G, translocation took place, and puromycin could then bind to the ribosome. Puromycin's ability to bind only after translocation indicates that a second site on the ribosome becomes available after translocation.

STREPTOMYCIN, TETRACYCLINE, AND CHLORAMPHENICOL

Streptomycin, which binds to one of the proteins (protein S12) of the 30S subunit of the prokaryotic ribosome, inhibits initiation of protein synthesis. Streptomycin also causes misread-

Biomedical Applications

Antibiotics

ing of codons if chain initiation has already begun, presumably by altering the conformation of the ribosome so that transfer RNAs are less firmly bound to it. Bacterial mutants that are streptomycin resistant, as well as mutants that are streptomycin dependent (they cannot survive without the antibiotic), occur. Both types of mutants have altered 30S subunits, specifically changed in protein S12.

Tetracycline blocks protein synthesis by preventing an aminoacyl-tRNA from binding to the A site on the ribosome. Chloramphenicol blocks protein synthesis by binding to the 50S subunit of the prokaryotic ribosome, where it blocks the peptidyl transfer reaction. Chloramphenicol does not affect the eukaryotic ribosome. However, chloramphenicol, as well as several other antibiotics, is used cautiously because the mitochondrial ribosomes within eukaryotic cells are very similar to prokaryotic ribosomes. Some of the antibiotics that affect prokaryotic ribosomes thus also affect mitochondria. As was mentioned, the similarity between bacteria and mitochondria implies that mitochondria have a prokaryotic origin. (Similarities between cyanobacteria and chloroplasts also support the idea that chloroplasts have a prokaryotic origin.)

THE TROUBLE WITH ANTIBIOTICS

Over the years, antibiotics have virtually eliminated certain diseases from the industrialized world. They have also made modern surgery possible by preventing most serious infections

that tend to follow operations. Antibiotics have been so successful that, in the 1980s, many pharmaceutical companies drastically cut back the development of new antibiotics. However, a disaster was in the making as we overprescribed antibiotics to people and farm animals: bacteria are not prepared to take this onslaught without fighting back.

Mutation takes place all the time at a low but dependable rate. Thus, resistant bacteria are constantly arising from sensitive strains. We can select for penicillin- and streptomycin-resistant strains of bacteria in the laboratory by allowing the antibiotic to act as a selective agent, removing all but the resistant individuals. The same sort of artificial selection that we can apply in the lab applies every time a person or animal takes an antibiotic. We may be at a point now where the ability of bacteria to develop resistance, and to pass that resistance to other strains, has put us on the verge of disaster. The process of evolution works amazingly fast in bacteria because of their ubiquity, large population sizes, and ability to transfer genetic material between individuals. We may shortly find ourselves as we were before World War II, when simple infections in hospitals were often lethal. Right now, only one antibiotic can keep the common—and potentially deadly—infectious bacterium *Staphylococcus* under control: vancomycin. Several types of disease-causing bacteria have already evolved a tolerance to vancomycin.

The answer to this potentially disastrous problem is to develop new antibiotics and reduce the irresponsible use of antibiotics in people and animals. Hopefully, the warning bell has sounded. At least a dozen new antibiotics that show promise are in the early stages of development by pharmaceutical companies.

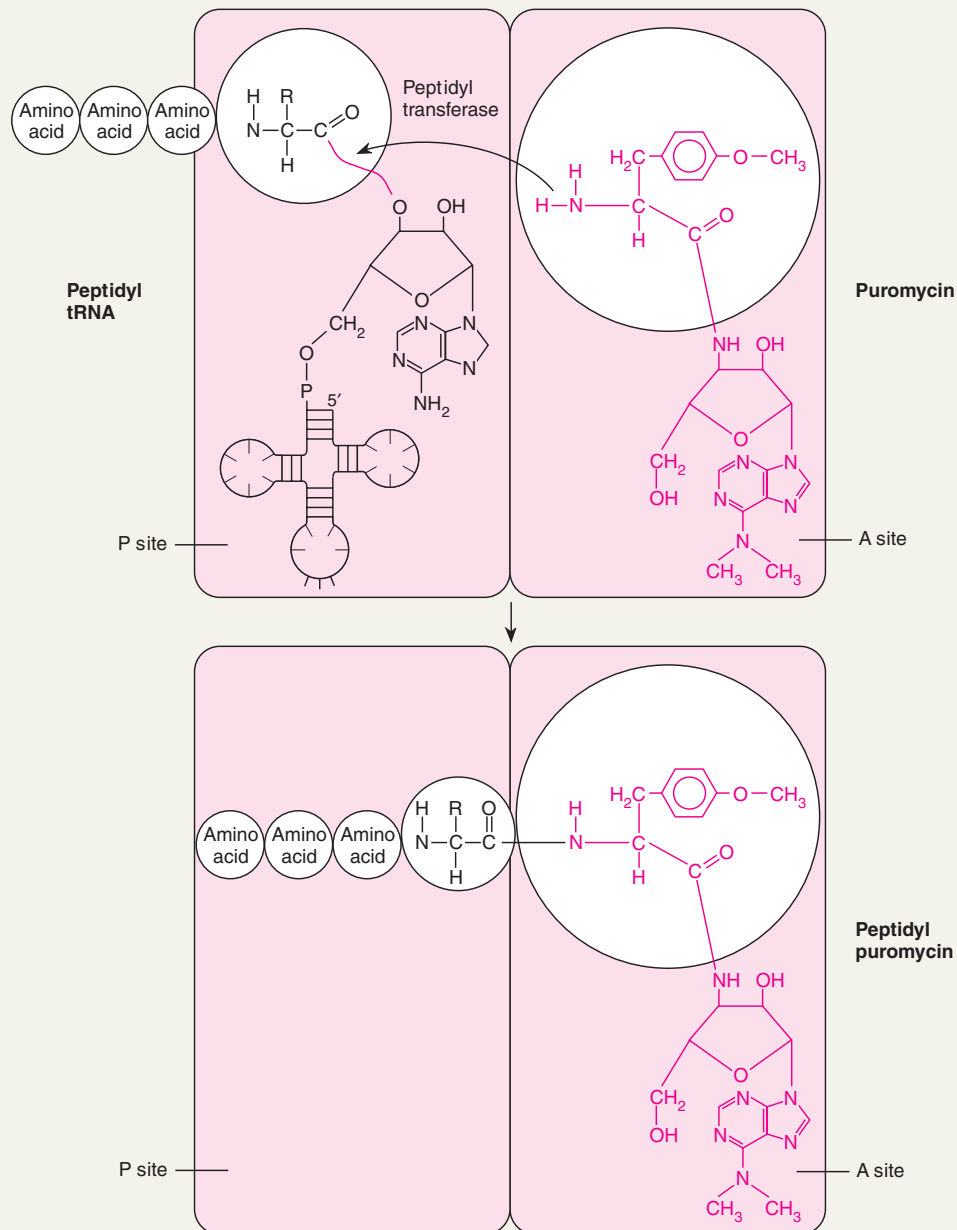


Figure 1 Puromycin is bound to the A site of the ribosome. A peptide bond then forms. Further elongation is prevented, and the chain is terminated.

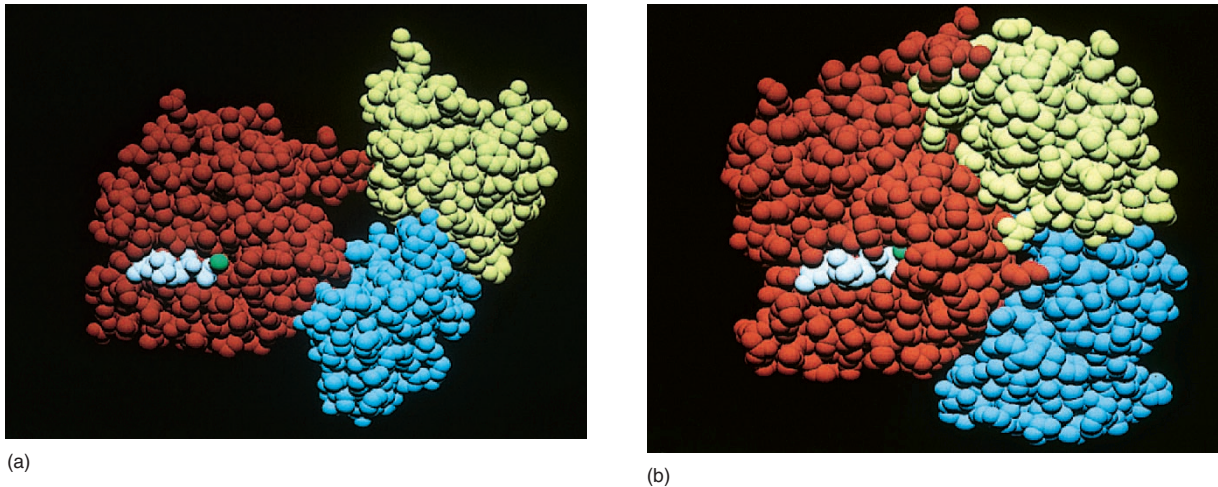


Figure 11.15 Space-filling model of EF-Tu bound with (a) GDP and (b) GTP, showing the change in the protein's structure. Yellow, blue, and red are domains of the protein. The GTP and GDP are in white, with a magnesium ion, Mg^{2+} , in green. When EF-Tu is bound with GDP, there is a visible hole in the molecule. The hole disappears when GTP is bound. The aminoacyl-transfer RNA is believed to bind between the red and yellow domains. (Courtesy of Rolf Hilgenfeld.)

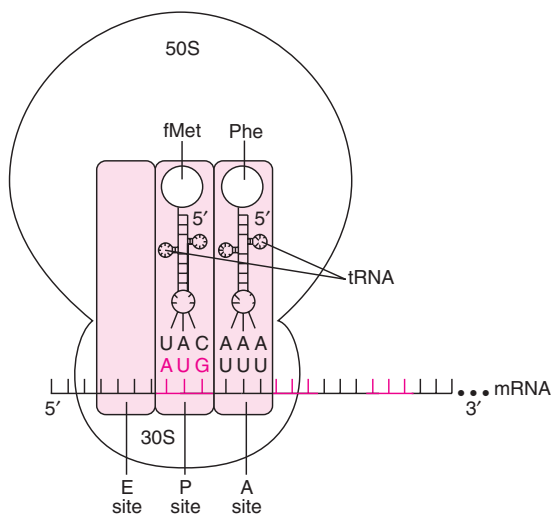


Figure 11.16 A ribosome with two transfer RNAs attached. In this case, the second codon (UUU) is for the amino acid phenylalanine. The two amino acids are next to each other.

protein synthesis, two GTPs are hydrolyzed per peptide bond: one GTP in the release of EF-Tu from the A site, and one GTP in the translocational process of the ribosome after the peptide bond has formed. In addition, every

charged transfer RNA has had an amino acid attached at the expense of the hydrolysis of an ATP to AMP + PP. There is some evidence that the action of EF-Tu hydrolyzes two GTPs.

Termination



Nonsense Codons

Termination of protein synthesis in both prokaryotes and eukaryotes occurs when one of three **nonsense codons** appears in the A site of the ribosome. These codons are UAG (sometimes referred to as *amber*), UAA (*ochre*), and UGA (*opal*). ("Amber," or brown stone, is the English translation of the name Bernstein, a graduate student who took part in the discovery of UAG in R. H. Epstein's lab at the California Institute of Technology. "Ochre" and "opal" are tongue-in-cheek extensions of the first label.) In prokaryotes, three proteins called **release factors (RF)** are involved in termination, and a GTP is hydrolyzed to GDP + P_i .

When a nonsense codon enters the A site on the ribosome, a release factor recognizes it. RF1 and RF2 are class 1 release factors: They recognize stop codons and then promote hydrolysis of the bond between the terminal amino acid and its tRNA in the P site. Class 2 release factors (RF3) do not recognize stop codons, but they stimulate class 1 release factors to act. RF1 recognizes the stop codons UAA and UAG, and RF2 recognizes UAA and UGA (fig. 11.20). Both do so because they have tripep-

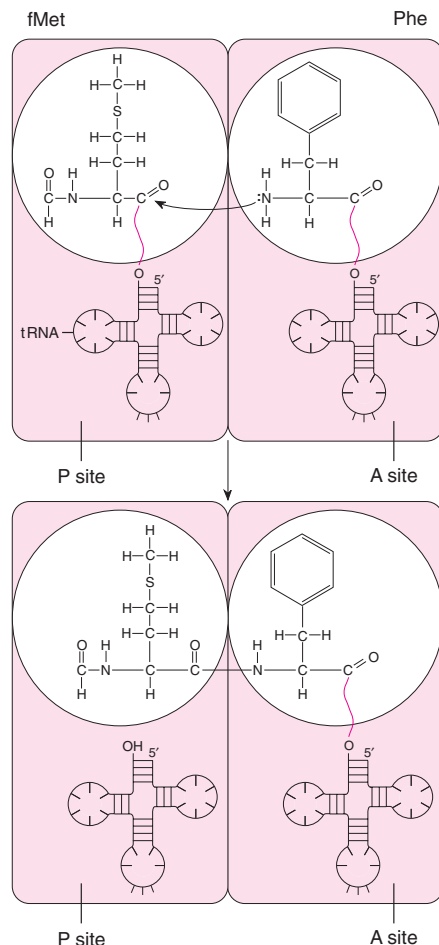


Figure 11.17 Peptide bond formation on the ribosome between N-formyl methionine and phenylalanine. The bond attaching the carboxyl end of the first amino acid to its tRNA is transferred to the amino end of the second amino acid. The first tRNA is now uncharged, whereas the second tRNA has a dipeptide attached.

tides that mimic anticodons to recognize the stop codons: proline-alanine-threonine in RF1 and serine-proline-phenylalanine in RF2. In this **molecular mimicry**, a protein mimics the shape of a nucleic acid in order to function properly.

The next base in the messenger RNA past the stop codon is usually an adenine, required for efficient termination. After the release factors act, with the hydrolysis of a GTP, the ribosome has completed its task of translating mRNA into a polypeptide. Final release of all factors and dissociation of the two subunits of the ribosome take

place with the help of IF3, which rebinds to the 30S subunit, and a **ribosome recycling factor (RRF)**. Table 11.1 compares prokaryotic and eukaryotic translation.

Rate and Cost of Translation

As mentioned, the average speed of protein synthesis is about fifteen peptide bonds per second in prokaryotes. Discounting the time for initiation and termination, an average protein of three hundred amino acids is synthesized in about twenty seconds (the released protein forms its final structure spontaneously or is modified with the aid of other proteins, as we shall see). An equivalent eukaryotic protein takes about 2.5 minutes to be synthesized. The energy cost is at least four high-energy phosphate bonds per peptide bond (two from an ATP during transfer RNA charging, and two from GTP hydrolysis during transfer RNA binding at the A site and translocation), or about twelve hundred high-energy bonds per protein. This cost is very high—about 90% of the energy production of an *E. coli* cell goes into protein synthesis. A high energy cost is presumably the price a living system has to pay for the speed and accuracy of its protein synthesis.

Coupling of Transcription and Translation

In prokaryotes, such as *E. coli*, in which no nuclear envelope exists, translation begins before transcription is completed. Figure 11.21 shows a length of an *E. coli* chromosome. An RNA polymerase is visible on the DNA, transcribing a gene. The messenger RNA, still being synthesized, can be seen extending away from the DNA. Attached to the messenger RNA are about a dozen ribosomes. Since translation starts at the end of the messenger that is synthesized first (5'), an initiation complex can form and translation can begin shortly after transcription begins. As translation proceeds along the messenger, its 5' end will again become exposed, and a new initiation complex can form. The occurrence of several ribosomes translating the same messenger is referred to as a **polyribosome**, or simply a **polysome** (fig. 11.22).

In prokaryotes, most messenger RNAs contain the information for several genes. These RNAs are said to be **polycistronic** (fig. 11.23). (*Cistron*, another term for gene, is defined in chapter 12.) Each gene on the messenger RNA is translated independently: each has a Shine-Dalgarno sequence for ribosome recognition (see fig. 11.12) and an initiation codon (AUG) for fMet. The ribosome that completes the translation of the first gene may or may not continue to the second gene after dissociation. The translation of any gene follows all the steps we have outlined.

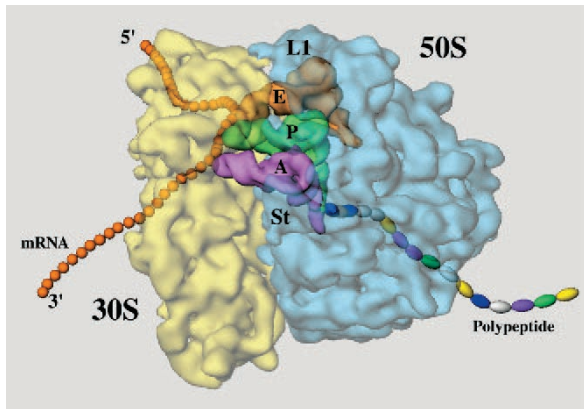
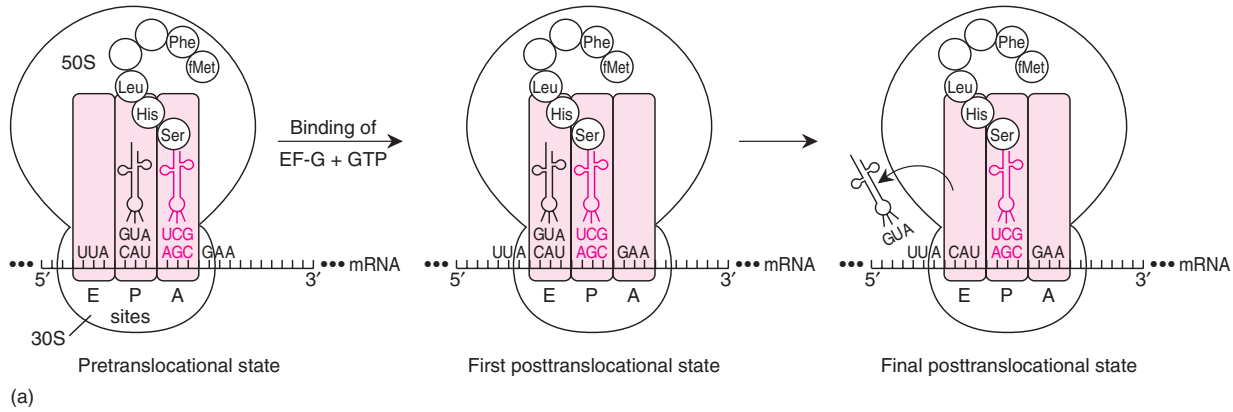
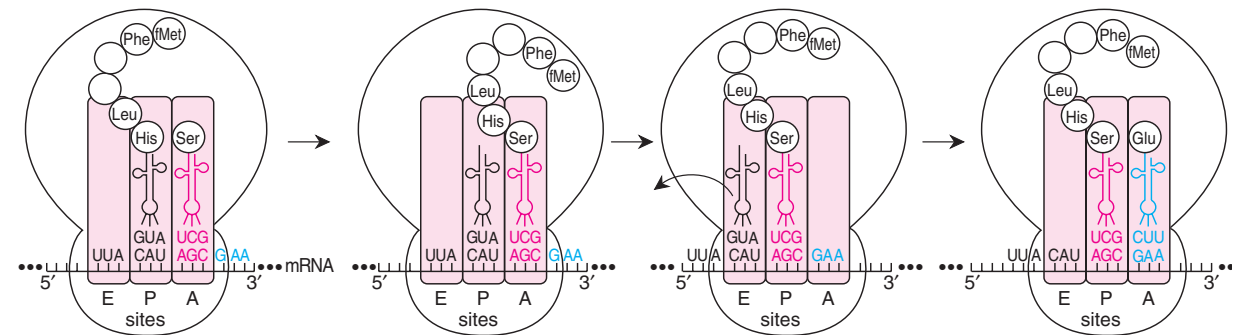


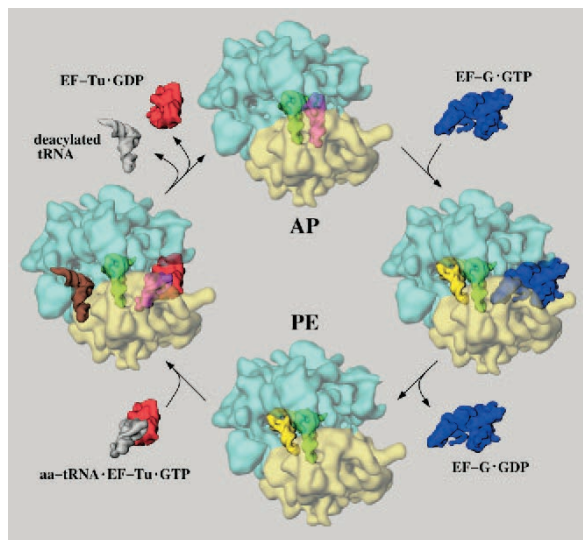
Figure 11.18 (a) EF-G's translocation of the ribosome converts it from a pretranslocational state (P and A sites occupied) to a posttranslocational state (E and P sites occupied). The uncharged transfer RNA in the E site is then ejected. (b) A see-through model of the 70S ribosome of *E. coli* with transfer RNAs in the A, P, and E sites. The structure was determined by cryoEM mapping, an electron microscopic technique using rapidly frozen specimens. The position of the messenger RNA is shown, as well as the stalk of the 50S subunit (St) and one of the polypeptides of the large subunit, L1. ([b] Courtesy of Joachim Frank, Howard Hughes Medical Institute.)

Table 11.1 Some Comparisons Between Prokaryotic and Eukaryotic Translation

	Prokaryotes	Eukaryotes
Initiation codon	AUG, occasionally GUG, UUG	AUG, occasionally GUG, CUG
Initiation amino acid	N-formyl methionine	Methionine
Initiation tRNA	tRNA _f ^{Mct}	tRNA _i ^{Mct}
Interior methionine tRNA	tRNA _m ^{Mct}	tRNA _m ^{Mct}
Initiation factors	IF1, IF2, IF3	eIF factors
Elongation factor	EF-Tu	eEF1α
Elongation factor	EF-Ts	eEF1βγ
Translocation factor	EF-G	eEF2
Release factors	RF1, RF2, RF3, RRF	eRF1, eRF3



(a)



(b)

Figure 11.19 Cycle of peptide bond formation and translocation on the ribosome. (a) After the peptide bond is transferred (fig. 11.17), the ribosome and messenger RNA move over one codon. Now the transfer RNA with the peptide is in the P site, and the A site is again open. In this example, the next transfer RNA that moves into the A site carries glutamic acid. (b) Three-dimensional model of the translocation process minus the mRNA and amino acids. The tRNA in the A site is *pale blue*, the tRNA in the P site is *green*, and the tRNA in the E site is *yellow*, then *brown* when ready to leave. Going clockwise from *a*, in which the A and P sites are occupied: EF-G translocates the ribosome after peptide bond formation and then evacuates the A site. Ef-Tu brings a new charged tRNA to the A site while the E site is emptied. ([b] Courtesy of Joachim Frank.)

In eukaryotes, however, almost all messenger RNAs contain the information for only one gene (**monocistronic**). Since most ribosomal recognition of eukaryotic genes depends on the 5' cap, and since each eukaryotic messenger RNA has only one cap, usually only one polypeptide can be translated for any given messenger RNA. Exceptions occur when the messenger RNAs contain internal ribosome entry sites. Although it is certainly not the rule, the translated peptide can be modified or cleaved into smaller functional peptides. For example, in mice, a single messenger RNA codes for a protein that is later cleaved into epidermal growth factor and at least seven other related peptides. In addition, the same sequence can, in some cases, give rise to alternative proteins through alterna-

tive start codons, termination read-through, or alternative splicing.

More on the Ribosome

In the last chapter, we briefly discussed the shape and composition of the ribosomal subunits. All of the protein and RNA components have been isolated. Assembly pathways are known. We know approximately where the messenger RNA, initiation factors, and EF-Tu are located on the 30S subunit during translation (fig. 11.24; cf. fig. 11.18). We also know where peptidyl transferase activity and EF-G reside on the 50S subunit, which has a cleft leading into a tunnel that passes through the structure. At present, it seems that the nascent peptide passes through this

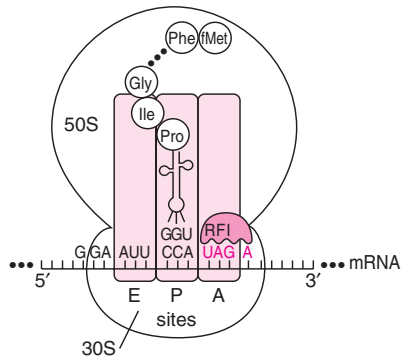


Figure 11.20 Chain termination at the ribosome. One of two release factors recognizes a nonsense codon in the A site. In this case, RF1 recognizes UAG. The complex then falls apart, releasing the peptide.

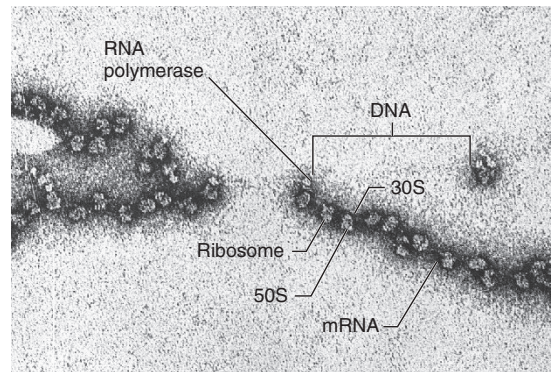
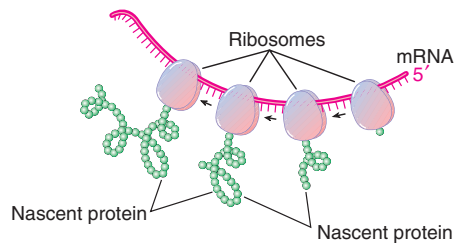
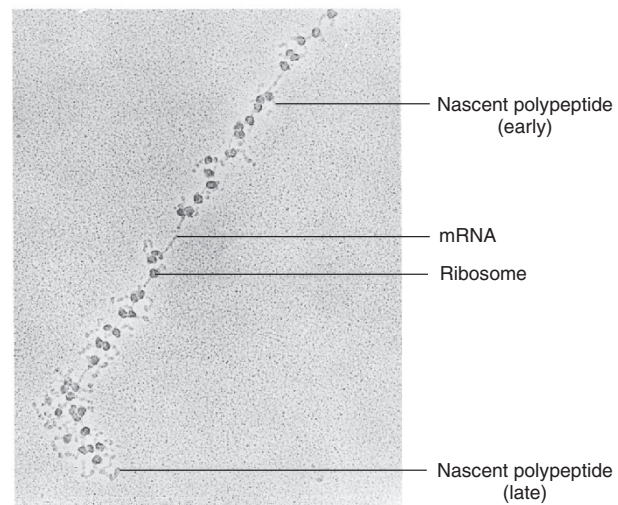


Figure 11.21 A polysome (i.e., multiple ribosomes on the same strand of mRNA). Each ribosome is approximately 250 Å units across. Also visible in this illustration are DNA and RNA polymerase. (Reproduced courtesy of Dr. Barbara Hamkalo, *International Review of Cytology*, (1972) 33:7, fig. 5. Copyright by Academic Press, Inc., Orlando, Florida.)



(a)

Figure 11.22 (a) Protein synthesis at a polysome. Nascent proteins exit from a tunnel in the 50S subunit. Messenger RNA is being translated by the ribosomes while the DNA is being transcribed. (b) A messenger RNA from the midge, *Chironomus tentans*, showing attached ribosomes and nascent polypeptides emerging from the ribosomes. Note the 5' end of the messenger RNA at the upper right (small peptides). Magnification 165,000 $\times$. ([b] Courtesy of S. L. McKnight and O. L. Miller, Jr.)



(b)

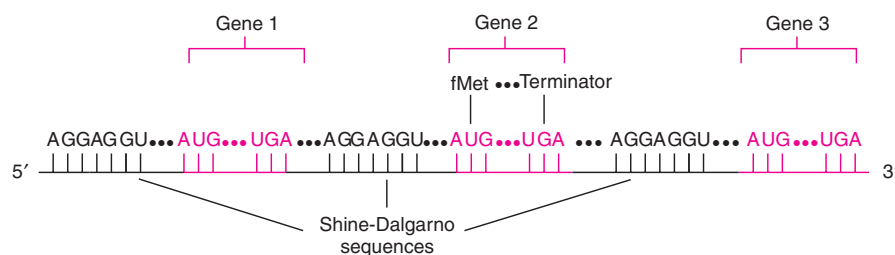


Figure 11.23 A prokaryotic polycistronic mRNA. Note the several Shine-Dalgarno sequences for ribosomal attachment and the initiation and termination codons marking each gene.

tunnel, emerging close to a membrane-binding site (fig. 11.24). The tunnel can hold a peptide length of about forty amino acids. Note that although every ribosome has a membrane-binding site, not all active ribosomes are bound to membranes.

The Signal Hypothesis

Ribosomes are either free in the cytoplasm or associated with membranes, depending on the type of protein being synthesized. Membrane-bound ribosomes, indistinguishable from free ribosomes, synthesize proteins that enter membranes. These proteins either become a part of the membrane or, in eukaryotes, either pass into membrane-bound organelles (e.g., the Golgi apparatus, mitochondria, chloroplasts, vacuoles) or are transported outside



Gunter Blobel (1936–).

(Courtesy of Dr. Gunter
Blobel, Dept. of Cell Biology,
Rockefeller University.)

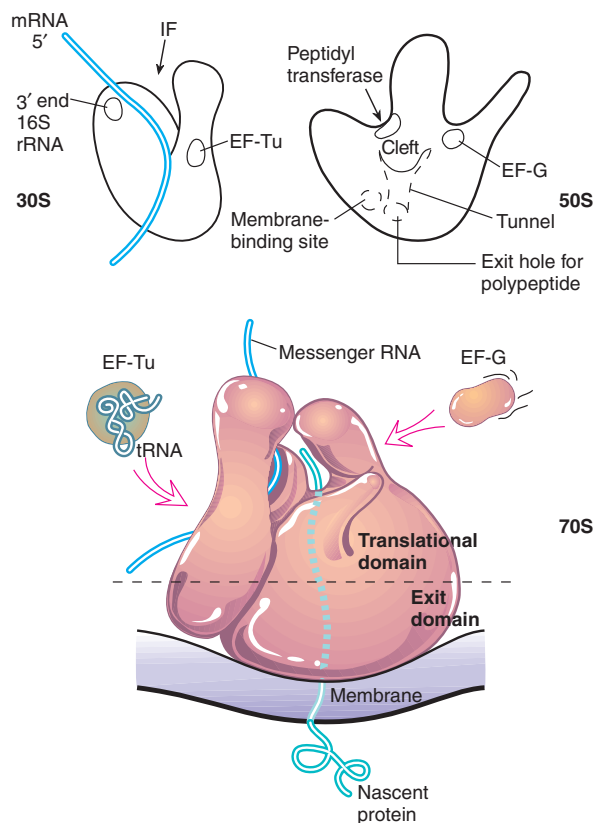


Figure 11.24 Functional sites on the prokaryotic ribosome. The ribosome is synthesizing a protein involved in membrane passage. Note the position of the messenger RNA on the 30S subunit and the cleft, tunnel, and membrane-binding site on the 50S subunit. (From C. Bernabeu and J. A. Lake, *Proceedings of the The National Academy of Sciences*; 79:3111–15, 1982. Reprinted by permission.)

the cell membrane. The **signal hypothesis** of G. Blobel, a 1999 Nobel laureate, and his colleagues, explains the mechanism for membrane attachment. The mechanism applies to both prokaryotes and eukaryotes. Here, we describe it in mammals.

The signal for membrane insertion is coded into the first one to three dozen amino acids of membrane-bound proteins. This **signal peptide** takes part in a chain of events that leads the ribosome to attach to the membrane and to the insertion of the protein. The first step occurs when the signal peptide becomes accessible outside of the ribosome. A ribonucleoprotein particle called the **signal recognition particle (SRP)**, which consists of six different proteins and a 7S RNA about three hundred nucleotides long, recognizes the signal peptide. The complex of signal recognition particle, ribosome, and signal peptide then passes, or diffuses, to a membrane, where the SRP binds to a receptor called a **docking protein (DP)** or signal recognition particle receptor (fig. 11.25). During this time, protein synthesis halts. The ribosome is brought into direct contact with the membrane, and other proteins of the membrane help anchor the ribosome. Protein synthesis then resumes, with the nascent protein usually passing directly into a **translocation channel (translocon)**. Once through the membrane, the signal peptide is cleaved from the protein by an enzyme called **signal peptidase**. A striking verification of this hypothesis came about through recombinant DNA techniques (chapter 13). A signal sequence was placed in front of the α -globin gene, whose protein product is normally not transported through a membrane. When this gene was translated, the ribosome became membrane bound, and the protein passed through the membrane.

Since different proteins enter different membrane-bound compartments (e.g., the Golgi apparatus), some mechanism must direct a nascent protein to its proper

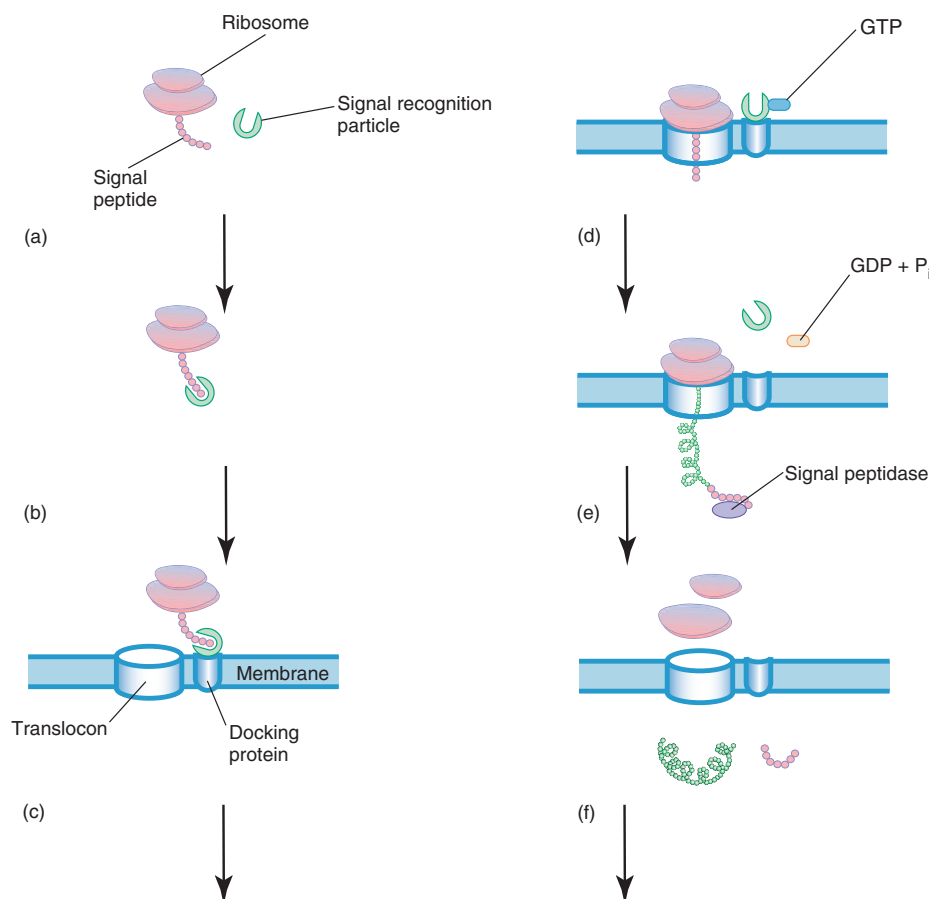


Figure 11.25 The signal hypothesis. A signal recognition particle recognizes a ribosome with a signal peptide, then draws the ribosome to a docking protein located near a translocon in the membrane. With the addition of GTP, the signal recognition particle releases the signal peptide; hydrolysis of the GTP to GDP + P_i causes the signal recognition particle to leave the docking protein. Peptide synthesis then resumes, with the newly synthesized peptide passing through the translocon in the membrane. A signal peptidase on the other side of the membrane removes the signal peptide. When translation is completed, the ribosome dissociates and drops free of the translocon.

membrane. This specificity seems to depend on the exact signal sequence and membrane-bound glycoproteins called *signal-sequence receptors*. Apparently, after the ribosome binds to the docking protein, the signal peptide interacts with a signal-sequence receptor, which presumably determines whether that protein is specific for that membrane. If it is, the remaining processes continue. If not, the ribosome may be released from the membrane.

The signal peptide does not seem to have a consensus sequence like the transcription or translation recognition boxes. Rather, similarities (at least for the endoplasmic reticulum and bacterial membrane-bound proteins) include a positively charged (basic) amino acid (commonly lysine or arginine) near the beginning

(N-terminal end), followed by about a dozen hydrophobic (nonpolar) amino acids, commonly alanine, isoleucine, leucine, phenylalanine, and valine (table 11.2).

Table 11.2 The Signal Peptide of the Bovine Prolactin Protein*

NH ₂ – Met Asp Ser Lys Gly Ser Ser Gln Lys Gly Ser Arg Leu
Leu Leu Leu Leu Val Val Ser Asn Leu Leu Leu Cys Gln Gly Val
Val Ser Thr Pro Val...Asn Asn Cys – COOH

Source: From Sasavage et al., *Journal of Biological Chemistry*, 257:678–81, 1982. Reprinted with permission.

* The vertical line separates the signal peptide from the rest of the protein, which consists of 199 residues.



Figure 11.26 Electron micrograph of a chaperone protein (GroEL) from *E. coli*. Note the hollow, barrel shape of the protein. (Courtesy of Dr. R. W. Hendrix.)

The mitochondrion, which needs to import upwards of one thousand proteins through both inner and outer membranes, poses a specific problem. Recent research has revealed a family of translocation proteins (called Tom proteins) in the outer membrane and a different set of translocation proteins (called Tim proteins) in the inner membrane. These proteins control the passage of proteins synthesized in the cytoplasm into the mitochondrion.

The Protein-Folding Problem

Since biochemist Christian Anfinsen won a 1972 Nobel Prize for showing that the enzyme ribonuclease refolds to its original shape after denaturation in vitro, scientists have believed that the final protein shape (secondary and tertiary structure) forms spontaneously. Recently it has been shown, however, that many proteins do not normally form their final active shape in vivo without the help of proteins called **chaperones** or **molecular chaperones**. The chaperones do not provide the three-dimensional structure of the proteins they help, but rather bind to a protein in the early stages of folding to prevent unproductive folding or to allow denatured proteins to refold correctly. Like human chaperones, they prevent or undo “incorrect interactions,” according to J. Ellis. That is, many proteins have a large number of different structures they could fold into. Many of these structures would have no enzymatic activity or would form functionless aggregates with other proteins. Molecular chaperones allow proteins to fold into a thermodynamically stable and functional configuration. Each cycle of refolding requires ATP energy.

A well-studied class of chaperones is known as the *chaperonins*, or Hsp60 proteins, because they are heat shock proteins about 60 kilodaltons (60,000 daltons) in size. They occur in bacteria, chloroplasts, and mitochondria. One of the best studied of these chaperonins is the protein GroE of *E. coli*. This protein in its active form is composed of two components, GroEL and GroES. GroEL (Hsp60) is made up of two disks, each composed of seven copies of a polypeptide. GroES (Hsp10) is a smaller component composed of seven copies of a small subunit. GroEL forms a barrel in which protein folding takes place (fig. 11.26). The barrel is shaped in such a way that entering proteins of a certain size make contact at interior points in either the upper or lower ring of GroEL (upper ring shown in fig. 11.27). The attachment of GroES, the cap, causes the ring to open outward at the top, stretching the protein inside. This stretching takes energy from the hydrolysis of ATP molecules located inside the rings. When GroES dissociates, the protein can fold into a new, more functional, configuration. If it doesn't, the cycle repeats.

There are several classes of molecular chaperones, proteins of different sizes and shapes that recognize different groups of proteins or protein conformations. GroEL recognizes about 300 different proteins, small enough to fit into the barrel (20–60 kilodaltons) and having hydrophobic surfaces. These include many proteins in the transcription and translation machinery of the cell. Hsp90, another heat shock protein, recognizes proteins involved in signal transduction, discussed in chapter 16. Hsp70 recognizes hydrophobic regions in polypeptide side chains, many of which extend across membranes.

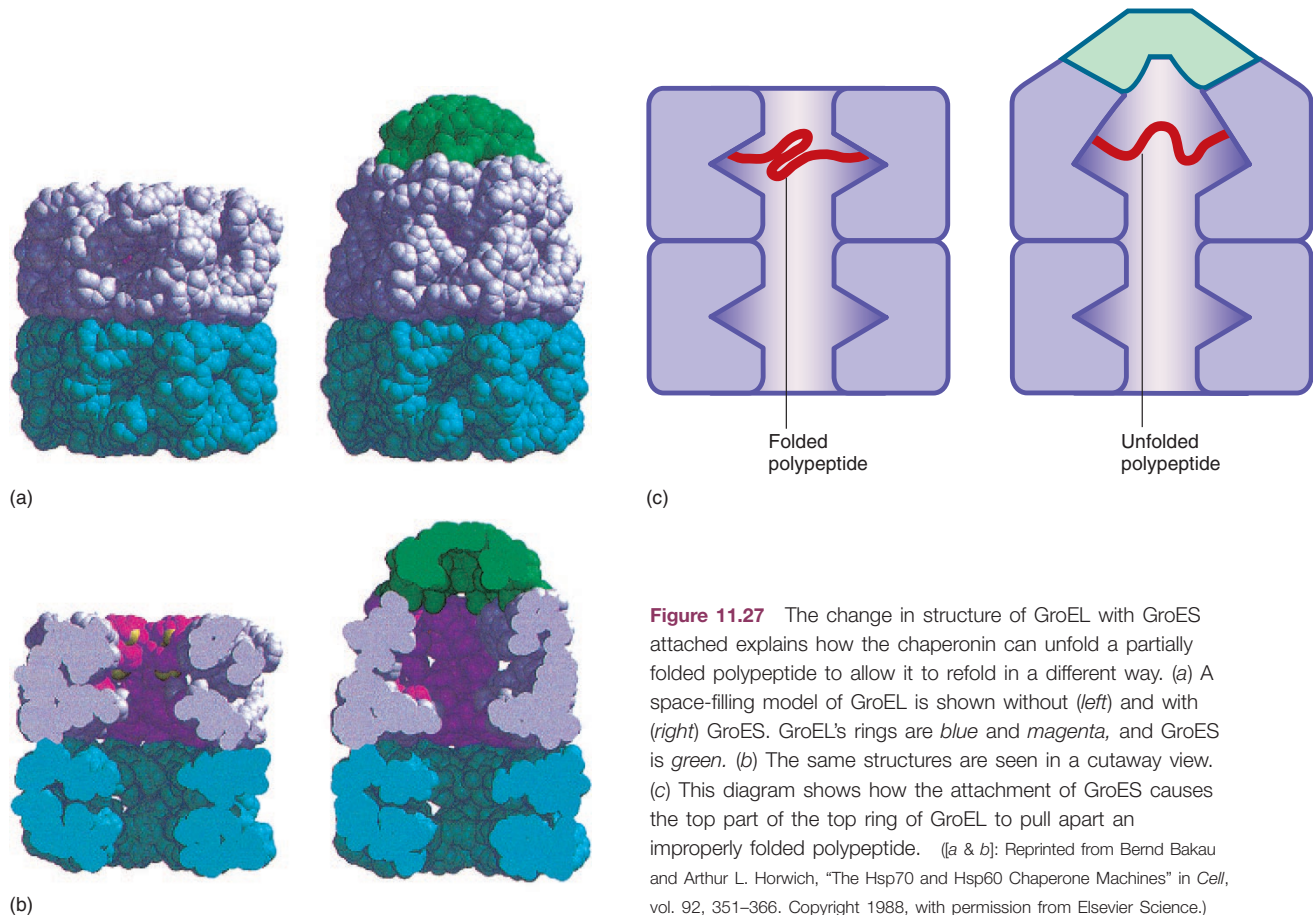


Figure 11.27 The change in structure of GroEL with GroES attached explains how the chaperonin can unfold a partially folded polypeptide to allow it to refold in a different way. (a) A space-filling model of GroEL is shown without (left) and with (right) GroES. GroEL's rings are blue and magenta, and GroES is green. (b) The same structures are seen in a cutaway view. (c) This diagram shows how the attachment of GroES causes the top part of the top ring of GroEL to pull apart an improperly folded polypeptide. ([a & b]: Reprinted from Bernd Bakau and Arthur L. Horwich, "The Hsp70 and Hsp60 Chaperone Machines" in *Cell*, vol. 92, 351–366. Copyright 1988, with permission from Elsevier Science.)

THE GENETIC CODE

Researchers in the mid-1950s assumed that the genetic code consisted of simple sequences of nucleotides specifying particular amino acids. They sought answers to questions such as: Is the code overlapping? Are there nucleotides between code words (punctuation)? How many letters make up a code word (codon)? Logic, along with genetic experiments, supplied some of the answers, but only with the rapidly improving techniques of biochemistry did they eventually decode the genetic language.

Triplet Nature of the Code

Several lines of evidence seemed to indicate that the nature of the code was triplet (three bases in messenger RNA specifying one amino acid). If codons contained only one base, they would only be able to specify four amino acids since there are only four different bases in DNA (or messenger RNA). A couplet code would have $4 \times 4 = 16$ two-base words, or codons, which is still not enough to specify uniquely twenty different amino acids. A triplet

code would allow for $4 \times 4 \times 4 = 64$ codons, which are more than enough to specify twenty amino acids.

Evidence for the Triplet Nature of the Code

The experimental manipulation of mutant genes, primarily by Francis Crick and his colleagues, reinforced the triplet code concept. In these experiments, a chemical mutagen, the acridine dye proflavin, was used to cause inactivation of the rapid lysis (*rII*B) gene of the bacteriophage T4. Proflavin inactivates the gene by either adding or deleting a nucleotide from the DNA (see chapter 12). The *rII* gene controls the plaque morphology of this bacteriophage growing on *E. coli* cells. Rapid-lysis mutants produce large plaques; the wild-type form of the gene, *rII*⁺, results in normal plaque morphology.

Figure 11.28 shows the consequences of adding or deleting a nucleotide. From the point of addition or deletion onward, a **frameshift** causes codons to be read in different groups of three. If a deletion is combined with an addition to produce a double-mutant gene, the frameshift occurs only in the region between the two mu-

tants. If this region is small enough or does not contain coding for vital amino acids, the function of the gene may be restored. Two deletions or two insertions combined will not restore the reading frame. However, Crick and his colleagues found that the combination of three additions or three deletions did restore gene function. This finding led to the conclusion that the genetic code was triplet, be-

cause a triplet code would be put back into the reading frame by three additions or three deletions (fig. 11.29).

Overlap and Punctuation in the Code

Questions still remained: was the code overlapping? Did it have punctuation? Several logical arguments favored a nonpunctuation, nonoverlapping model (fig. 11.30). An overlapping code would be subject to two restrictions. First, a change in one base (a mutation) could affect more than one codon and thus affect more than one amino acid. But studies of amino acid sequences almost always showed that only one amino acid was changed, which argued against codon overlap. Second, certain restrictions affected which amino acids occurred next to each other in proteins. For example, the amino acid UUU coded could never be adjacent to the amino acid coded by AAA because one or both (depending on the number of bases overlapped) of the overlap codons UUA and UAA would always insert other amino acids between them. Overlap, then, seemed to be ruled out since every amino acid appears next to every other amino acid in one protein or another.

Punctuation between codons was also tentatively ruled out. The messenger RNA in the tobacco necrosis satellite virus has just about enough codons to specify its coat protein with no room left for a punctuating base or bases between each codon.

Breaking the Code

Once geneticists had figured out that the genetic code is in nonoverlapping triplets, they turned their attention to the sixty-four codons. They wondered which amino acid, for example, does ACU specify? The work was done in two stages. In the first stage, M. W. Nirenberg, S. Ochoa, and their colleagues made long artificial messenger RNAs and determined which amino acids these messenger RNAs incorporated into protein. In the second stage, specific triplet RNA sequences were synthesized. The researchers then determined the amino acid-transfer RNA complex that was bound by each sequence.

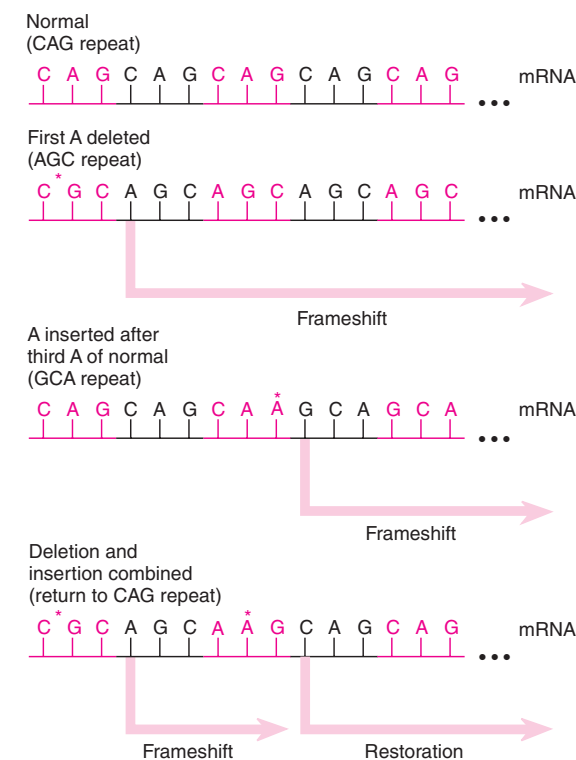


Figure 11.28 Frameshift mutations in a gene result from the addition or deletion of one or several nucleotides (any number other than a multiple of three) in the DNA. The messenger RNA shown here normally has a CAG repeat. A single-base deletion shifts the three-base reading frame to a series of AGC repeats. A later insertion restores the reading frame. Asterisks (*) indicate points of deletion or insertion.

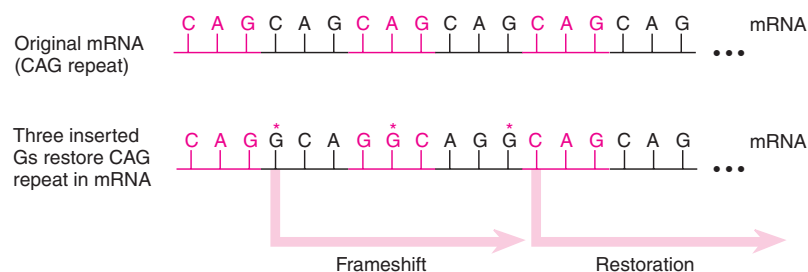


Figure 11.29 The coding frame of CAG repeats is first shifted and then restored by three additions (insertions). Asterisks (*) indicate insertions.



Severo Ochoa (1905–1993).
(Courtesy of Dr. Severo Ochoa.)



Marshall W. Nirenberg
(1934–). (Courtesy of Dr.
Marshall W. Nirenberg.)

Synthetic Messenger RNAs

The ability to synthesize long-chain messenger RNAs resulted from the 1955 discovery of M. Grunberg-Manago and Ochoa of the enzyme **polynucleotide phosphorylase**, which joins diphosphate nucleotides into long-chain, single-stranded polynucleotides. Unlike a polymerase, polynucleotide phosphorylase does not need a primer on which to act. This enzyme is found in all bacteria. (Its main function in the cell is probably the reverse of its use here. It most likely serves as an exonuclease, degrading messenger RNA.) In 1961, Nirenberg and J. H. Matthai added artificially formed RNA polynucleotides of known composition to an *E. coli* ribosomal system and looked for the incorporation of amino acids into proteins.

The system just described is called a **cell-free system**, a mixture primarily of the cytoplasmic components of cells, such as *E. coli*, but missing nucleic acids and membrane components. These systems are relatively easy to create by disrupting and then fractionating whole cells. The systems hold the advantage of containing virtually all the components needed for protein synthesis except the messenger RNAs. Their disadvantages are that they are relatively short-lived (several hours) and are relatively inefficient in translation. However, an added benefit to the *E. coli* cell-free system is that it will translate, albeit inefficiently, RNAs that normally are not translated *in vivo* because they lack translation initiation signals. This feature allowed these scientists to use artificial messenger RNAs that contained no Shine-Dalgarno sequence for ribosomal binding.

Nirenberg and Matthai found that when the enzyme polynucleotide phosphorylase in the *E. coli* cell-free system made uridine diphosphates into a poly-U messenger RNA, phenylalanine residues were incorporated into a polypeptide. Thus, the first code word established was UUU for phenylalanine. Nirenberg and Ochoa and their associates continued the work. They found that AAA was the code word for lysine, CCC was the code word for proline, and GGG was the code word for glycine.

They then made synthetic messenger RNAs by using mixtures of the various diphosphate nucleotides in known proportions. Table 11.3 gives an example. From their experiments, it was possible to determine the bases used in many of the code words, but not their specific order. For example, cysteine, leucine, and valine are all coded by two Us and a G, but the experiment could not sort out the order of these bases (5'-UUG-3', 5'-UGU-3', or 5'-GUU-3') for any one of them. Determining the order required an extra step in sophistication—that is, being able to synthesize known trinucleotides.

Synthetic Codons

Once trinucleotides of known composition could be manufactured, Nirenberg and P. Leder in 1964 developed a “binding assay.” They found that isolated *E. coli* ribosomes, in the presence of high-molarity magnesium chlo-

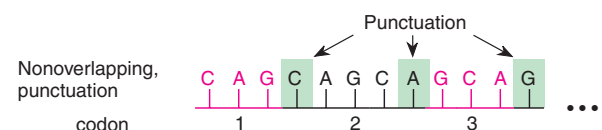
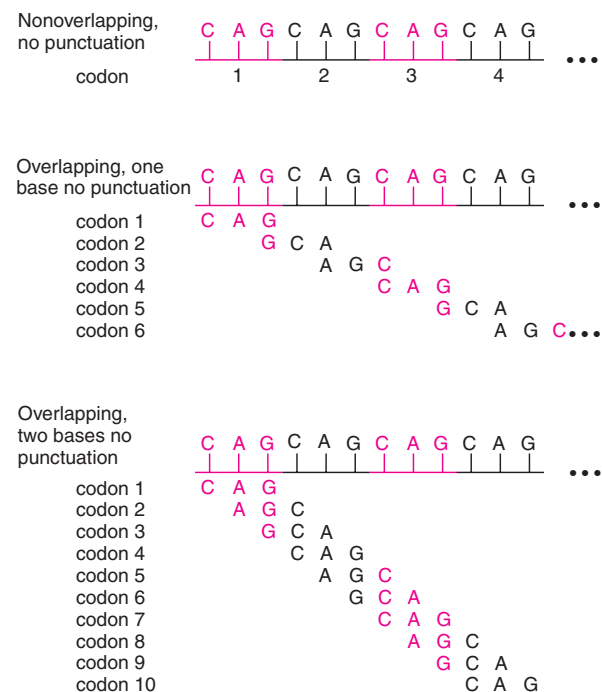


Figure 11.30 The genetic code is read as a nonoverlapping code with no punctuation (top). Before that was proven, it was suggested that the code could overlap by one or two bases (middle) or have noncoding bases (punctuation) between code words (bottom).

Table 11.3 Structure of Artificial mRNA Made by Randomly Assembling Uracil- and Guanine-Containing Ribose Diphosphate Nucleotides with a Ratio of 5U:1G

Codon	Frequency of Occurrence
UUU	$(5/6)^3 = 0.58$
UUG	$(5/6)^2(1/6) = 0.12$
UGU	$(5/6)^2(1/6) = 0.12$
GUU	$(5/6)^2(1/6) = 0.12$
UGG	$(5/6)(1/6)^2 = 0.02$
GUG	$(5/6)(1/6)^2 = 0.02$
GGU	$(5/6)(1/6)^2 = 0.02$
GGG	$(1/6)^3 = 0.005$

ride, could bind trinucleotides as if they were messenger RNAs. Also bound was the transfer RNA that carried the anticodon complementary to the trinucleotide. It was thus possible, using radioactive amino acids, to determine which messenger RNA trinucleotide coded for a particular amino acid. A given synthetic trinucleotide was mixed with ribosomes and aminoacyl-tRNAs, including one radioactively labeled amino acid. The reaction mixture was passed over a filter that would allow everything except the large trinucleotide + ribosome + aminoacyl-tRNA complex to pass through. If the radioactivity passed through the filter, it meant that the radioactive amino acid was not associated with the ribosome. The experiment was then repeated with another labeled amino acid. When the radioactivity appeared on the filter, the investigators knew that the amino acid was affiliated with the ribosome. Thus, that amino acid was coded by the selected trinucleotide codon. In other words, the radioactive amino acid was attached to a transfer RNA whose anticodon was complementary to the trinucleotide codon and thus bound at the ribosome.

Figure 11.31 shows an example. In the figure, the trinucleotide is 5'-CUG-3'. The transfer RNA with the anti-

codon 3'-GAC-5' is charged with leucine. The mixture is passed through a filter. If threonine, or any other amino acid except leucine, is radioactive, the radioactivity passes through the filter. When the experiment is repeated with radioactive leucine, the leucine, and hence the radioactivity, is trapped by the filter. In a short period of time, all of the codons were deciphered (table 11.4).

Wobble Hypothesis

The genetic code is a **degenerate code**, meaning that a given amino acid may have more than one codon. As you can see from table 11.4, eight of the sixteen boxes contain just one amino acid per box. (A box is determined by the first and second positions; e.g., the UUX box, in which X is any of the four bases.) Therefore, for these eight amino acids, the codon need only be read in the first two positions because the same amino acid will be represented regardless of the third base of the codon. These eight groups of codons are termed **unmixed families** of codons. An unmixed family is the four codons beginning with the same two bases that specify a single amino acid. For example, the codon family GUX codes for valine. **Mixed families** code for two amino acids or for stop signals and one or two amino acids.

Six of the mixed-family boxes are split in half so that the codons are differentiated by the presence of a purine or a pyrimidine in the third base. For example, CAU and



Phillip Leder (1934–).
(Courtesy of Dr. Phillip Leder.)

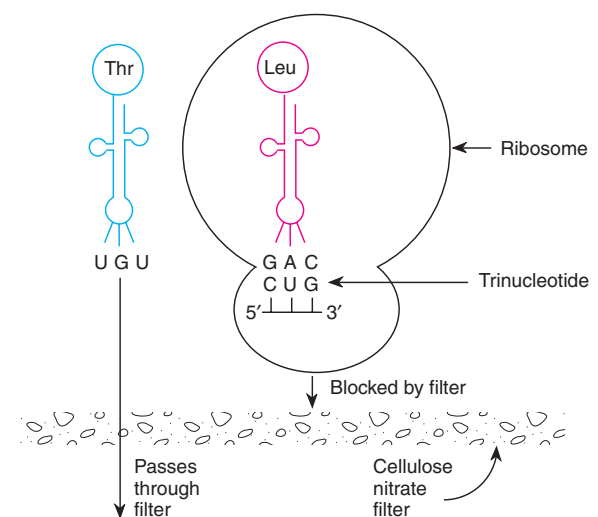


Figure 11.31 The binding assay determines the amino acid associated with a given trinucleotide codon. Transfer RNAs with noncomplementary codons pass through the membrane. Transfer RNAs with anticodons complementary to the trinucleotide bind to the ribosome and do not pass through the filter. When the transfer RNA is charged with a radioactive amino acid, the radioactivity is trapped on the filter.

Table 11.4 The Genetic Code

First Position (5' End)	Second Position				Third Position (3' End)
	U	C	A	G	
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	stop	stop	A
	Leu	Ser	stop	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met (<i>start</i>)	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

CAC both code for histidine; in both, the third base, U (uracil) or C (cytosine), is a pyrimidine. Only two of the families of codons are split differently.

The lesser importance of the third position in the genetic code ties in with two facts about transfer RNAs. First, although there would seem to be a need for sixty-two transfer RNAs—since there are sixty-one codons specifying amino acids and an additional codon for initiation—there are actually only about fifty different transfer RNAs in an *E. coli* cell. Second, a rare base such as inosine can appear in the anticodon, usually in the position that is complementary to the third position of the codon. These two facts lead researchers to believe that some kind of conservation of transfer RNAs is occurring and that rare bases may be involved.

We should mention, to avoid confusion, that both messenger RNA and transfer RNA bases are usually numbered from the 5' side. Thus, the number-one base of the codon is complementary to the number-three base of the anticodon (fig. 11.32). Thus, the codon base of lesser importance is the number-three base, whereas its complement in the anticodon is the number-one base.

Since the first position of the anticodon (5') is not as constrained as the other two positions, a given base at that position may be able to pair with any of several bases in the

third position of the codon. Crick characterized this ability as **wobble** (fig. 11.33). Table 11.5 shows the possible pairings that would produce a transfer RNA system compatible with the known code. For example, if an isoleucine transfer RNA has the anticodon 3'-UAI-5', it is compatible with the three codons for that amino acid (see table 11.4): 5'-AUU-3', 5'-AUC-3', and 5'-AUA-3'. That is, inosine in the first (5') position of the anticodon can recognize U, C, or A in the third (3') position of the codon, and thus one transfer RNA complements all three codons for isoleucine.

Universality of the Genetic Code

Until 1979, scientists concluded that the genetic code was universal. That is, the codon dictionary (see table 11.4) was the same for *E. coli*, human beings, and oak trees, as well as all other species studied up to that time. The universality of the code was demonstrated, for example, by taking the ribosomes and messenger RNA from rabbit reticulocytes and mixing them with the aminoacyl-tRNAs and other translational components of *E. coli*. Rabbit hemoglobin was synthesized.

In 1979 and 1980, however, researchers noted discrepancies when sequencing mitochondrial genes for struc-

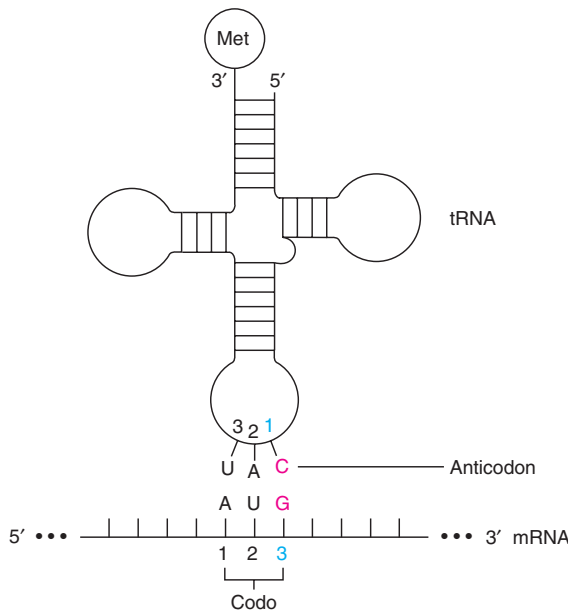


Figure 11.32 Codon and anticodon base positions are numbered from the 5' end. The 3' position in the codon (5' in the anticodon) is the wobble base.

tural proteins (see chapters 13 and 17). It was discovered that there were two kinds of deviations from universality in the way mitochondrial transfer RNAs read the code. First, fewer transfer RNAs were needed to read the code. Second, there were several instances in which the mitochondrial and cellular systems interpreted a codon differently.

According to Crick's wobble rules (see table 11.5), thirty-two transfer RNAs (including one for initiation) can complement all sixty-one nonterminating codons. Unmixed families require two transfer RNAs, and mixed families require one, two, or three transfer RNAs, depending on the family. The yeast mitochondrial coding system apparently needs only twenty-four transfer RNAs. The reduction in numbers is accomplished primarily by having only one transfer RNA recognize each unmixed family

(table 11.6; cf. table 11.4). Because mitochondrial transfer RNAs for unmixed families of codons have a U in the first (wobble) position of the anticodon, apparently, given the structure of the mitochondrial transfer RNAs, the U can pair with U, C, A, or G. Presumably, evolutionary pressure has minimized the number of transfer RNA genes in the DNA of the mitochondrion, in keeping with its small size. Reduction from thirty-two to twenty-four is a 25% savings. (Recent evidence suggests that mammalian mitochondria may need only twenty-two transfer RNAs.)

It has also been found that yeast mitochondria read the CUX family as threonine rather than as leucine (tables 11.4 and 11.6) and the terminator UGA (opal) as tryptophan rather than as termination. However, there appear to be differences among different groups of organisms reading the CUX family. Human and *Neurospora* mitochondria appear to read the CUX codons as leucine, just as cellular systems do. Of the groups so far analyzed, only yeast reads the CUX family as threonine. Similarly, human and *Drosophila* mitochondria read AGA and AGG as stop signals rather than as arginine (table 11.7).

In 1985, it was discovered that *Paramecium* species read the UAA and UAG stop codons as glutamine within the cell. In addition, a prokaryote (*Mycoplasma capricolum*) reads UGA as tryptophan. We do not yet know how general this finding is: scientists have scrutinized the genetic code of very few species. We can thus conclude that the genetic code seems to have universal tendencies among prokaryotes, eukaryotes, and viruses. Mitochondria, however, read the code slightly differently: different wobble rules apply, and mitochondria and cells read at least one terminator and one unmixed family of codons differently. Also, the mitochondrial discrepancies are not universal among all types of mitochondria. Further work, involving the sequencing of more mitochondrial DNAs, should elucidate the pattern of discrepancies among the mitochondria of diverse species. We also now know that not every organism reads all codons in the same way. Ciliated protozoa and a mycoplasma read some stop signals as coding for amino acids. Nuclear variants are known in the following codons: CUG, AUA, UAA, UAG, UGA, CGG, and AGA. Mitochondrial variants are known in CUX, AUA, UAA, UAG, AAA, UGA, CGX, AGA, and AGG.

One other type of variation of codon reading occurs: *site-specific variation*, in which the interpretation of a codon depends on its specific location. We are already familiar with the fact that GUG and, rarely, UUG can serve as prokaryotic initiation codons. This means that they are recognized by $\text{tRNA}_{\text{f}}^{\text{Met}}$. However, they are not recognized by $\text{tRNA}_{\text{m}}^{\text{Met}}$ (i.e., GUG and UUG are not misread internally in messenger RNAs). In some cases, two of the termination codons (UGA and UAG, but not UAA) are misinterpreted as codons for amino acids. That is, termination will not occur at the normal place, resulting in a longer-than-usual protein. In some cases, these "read-through" proteins are vital—the

Table 11.5 Pairing Combinations at the Third Codon Position

Number-one Base in tRNA (5' End)	Number-three Base in mRNA (3' End)
G	U or C
C	G
A	U
U	A or G
I	A, U, or C

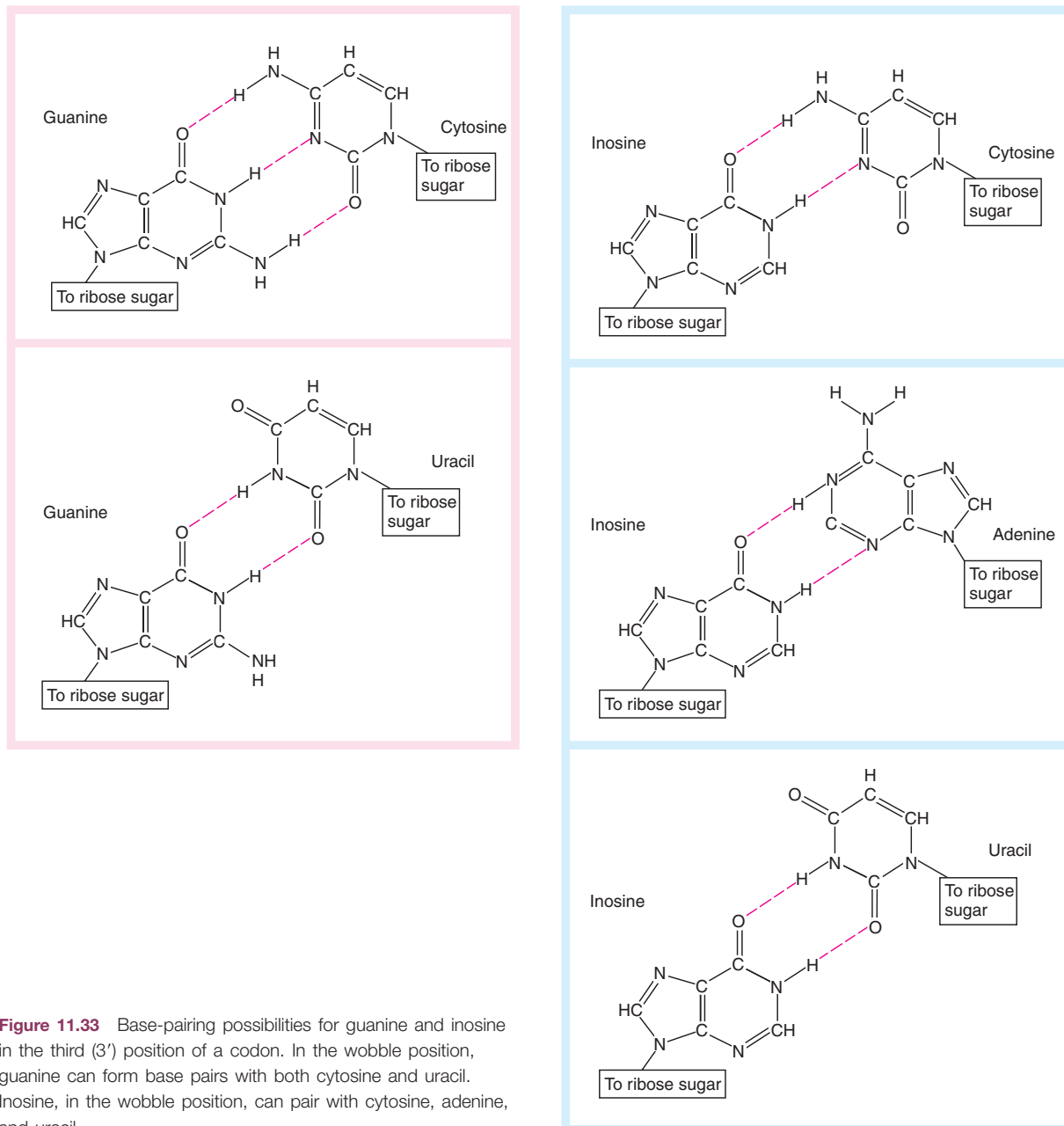


Figure 11.33 Base-pairing possibilities for guanine and inosine in the third (3') position of a codon. In the wobble position, guanine can form base pairs with both cytosine and uracil. Inosine, in the wobble position, can pair with cytosine, adenine, and uracil.

organism depends on their existence. For example, in the phage Q β , the coat-protein gene is read through about 2% of the time. Without this small number of read-through proteins, the phage coat cannot be constructed properly.

One last example of site-specific variation involves the amino acid selenocysteine (cysteine with a selenium atom replacing the sulfur; see fig. 11.1). Although many proteins have unusual amino acids, almost all are due to posttranslational modifications of normal amino acids.

However, the amino acid selenocysteine is inserted directly into some proteins, such as formate dehydrogenase in *E. coli*, which has selenium in its active site. Selenocysteine is inserted into the protein by a novel transfer RNA that recognizes the termination codon, UGA, if that codon is involved in a particular stem-loop secondary structure in the messenger RNA. The selenocysteine transfer RNA is originally charged with a serine that is then modified to a selenocysteine. In addition to the

Table 11.6 The Genetic Code Dictionary of Yeast Mitochondria*

First Position (5' End)	Second Position				Third Position (3' End)
	U	C	A	G	
U	Phe AAG	Ser AGU	Tyr AUG	Cys ACG	U
	Leu AAU		stop	Trp ACU	C
C	Thr GAU	Pro GGU	His GUG	Arg GCA	A
			Gln GUU		G
A	Ile UAG	Thr UGU	Asn UUG	Ser UCG	U
	Met UAC		Lys UUU	Arg UCU	C
G	Val CAU	Ala CGU	Asp CUG	Gly CCU	A
			Glu CUU		G

Source: Data from S. Bonitz, et al., "Codon recognition rules in yeast mitochondria," *Proceedings of the National Academy of Sciences* 77:3167–70, 1980.

* Anticodons (3' → 5') are given within boxes. (The ACU Trp anticodon is predicted.)

Table 11.7 Common and Alternative Meanings of Codons

Codon	General Meaning	Alternative Meaning
CUX	Leu	Thr in yeast mitochondria
AUA	Ile	Met in mitochondria of yeast, <i>Drosophila</i> , and vertebrates
UGA	Stop	Trp in mycoplasmas and mitochondria other than higher plants
AGA/AGG	Arg	Stop in mitochondria of yeast and vertebrates
		Ser in mitochondria of <i>Drosophila</i>
CGG	Arg	Trp in mitochondria of higher plants
UAA/UAG	Stop	Gln in ciliated protozoa
UAG	Stop	Ala or Leu in mitochondria of some higher plants

stem-loop structure 3' (downstream) from the amber codon (UAG), a selenocysteine elongation factor (SELB) is also needed at the ribosome. This same mechanism may occur in eukaryotes, but not all of the components have yet been identified.

Evolution of the Genetic Code

It has been theorized that the genetic code has wobble in it because it originally arose from a code in which only the first two bases were needed for the small number of amino acids in use several billion years ago. As new amino acids with useful properties became available, they were incorporated into proteins by a code modified by the third base, albeit with less specificity. This view has support from the fact that codons starting with the same

nucleotide come from the same biosynthetic pathway. This indicates that in early evolution, as biosynthetic pathways were extended to new amino acids, the newcomers were incorporated by use of the second and third bases of the code.

However, the question remains as to whether the genetic code is highly evolved or just a "frozen accident." In other words, is there a relationship between the codons and the amino acids they code for, or is the code just one of many random possibilities? Recent computer simulations of random codes indicate that the current genetic code is far outside the range of random in its ability to protect the organism from mutation. This suggests that the genetic code is not a frozen accident, but rather is highly evolved. Numerous examples in the current code support this view.

For example, in the unmixed codon family 5'-CUX-3', any mutation in the third position produces another codon for the same amino acid. Wobble in the third position and codon arrangement ensures that less than half of the mutations in the third codon position result in the specification of a different amino acid.

There are also patterns in the genetic code in which the mutation of one codon to another results in an amino acid of similar properties. A high probability exists that such a mutation will produce a functional protein. All the codons with U as the middle base, for example, are for amino acids that are hydrophobic (phenylalanine, leucine, isoleucine, methionine, and valine). Mutation in

the first or third positions for any of these codons still codes a hydrophobic amino acid. Both of the two negatively charged amino acids, aspartic acid and glutamic acid, have codons that start with GA. All of the aromatic amino acids—phenylalanine, tyrosine, and tryptophan (see fig. 11.1)—have codons that begin with uracil. Such patterns minimize the negative effects of mutation.

This chapter completes the discussion of the mechanics of gene expression. The next chapter deals with recombinant DNA technology, followed by several chapters concerned with the control of gene expression in both prokaryotes and eukaryotes.

S U M M A R Y

STUDY OBJECTIVE 1: To study the mechanism of protein biosynthesis, in which organisms, using the information in DNA, string together amino acids to form proteins 281–303

A charged transfer RNA has an anticodon at one end and a specific amino acid at the other end. The transfer RNAs are charged with the proper amino acid by aminoacyl-tRNA synthetase enzymes that incorporate the energy of ATP into amino acid-tRNA bonds. Hence, no additional source of energy is needed during peptide bond formation. During protein synthesis, the translation apparatus at the ribosome recognizes the transfer RNA. Through complementarity, the anticodon pairs with a messenger RNA codon.

An initiation complex forms at the start of translation. In prokaryotes, this complex consists of the messenger RNA, the 30S subunit of the ribosome, the initiator transfer RNA with N-formyl methionine (fMet-tRNA^{fMet}), and the initiation factors IF1, IF2, and IF3. The 50S ribosomal subunit is then added and A and P sites form in the resulting 70S ribosome. The charged N-formyl methionine transfer RNA is in the P site. A GTP is hydrolyzed, and the initiation factors are released.

A transfer RNA enters the A site, which requires the involvement of elongation factors EF-Ts and EF-Tu (in *E. coli*). At least one GTP hydrolysis releases the elongation factor, EF-Tu, which had originally brought the charged transfer RNA to the ribosome. Peptidyl transferase, which appears to be a ribozymic component of the 50S ribosomal subunit, transfers the amino acid from the transfer RNA in the P site to the amino end of the amino acid on the transfer RNA in the A site.

With the help of elongation factor G (EF-G), the ribosome translocates in relation to the messenger RNA. The depleted transfer RNA is moved from the P site to the E site, where it is released; the transfer RNA with the growing peptide is moved into the P site. EF-G is then released. Elongation and translocation continue until a nonsense codon enters the A site. With the aid of the release factors RF1 and

RF2, the protein is released, and the messenger RNA-ribosome complex dissociates. Eukaryotes have slightly more complex processes involving several more proteins.

Proteins pass through membranes with the help of a signal peptide synthesized at their N-terminal ends. Proteins fold into their final, functional configurations with the help of molecular chaperones, proteins that aid the folding process.

Many antibiotics interfere with translation in prokaryotes. Puromycin, streptomycin, tetracycline, and chloramphenicol all act at the ribosome. Studying the mode of action of these antibiotics has provided insights into the mechanism of the translation process.

STUDY OBJECTIVE 2: To examine the genetic code 304–312

The genetic code was first assumed to be triplet because of logical arguments regarding the minimum size of codons. With his work on deletion and insertion mutants, Crick provided evidence that the code was triplet. Part of the code was worked out initially with the synthesis of long, artificial messenger RNAs and then the synthesis of specific trinucleotide codons. Crick's wobble hypothesis accounts for the fact that fewer than sixty-one transfer RNAs can read the entire genetic code. Fewer transfer RNAs are needed because additional complementary base pairings occur in the third position (3') of the codon.

The rule of universality of the genetic code has to be modified in light of findings regarding mitochondrial transfer RNAs; only twenty-four are needed to read the code. In addition, some sense codons are interpreted differently in mitochondrial systems; some nonmitochondrial systems read stop codons differently (a mycoplasma and ciliated protozoan); and some site-specific variation in codon reading also occurs. The structure of the code in both cells and mitochondria seems to protect the cell against a good deal of potential mutation.

S O L V E D P R O B L E M S

PROBLEM 1: What is the energy requirement of protein biosynthesis?

Answer: The cost of adding one amino acid to a growing polypeptide is four or five high-energy bonds: two from an ATP during the charging of the transfer RNA, and two or three from the hydrolysis of GTPs during transfer RNA binding to the A site of the ribosome and during translocation. Thus, for an average protein of three hundred amino acids, there is a cost of 1,200 to 1,500 high-energy bonds.

PROBLEM 2: What are the start and stop signals of translation?

Answer: Once a messenger RNA is attached at the ribosome, the start signal is the methionine initiation codon (usually AUG), whereas the stop signal is one of the three nonsense codons (UAA, UAG, and UGA). Binding to the ribosome in order to position the messenger RNA in relation

to the A and P sites differs in prokaryotes and eukaryotes. In prokaryotes, the Shine-Dalgarno sequence allows the 16S ribosomal RNA and the messenger RNA to form hydrogen bonds, locating the beginning of the messenger RNA at the ribosome. In eukaryotes, the 5' cap is usually recognized by the ribosome, and the ribosome then proceeds to scan the messenger RNA for the initiation codon.

PROBLEM 3: What amino acids could replace methionine if a one-base mutation occurred?

Answer: The codon for methionine (internal as well as initiation) is AUG. If the A is replaced, we would get UUG (Leu), CUG (Leu), or GUG (Val); if the U is replaced, we would get AAG (Lys), ACG (Thr), or AGG (Arg); and if the G is replaced, we would get AUA (Ile), AUU (Ile), or AUC (Ile). Hence, a one-base change in the codon for methionine could result in any of six different amino acids.

E X E R C I S E S A N D P R O B L E M S*

INFORMATION TRANSFER

1. Given the following end part of a gene, which will be transcribed and then translated into a pentapeptide, provide the base sequence for its messenger RNA. Give the anticodons on the transfer RNAs by making use of wobble rules. What amino acids are incorporated? Draw the actual structure of the pentapeptide.

3'-TACAATGGCCCTTTTATC-5'

5'-ATGTTACCGGGAAAATAG-3'

2. Give an alternative translation mechanism that would require only one transfer RNA site on the ribosome.
3. Draw the details of a moment in time at the ribosome during the translation of the messenger RNA produced in problem 1. Include in the diagram the ribosomal sites, the transfer RNAs, and the various nonribosomal proteins involved.
4. How do prokaryotic and eukaryotic ribosomes recognize the 5' end of messenger RNAs? Could eukaryotic messenger RNAs be polycistronic?
5. How many aminoacyl-tRNA synthetases are there? What do they use for recognition signals?
6. What are the similarities and differences among the three nonsense codons? Using the wobble rules, what are their theoretical anticodons?

7. Describe an experiment that demonstrates that the transfer RNA, and not its amino acid, is recognized at the ribosome during translation.
8. Other than the antibiotics named in the chapter, suggest five "theoretical" antibiotics that could interfere with the prokaryotic translation process.
9. How many single-base deletions are required to restore the reading frame of a messenger RNA? Give an example.
10. A "nonsense mutation" is one in which a codon for an amino acid changes to one for chain termination. Give an example. What are its consequences?
11. The reverse situation to problem 10 is a mutation from a nonsense codon to a codon for an amino acid. Give an example. What are its consequences?
12. What are the consequences when an internal methionine codon recognizes a prokaryotic initiation transfer RNA?
13. What role does EF-Ts play in elongation? EF-Tu? What are their eukaryotic equivalents?
14. What roles do RF1 and RF2 play in chain termination? What are their eukaryotic equivalents?
15. What is a signal peptide? What role does it play in eukaryotes? What is its fate?
16. Why doesn't puromycin disrupt eukaryotic translation?

*Answers to selected exercises and problems are on page A-12.

17. A peptide, fifteen amino acids long, is digested by two methods, and each segment is sequenced according to the Edman degradation technique (see box 11.1). The fifteen amino acids are denoted by the letters A through O, with F as the N-terminal amino acid. If the segments are as follows, what is the sequence of the original peptide?

Method 1: CABHLN; FGKI; OEDJM

Method 2: KICAB; JM; FG; HLNOED

18. In human hemoglobin, the β chain is 146 amino acids long. What is the minimum length of RNA needed to make this protein?
19. Part of a DNA strand to be transcribed has the following sequence:

3'-TACTAACTTACGCTCGCCTCA-5'

- What is the sequence of RNA transcribed from this part of the strand?
- What sequence of amino acids does the RNA produce?

THE GENETIC CODE

20. If DNA contained only the bases cytosine and guanine, how long would a code word have to be? How could we tell if this DNA were double-stranded?
21. If an artificial messenger RNA contains two parts uracil to one of cytosine, name the amino acids and the proportions in which they should be incorporated into protein.
22. What would be proved or disproved if an organism were discovered that did not follow any of the rules of the codon dictionary? Would we expect organisms from another galaxy (if they exist) to use our codon dictionary?
23. What would the genetic code dictionary (see table 11.4) look like if wobble occurred in the second position rather than the third (i.e., if an unmixed family of codons were of the form GXU)?

24. In experiments using repeating polymers, $(\text{GCGC})_n$ incorporates alanine and arginine into polypeptides, and $(\text{CGGCGG})_n$ incorporates arginine, glycine, and alanine. What codon can probably be assigned to glycine?
25. If poly-G is used as a messenger RNA in an incorporation experiment, glycine is incorporated into a polypeptide. If poly-C is used, proline is incorporated. If both poly-G and poly-C are used, no amino acids are incorporated into protein. Why?
26. A protein has leucine at a particular position. If the codon for leucine is CUC, how many different amino acids might appear as the result of a single-base substitution?
27. Polymers of $(\text{GUA})_n$ result in the incorporation of only two different amino acids rather than three, as for most other three-base polymers. Why?
28. The sixth amino acid in the β chain of normal human hemoglobin is glutamate. Two different mutations of this codon substitute valine and lysine. What is the likely codon for glutamate?
29. A normal protein has the following C-terminal amino acid sequence: *ser-thr-lys-leu*-COOH. A mutant is isolated with the following sequence: *ser-thr-lys-leu-leu-phe-arg*-COOH. What has probably happened to produce the mutant protein?
30. A segment of a normal protein and three different mutants appears as follows:

normal	_____gly-ala-ser-his-cys-leu-phe_____
mutant 1	_____gly-ala-ser-his_____
mutant 2	_____gly-ala-ser-leu-cys-leu-phe_____
mutant 3	_____gly-val-ala-ile-ala-ser_____

What is the probable sequence of bases in the normal RNA?

31. A normal protein has histidine in a given position. Four mutants are isolated and determined to have either tyrosine, glutamine, proline, or leucine in place of histidine. What are the possible codon assignments, and what codon is probably used for histidine?

CRITICAL THINKING QUESTIONS

- How do transcriptional and translational signals interact?
- Why is the E site necessary in translation?

12

DNA *Its Mutation, Repair, and Recombination*

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5. To investigate the processes of DNA repair and recombination 339

STUDY OUTLINE

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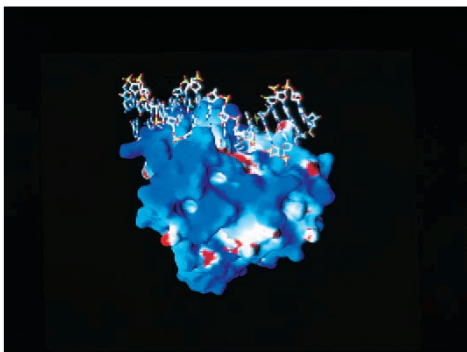
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Box 12.3 Adaptive Mutation 339



Computer-generated space-filling model of a DNA
enzyme repairing damaged DNA.

(© James King-Holmes/SPL/Photo Researchers, Inc.)

The mutation, repair, and recombination of DNA are treated together in this chapter because the three processes have much in common. The physical alteration of DNA is involved in each; repair and recombination share some of the same enzymes. We progress from mutation—the change in DNA—to repair of damaged DNA, and, finally, to recombination, the new arrangement of pieces of DNA.

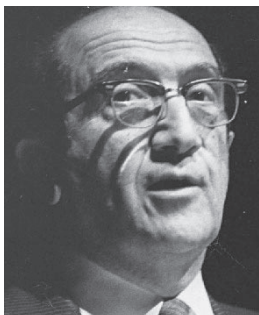
MUTATION



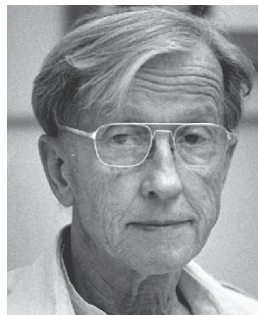
The concept of mutation (a term coined by de Vries, a re-discoverer of Mendel) is pervasive in genetics. **Mutation** is both the process by which a gene (or chromosome) changes structurally and the end result of that process. Without alternative forms of genes, the biological diversity that exists today could not have evolved. Without alternative forms of genes, it would have been virtually impossible for geneticists to determine which of an organism's characteristics are genetically controlled. Studies of mutation provided the background for our current knowledge in genetics.

Fluctuation Test

In 1943, Salvador Luria and Max Delbrück published a paper entitled “Mutations of Bacteria from Virus Sensitivity to Virus Resistance.” This paper ushered in the era of bacterial genetics by demonstrating that the phenotypic variants found in bacteria are actually attributable to mutations rather than to induced physiological changes. Very little work had previously been done in bacterial genetics because of the feeling that bacteria did not have “normal” genetic systems like the systems of fruit flies and corn. Rather, bacteria were believed to respond to environmental change by physiological adaptation, a



Salvador E. Luria
(1912–1991). (Courtesy of
Dr. S. E. Luria.)



Max Delbrück (1906–1981).
(Courtesy of Dr. Max Delbrück.)

non-Darwinian view. As Luria said, bacteriology remained “the last stronghold of Lamarckism” (the belief that acquired characteristics are inherited).

What Causes Genetic Variation?

Luria and Delbrück studied the Ton^r (phage T1-resistant) mutants of a normal Ton^s (phage T1-sensitive) *Escherichia coli* strain. They used an enrichment experiment, as described in chapter 7, wherein a petri plate is spread with *E. coli* bacteria and T1 phages. Normally, no bacterial colonies grow on the plate: all the bacteria are lysed. However, if one of the bacterial cells is resistant to T1 phages, it produces a bacterial colony, and all descendants of this colony are T1 resistant. There are two possible explanations for the appearance of T1-resistant colonies:

1. Any *E. coli* cell may be induced to be resistant to phage T1, but only a very small number actually are. That is, all cells are genetically identical, each with a very low probability of exhibiting resistance in the presence of T1 phages. When resistance is induced, the cell and its progeny remain resistant.
2. In the culture, a small number of *E. coli* cells exist that are already resistant to phage T1; in the presence of phage T1, only these cells survive.

If the presumed rates of physiological induction and mutation are the same, determining which of the two mechanisms is operating is difficult. Luria and Delbrück, however, developed a means of distinguishing between these mechanisms. They reasoned as follows: If T1 resistance was physiologically induced, the relative frequency of resistant *E. coli* cells in a culture of the normal (Ton^s) strain should be a constant, independent of the number of cells in the culture or the length of time that the culture has been growing. If resistance was due to random mutation, the frequency of mutant (Ton^r) cells would depend on when the mutations occurred. In other words, the appearance of a mutant cell would be a random event. If a mutation occurs early in the growth of the culture, then many cells descend from the mutant cell, and therefore many resistant colonies develop. If the mutation does not occur until late in the growth of the culture, then the subsequent number of mutant cells is few. Thus, if the mutation hypothesis is correct, there should be considerable fluctuation from culture to culture in the number of resistant cells present (fig. 12.1).

Results of the Fluctuation Test

To distinguish between these hypotheses, Luria and Delbrück developed what is known as the **fluctuation test**. They counted the mutants both in small (“individual”) cultures and in subsamples from a single large (“bulk”) culture. All subsamples from a bulk culture should have the same number of resistant cells, differing only because of

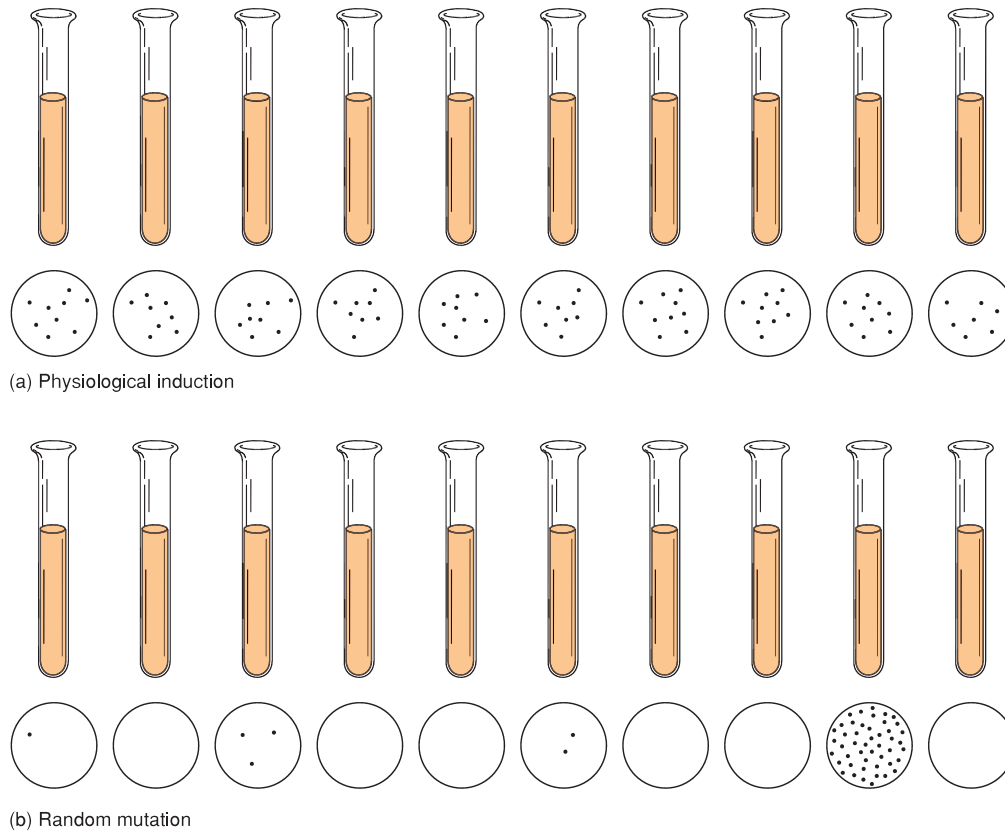


Figure 12.1 Occurrence of *E. coli* Ton^r colonies in Ton^s cultures. Ten cultures of *E. coli* cells were grown from a standard inoculum in separate test tubes in the absence of phage T1, then spread on petri plates in the presence of phage T1. The resistant cells grow into colonies on the plates. We expect a uniform distribution of resistant cells if the physiological induction hypothesis is correct (a) or a great fluctuation in the number of resistant cells if the random mutation hypothesis is correct (b).

random sampling error. If, however, mutation occurs, the number of resistant cells among the individual cultures should vary considerably from culture to culture; the number would be related to the time that the mutation occurred during the growth of each culture. If mutation arose early, there would be many resistant cells. If it arose late, there would be relatively few resistant cells. Under physiological induction, the distribution of resistant colonies should not differ between the individual and bulk cultures.

Luria and Delbrück inoculated twenty individual cultures and one bulk culture with *E. coli* cells and incubated them in the absence of phage T1. Each individual culture was then spread out on a petri plate containing a very high concentration of T1 phages; ten subsamples from the bulk culture were plated in the same way. We can see from the results (table 12.1) that there was minimal variation in the number of resistant cells among the bulk culture subsamples but a very large amount of variation, as predicted for random mutation, among the individual cultures.

If bacteria have “normal” genetic systems that undergo mutation, bacteria could then be used, along with higher organisms, to answer genetic questions. As we have pointed out, the modern era of molecular genetics began with the use of prokaryotic and viral systems in genetic research. In the next section, we turn our attention to several basic questions about the gene, questions whose answers were found in several instances only because prokaryotic systems were available.

Genetic Fine Structure

How do we determine the relationship among several mutations that cause the same phenotypic change? What are the smallest units of DNA capable of mutation and recombination? Are the gene and its protein product colinear? The answers to the latter two questions are important from a historical perspective. The answer to the first question is relevant to our current understanding of genetics.

Table 12.1 Results from the Luria and Delbrück Fluctuation Test

Individual Cultures*		Samples from Bulk Culture*	
Culture Number	Ton ^r Colonies Found	Sample Number	Ton ^r Colonies Found
1	1	1	14
2	0	2	15
3	3	3	13
4	0	4	21
5	0	5	15
6	5	6	14
7	0	7	26
8	5	8	16
9	0	9	20
10	6	10	13
11	107		
12	0		
13	0		
14	0		
15	1		
16	0		
17	0		
18	64		
19	0		
20	35		
Mean ($\bar{n}$)	11.4		16.7
Standard deviation	27.4		4.3

Source: From E. Luria and M. Delbrück, *Genetics*, 28: 491. Copyright © 1943 Genetics Society of America.

* Each culture and sample was 0.2 ml and contained about 2×10^7 *E. coli* cells.

Complementation

If two recessive mutations arise independently and both have the same phenotype, how do we know whether they are both mutations of the same gene? That is, how do we know whether they are alleles? To answer this question, we must construct a heterozygote and determine the **complementation** between the two mutations. A heterozygote with two mutations of the same gene will produce only mutant messenger RNAs, which result in mutant enzymes (fig. 12.2a). If, however, the mutations are not allelic, the gamete from the a_1 parent will also contain an a_2^+ allele, and the gamete from the a_2 parent will also contain the a_1^+ allele (fig. 12.2b). If the two mutant genes are truly alleles, then the phenotype of the heterozygote should be

mutant. If, however, the two mutant genes are nonallelic, then the a_1 mutant will have contributed the wild-type allele at the A_2 locus, and the a_2 mutant will have contributed the wild-type allele at the A_1 locus to the heterozygote. Thus, the two mutations will complement each other and produce the wild-type. Mutations that fail to complement each other are termed **functional alleles**. The test for defining alleles strictly on this basis of functionality is termed the ***cis-trans* complementation test**.

There are two different configurations in which a heterozygous double mutant of functional alleles can form (fig. 12.3). In the *cis-trans* complementation test, only the *trans* configuration is used to determine whether the two mutations were allelic. In reality, the *cis* configuration is not tested; it is the conceptual control, in which wild-type activity (with recessive mutations) is always expected. The test is thus sometimes simply called a *trans* test. Functional alleles produce a wild-type phenotype in the *cis* configuration but a mutant phenotype in the *trans* configuration. This difference in phenotypes is called a *cis-trans* position effect.

From the terms *cis* and *trans*, Seymour Benzer coined the term **cistron** for the smallest genetic unit (length of genetic material) that exhibits a *cis-trans* position effect. We thus have a new word for the gene, one in which function is more explicit. We have, in essence, refined Beadle and Tatum's one-gene-one-enzyme hypothesis to a more accurate one-cistron-one-polypeptide concept. The cistron is the smallest unit that codes for a messenger RNA that is then translated into a single polypeptide or expressed directly (transfer RNA or ribosomal RNA).

From functional alleles, we can go one step further in recombinational analysis by determining whether two allelic mutations occur at exactly the same place in the cistron. In other words, when two mutations prove to be functional alleles, are they also **structural alleles**? The methods used to analyze complementation can be used here also. Crosses are carried out to form a mutant heterozygote (*trans* configuration) whose offspring are then tested for recombination between the two mutational sites. If no recombination occurs, then the two alleles probably contain the same



Seymour Benzer (1921–).
(Courtesy of Dr. Seymour Benzer,
1970.)

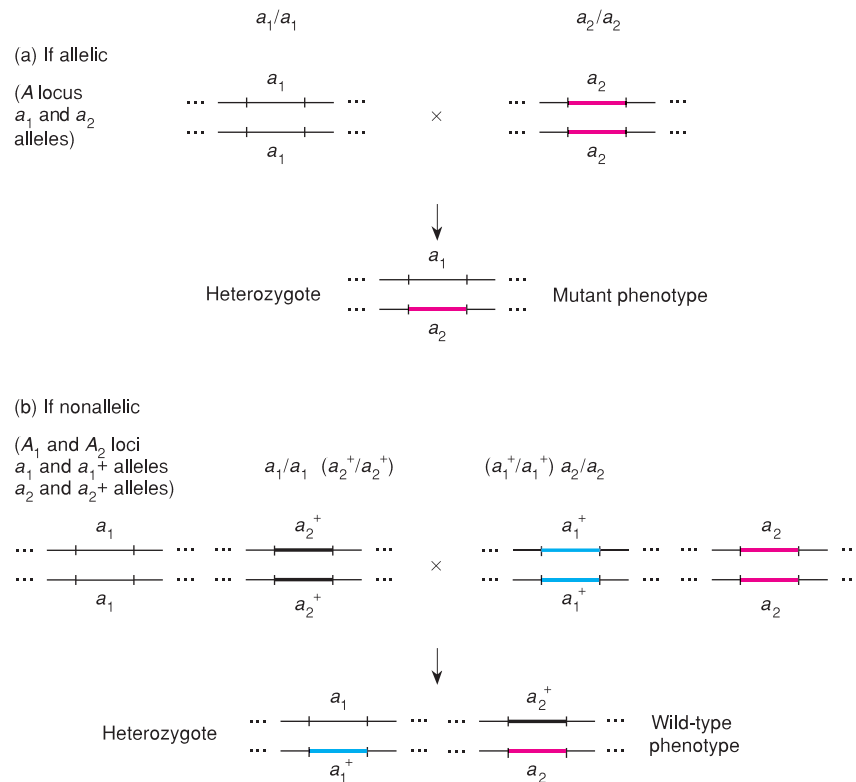


Figure 12.2 The complementation test defines allelism. Are two mutations (a_1 , a_2) allelic if they affect the same trait? To find out, mutant homozygotes are crossed to form a heterozygote. (a) If the mutations are allelic, then both copies of the gene in the heterozygote are mutant, resulting in the mutant phenotype. (b) If the mutations are nonallelic, then there is a wild-type allele of each gene present in the heterozygote, resulting in the wild-type phenotype. (The two loci need not be on the same chromosome.)

structural change (involving the same base pairs) and are thus structural alleles. If a small amount of recombination occurs that generates wild-type offspring, then the two alleles are not mutations at the same point (fig. 12.4). Alleles that were functional but not structural were first termed **pseudoalleles** because it was believed that loci were



Phenotype: Mutant Wild-type

Figure 12.3 A heterozygote of two recessive mutations can have either the *trans* or *cis* arrangement. In the *trans* position, functional alleles produce a mutant phenotype. (Red marks represent mutant lesions.) In the *cis* position, functional alleles produce a wild-type phenotype. The *cis-trans* position effect thus reveals functional alleles.

made up of subloci. Fine-structure analysis led to the understanding that a locus is a length of genetic material divisible by recombination rather than a “bead on a string.”

Eye-color mutants of *Drosophila melanogaster* can be studied by complementational analysis. The white-eye locus has a series of alleles producing varying shades of red. This locus is sex linked, at about map position 3.0 on the X chromosome. (Several other eye-color loci on the X chromosome are not relevant to this cross—e.g., prune and ruby.) If an apricot-eyed female is mated with a white-eyed male, the female offspring are all heterozygous and have mutant light-colored eyes (fig. 12.5). Thus, apricot and white are functional alleles: they do not complement (table 12.2). To determine whether apricot and white are structural alleles, light-eyed females are crossed with white-eyed males, and the offspring are observed for the presence of wild-type or light-eyed males. Though their rate of appearance is less than 0.001%, this is significantly above the background mutation rate. The conclusion is that apricot and white are functional, but not structural, alleles.

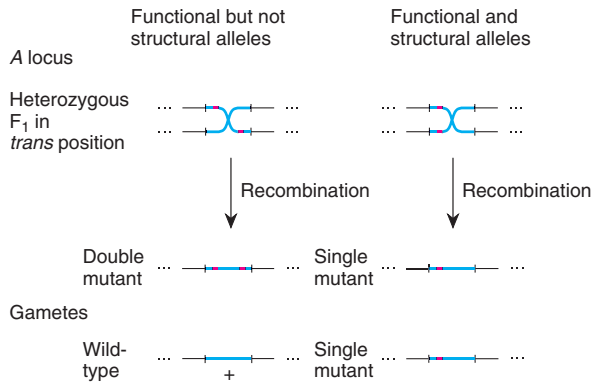


Figure 12.4 Functional alleles may or may not be structurally allelic. (Red marks represent mutant sites.) Functional alleles that are not also structural alleles can recombine between the mutant sites, resulting in occasional wild-type (and double mutant) offspring. Structural alleles (which are also always functional alleles) are defective at the same base pairs and cannot form either wild-type or double mutant offspring by recombination.

Fine-Structure Mapping

After Beadle and Tatum established in 1941 that a gene controls the production of an enzyme that then controls a step in a biochemical pathway, Benzer used analytical techniques to dissect the fine structure of the gene. Fine-structure mapping means examining the size and number of sites within a gene that are capable of mutation and recombination. In the late 1950s, when biochemical techniques were not yet available for DNA sequencing, Benzer used classical recombinational and mutational techniques with bacterial viruses to provide reasonable estimates on the details of fine structure and to give insight into the nature of the gene. He coined the terms **muton** for the smallest mutable site and **recon** for the smallest unit of recombination. It is now known that both muton and recon are a single base pair.

Before Benzer's work, genes were thought of as beads on a string. The very low rate of recombination between

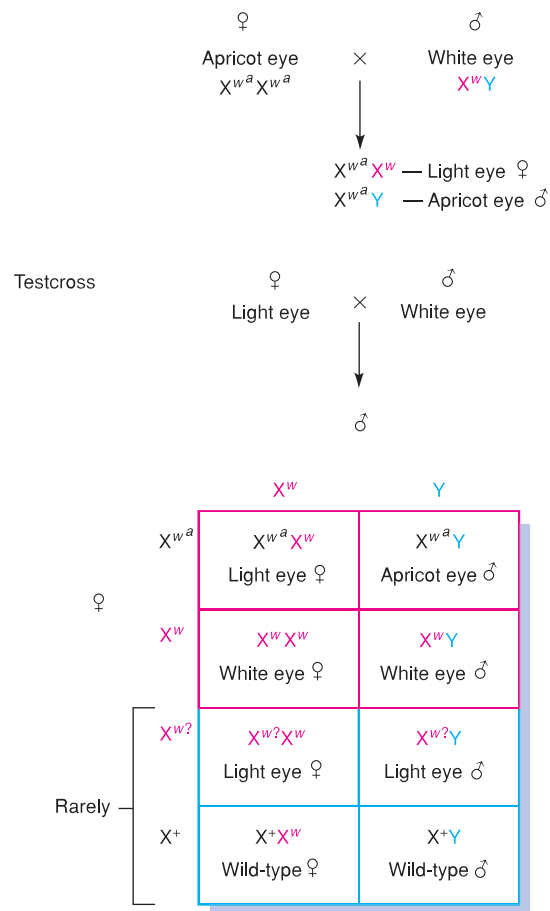


Figure 12.5 Crosses demonstrating that apricot and white eyes are functional, but not structural, alleles in *Drosophila*. Light-eyed females are heterozygous for both alleles. When testcrossed, they produce occasional offspring that are wild-type (X^+ allele) or light-eyed ($X^{w^?}$ allele). This indicates a crossover between the two mutant sites (white and apricot) in the heterozygous females, producing, reciprocally, an allele with both mutational sites and the wild-type.

Table 12.2 Complementation Matrix of X-Linked *Drosophila* Eye-Color Mutants

	white	prune	apricot	buff	cherry	eosin	ruby
white (<i>w</i>)	—	+	—	—	—	—	+
prune (<i>pn</i>)		—	+	+	+	+	+
apricot (<i>w^a</i>)			—	—	—	—	+
buff (<i>w^{bf}</i>)				—	—	—	+
cherry (<i>w^{ch}</i>)					—	—	+
eosin (<i>w^e</i>)						—	+
ruby (<i>rb</i>)							—

Note: Plus sign indicates that female offspring are wild-type; minus sign indicates that they are mutant.

sites within a gene hampered the analysis of mutational sites within a gene by means of recombination. If two mutant genes are functional alleles (involving different sites on the same gene), a distinct probability exists that we will get both mutant sites (and both wild-type sites) on the same chromosome by recombination (see fig. 12.4); but, in view of the very short distances within a gene, this probability is very low. Although it certainly seemed desirable to map sites within the gene, the problem of finding an organism that would allow fine-structure analysis remained until Benzer decided to use phage T4.

rII Screening Techniques. Benzer used the T4 bacteriophage because of the growth potential of phages, in which a generation takes about an hour and the increase in numbers per generation is about a hundredfold. Actually, any prokaryote or virus should suffice, but Benzer made use of other unique screening properties of the phage that made it possible to recognize one particular mutant in about a billion phages. Benzer used *rII* mutants of T4. These mutants produce large, smooth-edged plaques on *E. coli*, whereas the wild-type produces smaller plaques whose edges are not as smooth (see fig. 7.7).

The screening system that Benzer employed made use of the fact that *rII* mutants do not grow on *E. coli* strain K12, whereas the wild-type can. The normal host strain, *E. coli* B, allows growth of both the wild-type and *rII* mutants. Thus, various mutants can be crossed by mixed infection of *E. coli* B cells, and Benzer could screen for wild-type recombinants by plating the resultant progeny phages on *E. coli* K12 (fig. 12.6), on which only a wild-type recombinant produces a plaque. It is possible to detect about one recombinant in a billion phages, all in an afternoon's work. This ability to detect recombinants occurring at such a low level of frequency allowed Benzer to see recombinational events occurring very close together on the DNA, events that would normally occur at a frequency too low to detect in fruit flies or corn.

Benzer sought to map the number of sites subject to recombination and mutation within the *rII* region of T4. He began by isolating independently derived *rII* mutants and crossing them among themselves. The first thing he found was that the *rII* region was composed of two cistrons; almost all of the mutations belonged to one of two **complementation groups**. The *A*-cistron mutations would not complement each other but would complement the mutations of the *B* cistron. The exceptions were mutations that seemed to belong to both cistrons. These mutations were soon found to be deletions in which part of each cistron was missing (table 12.3).

Deletion Mapping. As the number of independently isolated mutations of the *A* and *B* cistrons increased, it became obvious that to make every possible pairwise cross would entail millions of crosses. To overcome this prob-

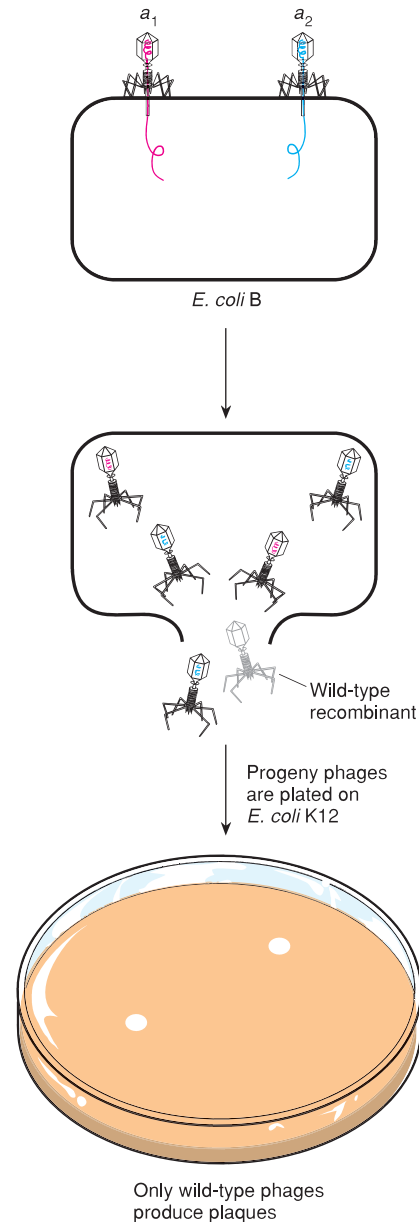


Figure 12.6 Using *E. coli* K12 and B strains to screen for recombination at the *rII* locus of phage T4. Two *rII* mutants are crossed by infecting the same B-strain bacteria with both phages. The offspring are plated on a lawn of K12 bacteria in which only wild-type phages can grow. The technique thus selects only wild-type recombinants.

lem, Benzer isolated mutants that had partial or complete deletions of each cistron. Deletion mutations were easy to discover because they acted like structural alleles to alleles that were not themselves structurally allelic. In other words, if mutations *a*, *b*, and *c* are functional—but not

structural—alleles of each other, and mutation *d* is a structural allele to *a*, *b*, and *c*, then *d* must contain a deletion of the bases mutated in *a*, *b*, and *c*. Once a sequence of deletion mutations covering the *A* and *B* cistrons was isolated, a minimal number of crosses was required to localize a new mutation to a portion of one of the cistrons. A second series of smaller deletions within each region was then isolated, further localizing the mutation (fig. 12.7).

Next, each new mutant was crossed with each of the other mutants isolated in its subregion to localize the relative position of the new mutation. If the mutation was structurally allelic to a previously isolated mutation, it was scored as an independent isolation of the same mutation. If it was not a structural allele to any of the known mutations of the subregion, it was added as a new mutation point. The exact position of each new mutation within the region was

determined by the relative frequency of recombination between it and the known mutations of this region (see chapter 7). Benzer eventually isolated about 350 mutations from eighty different subregions defined by deletion mutations. An abbreviated map is shown in figure 12.8.

What conclusions did Benzer draw from his work? First, he concluded that since all of the mutations in both *rII* cistrons can be ordered in a linear fashion, the original Watson-Crick model of DNA as a linear molecule was correct. Second, he concluded that reasonable inroads had been made toward saturating the map, localizing at least one mutation at every mutable site. Benzer reasoned that since many sites were represented by only one mutation, some sites must occur that were represented by zero mutations (i.e., not yet represented by a mutation). Since he had mapped about 350 sites, he calculated that there

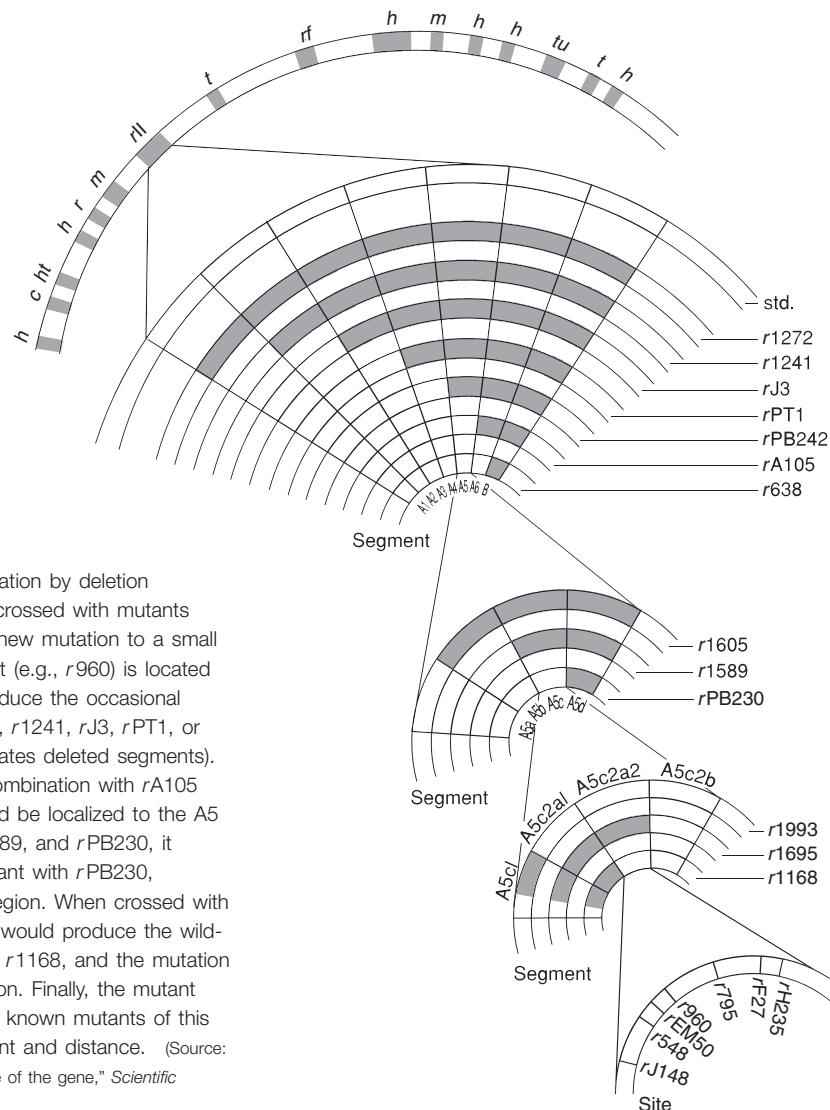


Figure 12.7 Localization of an *rII* mutation by deletion mapping. Newly isolated mutants are crossed with mutants with selected deletions to localize the new mutation to a small region of the cistron. If the new mutant (e.g., *r*960) is located in the A5c2a2 region, it would not produce the occasional wild-type by recombination with *r*1272, *r*1241, *r*J3, *r*PT1, or *r*PB242 (the solid part of the bar indicates deleted segments). It would produce the wild-type by recombination with *r*A105 and *r*638, and thus the mutation would be localized to the A5 region. When crossed with *r*1605, *r*1589, and *r*PB230, it would produce only the rare recombinant with *r*PB230, indicating the mutation is in the A5c region. When crossed with *r*1993, *r*1695, and *r*1168, the mutant would produce the wild-type by recombination with *r*1993 and *r*1168, and the mutation would be localized to the A5c2a2 region. Finally, the mutant would be crossed pairwise with all the known mutants of this region to determine relative arrangement and distance. (Source: Data from Seymour Benzer, "The fine structure of the gene," *Scientific American*, 206: 70–84, January 1962.)

Table 12.3 Complementation Matrix of Ten *rII* Mutants

	1	2	3	4	5	6	7	8	9	10
1	—	—	+	—	—	—	+	—	—	—
2		—	+	—	—	—	+	—	—	—
3			—	—	+	+	—	+	—	+
4				—	—	—	—	—	—	—
5					—	—	+	—	—	—
6						—	+	—	—	—
7							—	+	—	+
8								—	—	—
9									—	—
10										—

Note: Plus sign indicates complementation; minus sign indicates no complementation. The two cistrons are arbitrarily designated *A* and *B*. Mutants 4 and 9 must be deletions that cover parts of both cistrons. Alleles: *A* cistron: 1, 2, 4, 5, 6, 8, 9, 10; *B* cistron: 3, 4, 7, 9.

were at least another 100 sites still undetected by mutation. We now know that 450 sites is an underestimate. However, since the protein products of these cistrons were not isolated, there were no independent estimates of the number of nucleotides in these cistrons (number

of amino acids times three nucleotides per codon). Thus, although Benzer had not saturated the map with mutations, he certainly had made respectable progress in dissecting the gene and demonstrating that it was not an indivisible unit, a “bead on a string.”

Hot Spots. Benzer also looked into the lack of uniformity in the occurrence of mutations (note two major “hot spots” at B4 and A6c of fig. 12.8). Presuming that all base pairs are either AT or GC, this lack of uniformity was unexpected. Benzer suggested that spontaneous mutation is not just a function of the base pair itself, but is affected by the surrounding bases as well. This concept still holds.

To recapitulate, Benzer’s work supports the model of the gene as a linear arrangement of DNA whose nucleotides are the smallest units of mutation. The link between any adjacent nucleotides can break in the recombinational process. The smallest functional unit, determined by a complementation test, is the cistron. Mutagenesis is not uniform throughout the cistron, but may depend on the particular arrangement of bases in a given region.

Intra-Allelic Complementation

Benzer warned that certainty is elusive in the complementation test because sometimes two mutations of the same functional unit (cistron) can result in partial activity. The

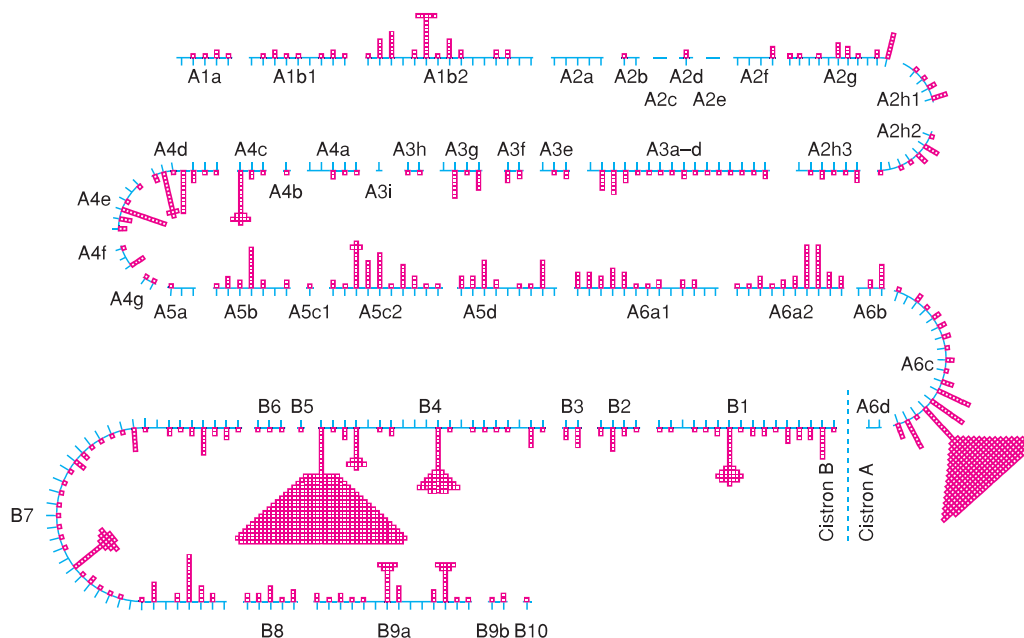


Figure 12.8 Abbreviated map of spontaneous mutations of the *A* and *B* cistrons of the *rII* region of T4. Each square represents one independently isolated mutation. Note the “hot spots” at A6c and B4. (From Seymour Benzer, “On the topography of the genetic fine structure”, *Proceedings of the National Academy of Sciences USA* 47:403–15, 1961. Reprinted by permission.)

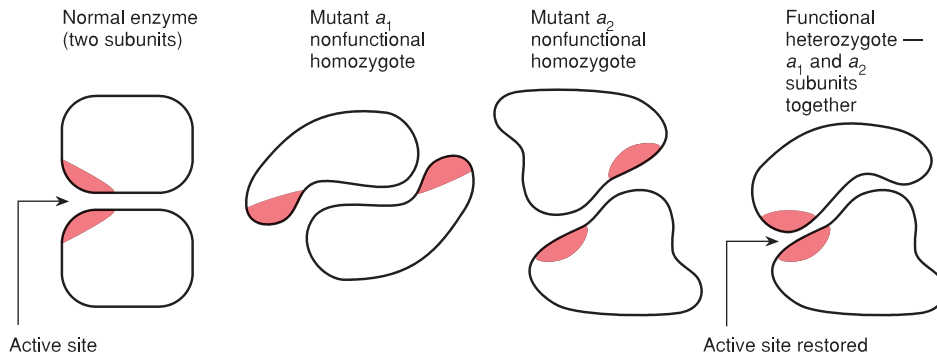


Figure 12.9 Intra-allelic complementation. With certain mutations, it is possible to get enzymatic activity in a heterozygote for two nonfunctional alleles, if the two polypeptides form a functional enzyme. (Active site is shown in color.)

problem can be traced to the interactions of subunits at the polypeptide level. Some proteins are made up of subunits, and it is possible that certain mutant combinations produce subunits that interact to restore the enzymatic function of the protein (fig. 12.9). This phenomenon is known as **intra-allelic complementation**. With this in mind, geneticists routinely use the complementation test to determine functional relationships among mutations.

Colinearity

Next we look at the colinearity of the gene and the polypeptide. Benzer's work established that the gene was a linear entity, as Watson and Crick had proposed. However, Benzer could not demonstrate the colinearity of the gene and its protein product. To do this, it is necessary to show that for every mutational change in the DNA, a corresponding change takes place in the protein product of the gene. Colinearity would be established by showing that nucleotide and amino acid changes occurred in a linear fashion and in the same order in the protein and in the cistron.

Ideally, Benzer himself might have solved the colinearity issue. He was halfway there, with his 350 or so isolated mutations of phage T4. However, Benzer did not have a protein product to analyze; no mutant protein had been isolated from *rII* mutants. In the midst of competition to find just the right system, Charles Yanofsky of Stanford University and his colleagues emerged in the mid-1960s with the required proof, showing that the order of a polypeptide's amino acids corresponded to the nucleotide sequence in the gene that specified it. Yanofsky's success rested with his choice of an amenable system, one using the enzymes from a biochemical pathway.

Yanofsky did his research on the tryptophan biosynthetic pathway in *E. coli*. The last enzyme in the pathway, tryptophan synthetase, catalyzes the reaction of indole-3-glycerol-phosphate plus serine to tryptophan and

3-phosphoglycerinaldehyde. The enzyme itself is made of four subunits specified by two separate cistrons, with each polypeptide present twice.

Yanofsky and his colleagues concentrated on the *A* subunit. They mapped *A*-cistron mutations with transduction (see chapter 7) using the transducing phage P1. They first tested each new mutant against a series of deletion mutants to establish the region where the mutation was. Then they crossed mutants for a particular region among themselves to establish relative positions and distances.

The protein products of the bacterial genes were isolated using electrophoresis and chromatography to establish the fingerprint patterns of the proteins (see chapter 11). Assuming a single mutation, a comparison of the mutant and the wild-type fingerprints would show a difference of just one polypeptide spot (fig. 12.10), avoiding the need to sequence the entire protein. The mutant amino acid was

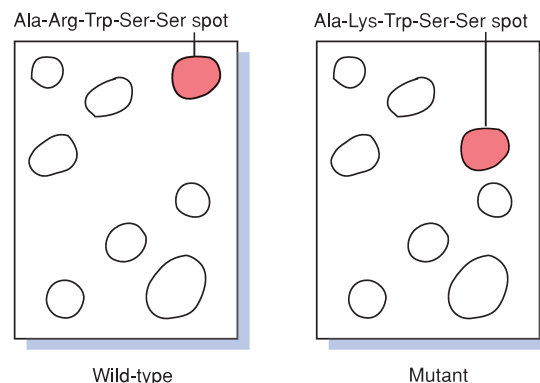


Figure 12.10 Difference in "fingerprints" between mutant and wild-type polypeptide digests. The single spot that differs in the mutant can be isolated and sequenced, eliminating the need to sequence the whole protein.

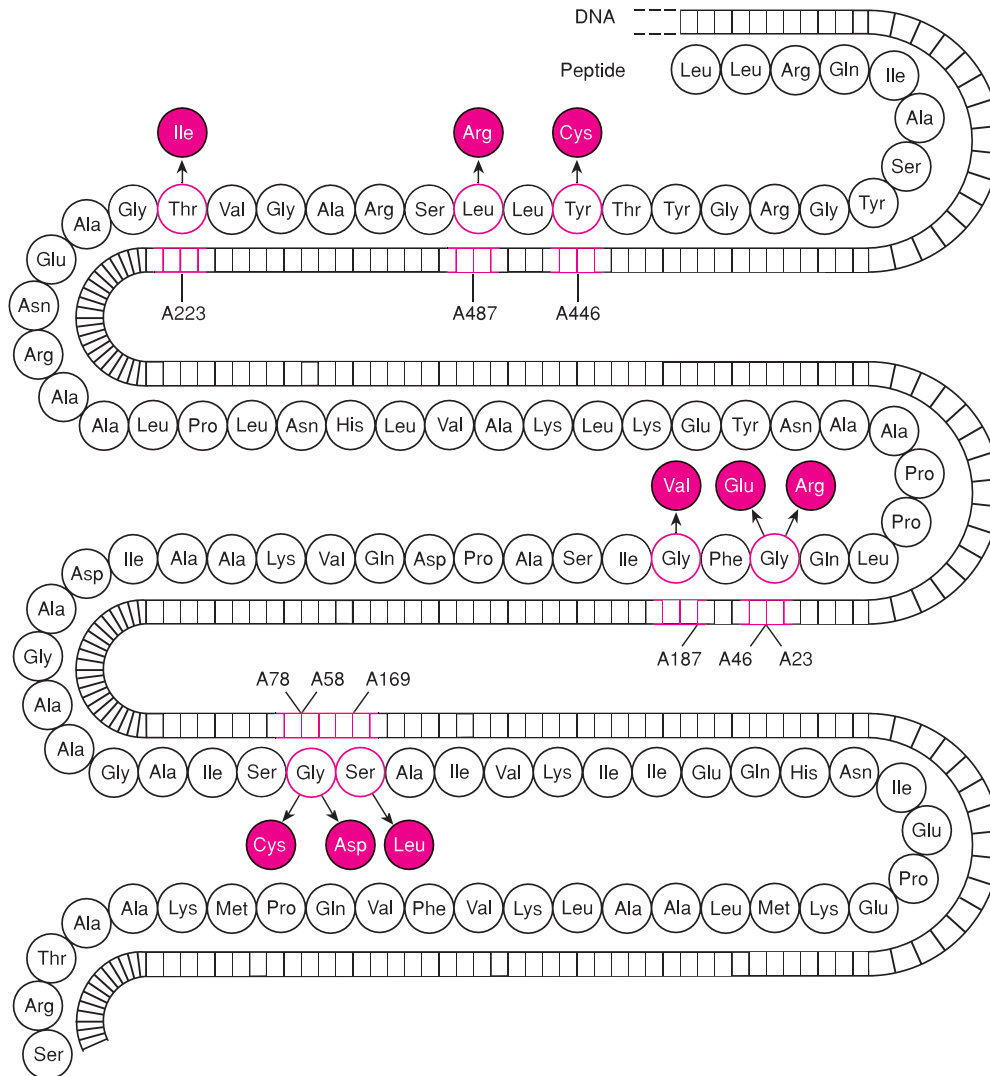


Figure 12.11 Amino acid sequence of the carboxyl terminal end of the tryptophan synthetase A protein and its DNA. Mutations are shown on the DNA (e.g., A446), as are the changed amino acids of those mutations (in color). DNA and protein changes are colinear.

identified by analysis of just this one spot. Figure 12.11 shows the details of nucleotide and amino acid changes for nine of the mutations in this 267-amino acid protein.

We can see from this figure that nine mutations in the linear *A* cistron of tryptophan synthetase are colinear with nine amino acid changes in the protein itself. In two cases, two mutations mapped so close as to be almost indistinguishable. In both cases, the two mutations proved to be in the same codon: the same amino acid position was altered in each (A23–A46, A58–A78). Thus, exactly as predicted and expected, colinearity exists between gene and protein. Brenner and his colleagues, using head-

protein mutants of phage T4, independently confirmed this work at the same time.

Spontaneous Versus Induced Mutation

H. J. Muller won the Nobel Prize for demonstrating that X rays can cause mutations. This work was published in 1927 in a paper entitled “Artificial Transmutation of the Gene.” At about the same time, L. J. Stadler induced mutations in barley with X rays. The basic impetus for their work was the fact that mutations occur so infrequently that genetic research was hampered by the inability to



Hermann J. Muller
(1890–1967). (Courtesy of
National Academy of Sciences.)



Lewis J. Stadler
(1896–1954). (*Genetics*, 41,
1956: frontispiece.)

obtain mutants. Muller exposed flies to varying doses of X rays and then observed their progeny. He came to several conclusions. First, X rays greatly increased the occurrence of mutations. Second, the inheritance patterns of X-ray-induced mutations and the resulting phenotypes of organisms were similar to those that resulted from natural, or “spontaneous,” mutations.

Mutation Rates

The **mutation rate** is the number of mutations that arise per cell division in bacteria and single-celled organisms, or the number of mutations that arise per gamete in higher organisms. Mutation rates vary tremendously depending upon the length of genetic material, the kind of mutation, and other factors. Luria and Delbrück, for example, found that in *E. coli* the mutation rate per cell division of Ton^s to Ton^r was 3×10^{-8} , whereas the mutation rate of the wild-type to the histidine-requiring phenotype (His⁺ to His[−]) was 2×10^{-6} . The rate of **reversion** (return of the mutant to the wild-type) was 7.5×10^{-9} . The mutation and reversion rates differ because many different mutations can cause the His phenotype, whereas reversion requires specific, and hence less probable, changes to correct the His phenotype back to the wild-type. The lethal mutation rate in *Drosophila* is about 1×10^{-2} per gamete for the total genome. This number is relatively large because, as with His, many different mutations produce the same phenotype (lethality, in this case).

Point Mutations



The mutations of primary concern in this chapter are **point mutations**, which consist of single changes in the nucleotide sequence. (In chapter 8 we discussed chromosomal mutations, changes in the number and visible structures of chromosomes.) If the change is a replacement of some kind, then a new codon is created. In many cases, this new codon, upon translation, results in a new amino acid.

As discussed in chapter 11, one of the outcomes of redundancy in the genetic code is partial protection of the cell from the effects of mutation; common amino acids have the most codons, similar amino acids have similar codons, and the wobble position of the codon is the least important position in translation. However, when base changes result in new amino acids, new proteins appear. These new proteins can alter the morphology or physiology of the organism and result in phenotypic novelty or lethality.

Frameshift Mutation

A point mutation may consist of replacement, addition, or deletion of a base (fig. 12.12). Point mutations that add

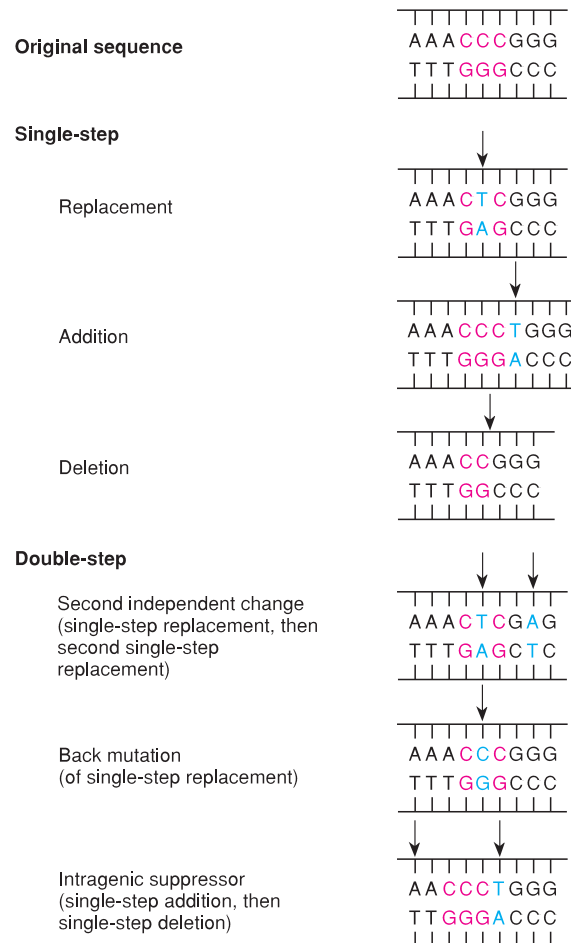


Figure 12.12 Types of DNA point mutations. Single-step changes are replacements, additions, or deletions. A second point mutation in the same gene can result either in a double mutation, reversion to the original, or intragenic suppression. In this case, intragenic suppression is illustrated by the addition of one base followed by the nearby deletion of a different base.

or subtract a base are, potentially, the most devastating in their effects on the cell or organism because they change the reading frame of a gene from the site of mutation onward (fig. 12.13). A frameshift mutation causes two problems. First, all the codons from the frameshift on will be different and thus yield (most probably) a useless protein. Second, stop-signal information will be misread. One of the new codons may be a nonsense codon, which causes translation to stop prematurely. Or, if the translation apparatus reaches the original nonsense codon, it is no longer recognized as such because it is in a different reading frame, and therefore, the translation process continues beyond the end of the gene.

Back Mutation and Suppression

A second point mutation in the same gene can have one of three possible effects (see fig. 12.12). First, the mutation can result in either another mutant codon or in one codon that has experienced two changes. Second, if the change is at the same site, the original sequence can be returned, an effect known as **back mutation**: the gene then becomes a revertant, with its original function restored. Third, **intragenic suppression** can take place. Intragenic suppression occurs when a second mutation in the same gene masks the occurrence of the original mutation without actually restoring the original sequence. The new sequence is a double mutation that appears to have the original (unmutated) phenotype. In figure 12.12, a T addition is followed by an A deletion

that substitutes the AACCT sequence for the original AAACCC. These sequences, when transcribed (UUGGGA, UUUGGG), are codons for leucine-glycine and phenylalanine-glycine, respectively. Intragenic suppression occurs whether the new codons are for different amino acids or the same amino acids, as long as the phenotype of the organism is reverted approximately to the original. Suppressed mutations can be distinguished from true back mutations either by subtle differences in phenotype, by genetic crosses, by changes in the amino acid sequence of a protein, or by DNA sequencing.

Conditional Lethality

A class of mutants that has been very useful to geneticists is the conditional-lethal mutant, a mutant that is lethal under one set of circumstances but not under another set. **Nutritional-requirement mutants** are good examples (see chapter 7). **Temperature-sensitive mutants** are conditional-lethal mutants that have made it possible for geneticists to work with genes that control vital functions of the cell, such as DNA synthesis. Many temperature-sensitive mutants are completely normal at 25° C but cannot synthesize DNA at 42° C. Presumably, temperature-sensitive mutations result in enzymes with amino acid substitutions that cause protein denaturation to occur at temperatures above normal. Thus, the enzyme has normal function at 25° C, the **permissive temperature**, but is nonfunctional at 42° C, the **restrictive temperature**.

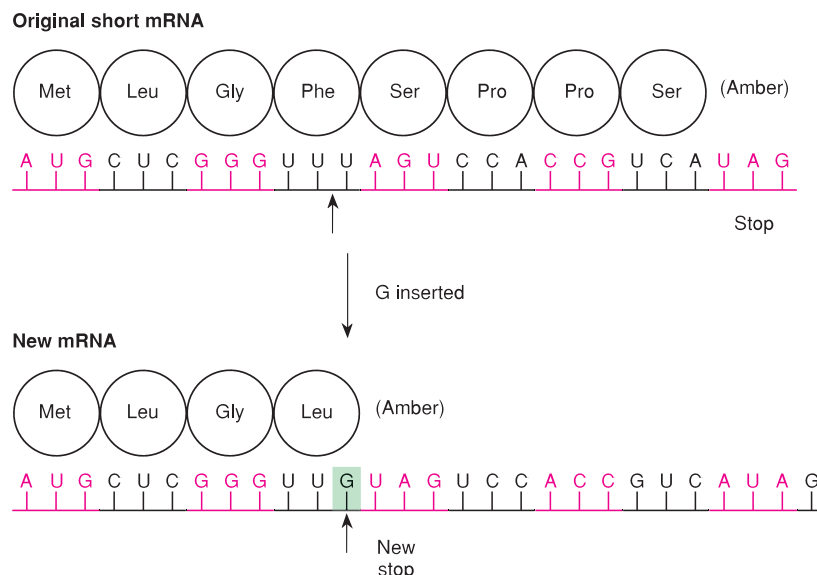


Figure 12.13 Possible effects of a frameshift mutation. The insertion of a single base results in the creation of a new stop sequence (*amber*). The result will be premature termination of translation.

The interesting thing about most conditional-lethal mutants of *E. coli* that cannot synthesize DNA at the restrictive temperature is that they have a completely normal DNA polymerase I. From this information, we infer that polymerase I is not the enzyme *E. coli* normally uses for DNA replication. When an organism with a conditional mutation of polymerase I was isolated, it was able to replicate its DNA normally, but unable to repair damage to the DNA. This led to the conclusion that polymerase I is primarily involved in repair rather than replication of DNA. Conditional-lethal mutants thus allow genetic analysis on genes otherwise impossible to study.

Spontaneous Mutagenesis

Watson and Crick originally suggested that mutation could occur spontaneously during DNA replication if pairing errors occurred. If a base of the DNA underwent a proton shift into one of its rare tautomeric forms (**tautomeric shift**) during the replication process, an inappropriate pairing of bases would occur. Normally, adenine and cytosine are in the amino (NH_2) form. Their tautomeric shifts are to the imino (NH) form. Similarly, guanine and thymine go from a keto ($\text{C}=\text{O}$) form to an enol (COH) form (fig. 12.14). Table 12.4 shows the new

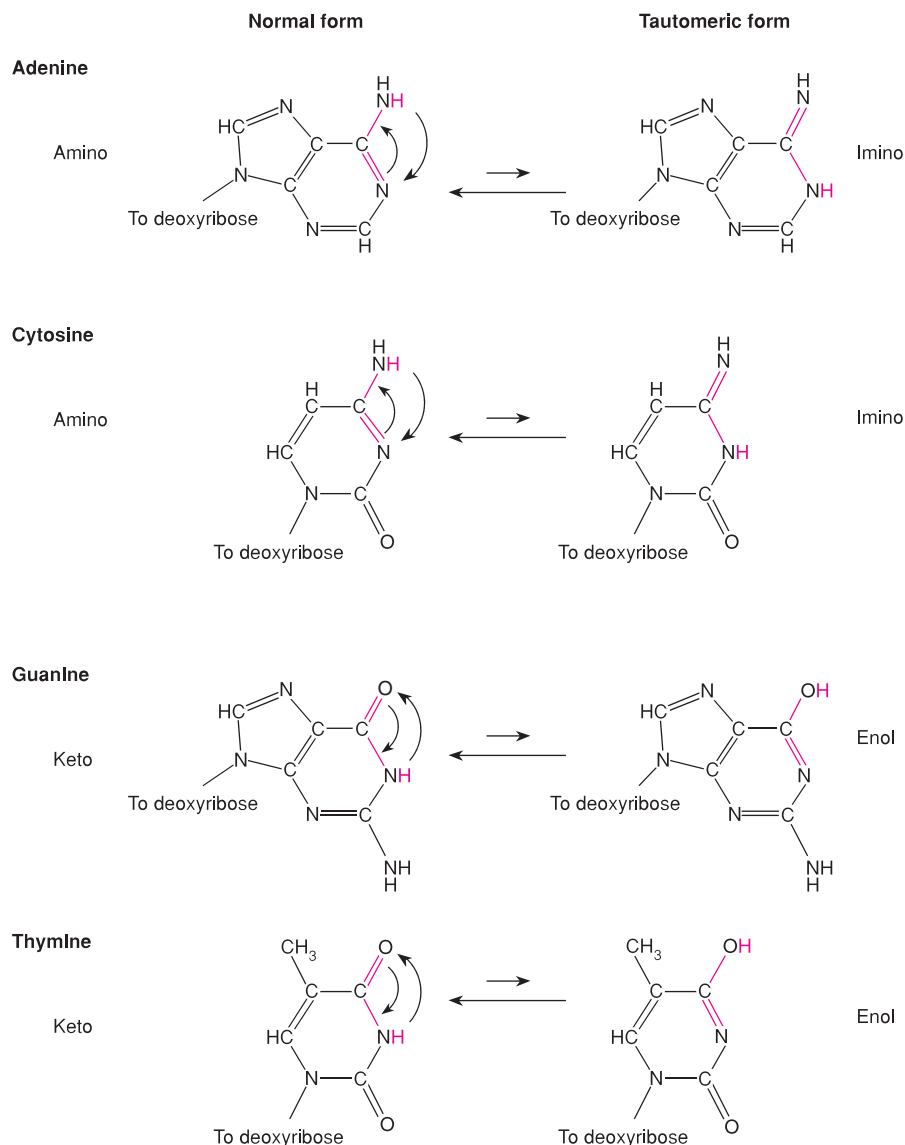


Figure 12.14 Normal and tautomeric forms of DNA bases. Adenine and cytosine can exist in the amino, or the rare imino, forms; guanine and thymine can exist in the keto, or rare enol, forms.

Table 12.4 Pairing Relationships of DNA Bases in the Normal and Tautomeric Forms

Base	In Normal State Pairs with	In Tautomeric State Pairs with
A	T	C
T	A	G
G	C	T
C	G	A

base pairings that would occur following tautomeric shifts of the DNA bases. Figure 12.15 illustrates the molecular structure of one of these tautomeric pairings.

During DNA replication, a tautomeric shift in either the incoming base (*substrate transition*) or the base already in the strand (*template transition*) results in mispairing. The mispairing will be permanent and result in a new base pair after an additional round of DNA replication. The original strand is unchanged (fig. 12.16).

In the example in figure 12.16, the replacement of one base pair maintains the same purine-pyrimidine relationship: AT is replaced by GC and GC by AT. In both examples, a purine-pyrimidine combination is replaced by a purine-pyrimidine combination. (Or, more specifically, a purine replaces another purine: guanine replaces

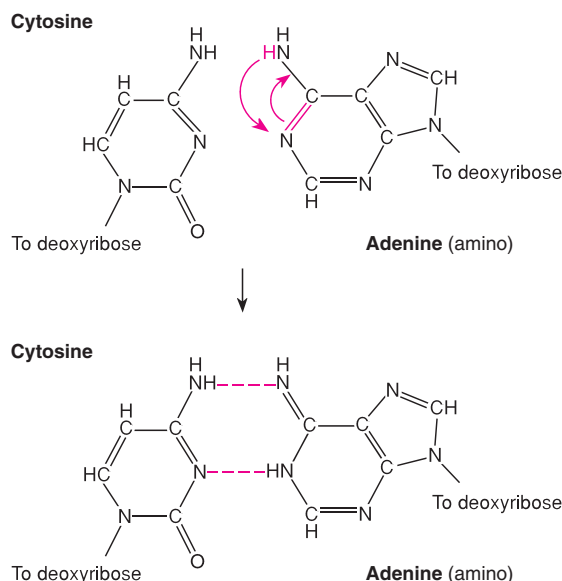


Figure 12.15 Tautomeric forms of adenine. In the common amino form, adenine does not base-pair with cytosine; in the tautomeric imino form, it can.

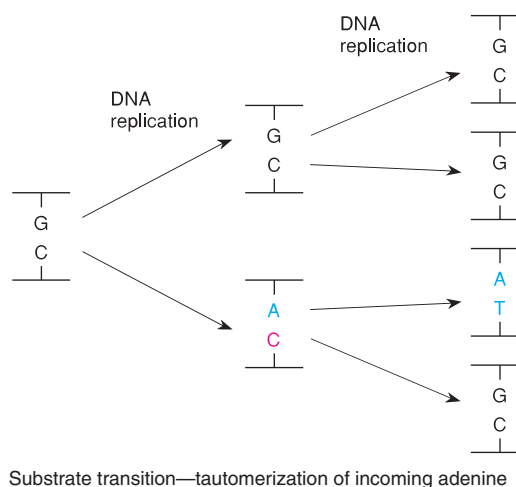
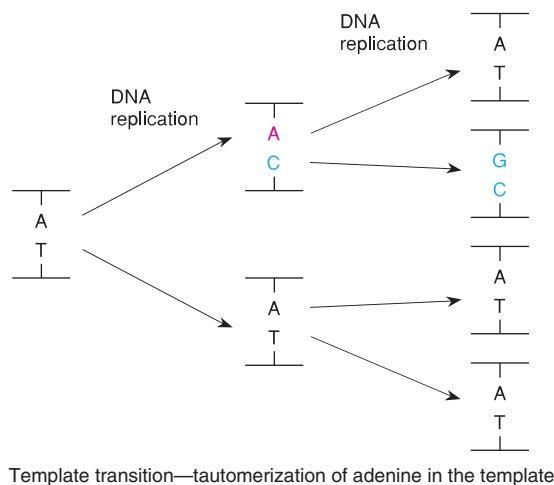


Figure 12.16 Tautomeric shifts result in transition mutations. The tautomerization can occur in the template base or in the substrate base. Tautomerizations are shown in red; the resulting transition in blue. The transition shows up after a second generation of DNA replication.

adenine in the first example and adenine replaces guanine in the second.) The mutation is referred to as a **transition mutation**: a purine (or pyrimidine) replaces another purine (or pyrimidine) through a transitional state involving a tautomeric shift. When a purine replaces a pyrimidine or vice versa, it is referred to as a **transversion mutation**.

Transversions may arise by a combination of two events, a tautomerization and a base rotation. (We saw base rotations in the formation of Z DNA in chapter 9.) For example, an AT base pair can be converted to a TA base pair (a transversion) by an intermediate AA pairing

(fig. 12.17). Adenine can pair with adenine if one of the bases undergoes a tautomeric shift while the other rotates about its base-sugar (glycosidic) bond (fig. 12.18). The normal configuration of the base is referred to as the *anti* configuration; the rotated form is the *syn* configuration. Since we now believe that as many as 10% of bases

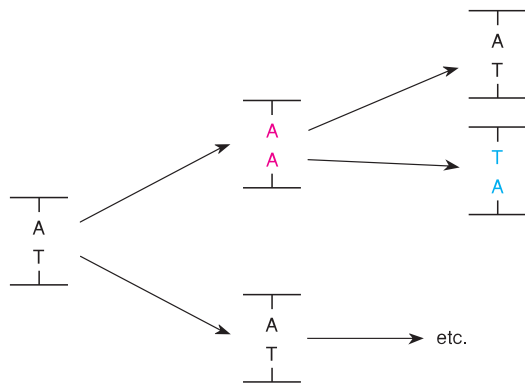


Figure 12.17 A model for transversion mutagenesis. An AT base pair can be converted to a TA base pair (a transversion) by way of an intermediate AA base pair. One of the red bases is in the rare tautomeric form, while the other is in the *syn* configuration. After a second round of DNA replication, one DNA duplex will have a transversion at that point (blue).

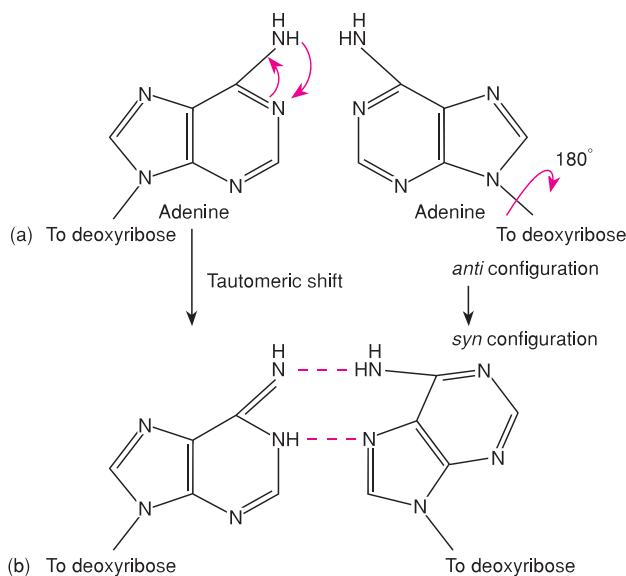


Figure 12.18 Transversion mutagenesis. An AA base pair can form if one base undergoes a tautomeric shift while the other rotates about its glycosidic (sugar) bond. In (a), both bases are in their normal configurations; no hydrogen bonding occurs. In (b), hydrogen bonds are possible.

may be in the *syn* configuration at any moment, the transversion mutagenesis rate should be about 10% of the transition mutagenesis rate, a value not inconsistent with current information.

Some base-pair mutations can have serious results. If guanine undergoes an oxidation to 8-oxoguanine (fig. 12.19), it pairs with adenine. A GC base pair is converted to a TA base pair through an 8-oxoguanine-adenine intermediate. This transversion has been found to be common in cancers.

Since 1953, when Watson and Crick first described the structure of DNA, tautomerization has been accepted as the obvious source of most transition mutations. However, recent structural data has cast some doubt on this assumption. X-ray crystallography and nuclear magnetic resonance (NMR) studies indicate that both bases in transition mismatches may be in their normal forms. Other mechanisms, similar to wobble base pairing (see chapter 11), may be responsible for most transition mutations. These studies also indicate that some transversions result from direct purine-purine or pyrimidine-pyrimidine base pairing during DNA synthesis. Much work needs to be done to clarify the nature of spontaneous mutagenesis.

Chemical Mutagenesis

Muller demonstrated that X rays can cause mutation. Certain chemical and temperature treatments can also cause mutation. Determining the mode of action of various chemical mutagens has provided insight into the mutational process as well as the process of carcinogenesis (box 12.1). In addition, knowing how chemical mutagens act has allowed geneticists to produce large numbers of specific mutations at will (box 12.2).

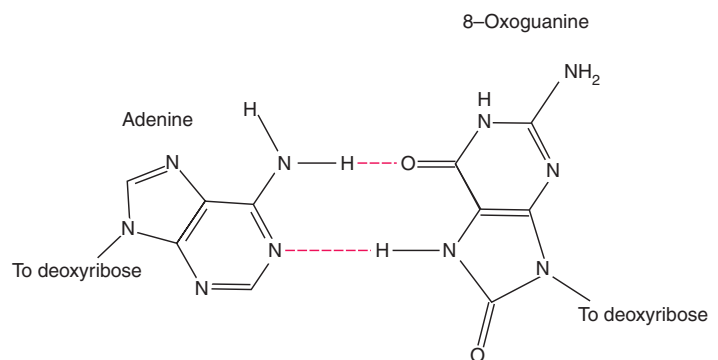


Figure 12.19 The structure of 8-oxoguanine, which base pairs with adenine, converting a GC base pair to a TA base pair—a transversion. (After C. Mol, et al., "DNA repair mechanisms for the recognition and removal of damaged DNA bases," *Annual Review of Biophysics and Biomolecular Structure*, 28:101–28, 1999, figure 6.)

Transitions

Transitions are routinely produced by base analogues. Two of the most widely used base analogues are the pyrimidine analogue 5-bromouracil (5BU) and the purine analogue 2-aminopurine (2AP; fig. 12.20). The mutagenic mechanisms of the two are similar. The 5-bromouracil is incorporated into DNA in place of thymine; it acts just like thymine in DNA replication and, since it doesn't alter the hydrogen bonding, should induce no mutation. However, it seems that the bromine atom causes 5-bromouracil to tautomerize more readily than thymine does. Thus, 5-bromouracil goes from the keto form (fig. 12.20) to the enol form more readily than thymine. Transitions frequently result when the enol form of 5-bromouracil pairs with guanine.

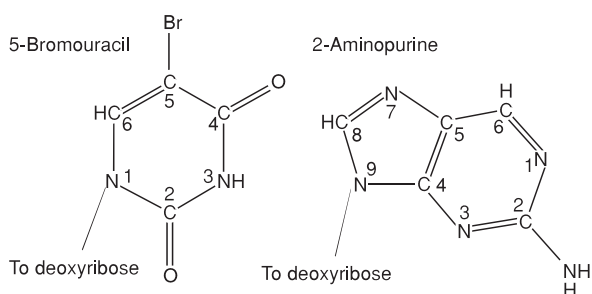


Figure 12.20 Structure of the base analogues 5-bromouracil (5BU) and 2-aminopurine (2AP).

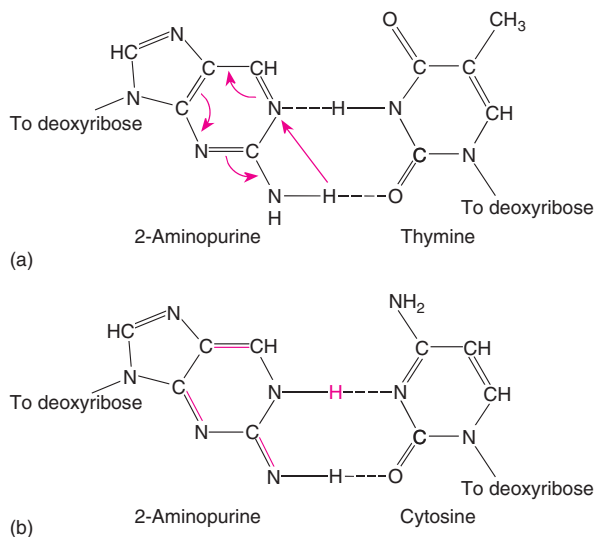


Figure 12.21 Two possible base pairs with 2-aminopurine. (a) In the normal state, 2-aminopurine acts like adenine and pairs with thymine. (b) In the rare state, 2-aminopurine acts like guanine and forms complementary base pairs with cytosine.

The 2-aminopurine is mutagenic by virtue of the fact that it can, like adenine, form two hydrogen bonds with thymine. When in the rare state, it can pair with cytosine (fig. 12.21). Thus, at times it replaces adenine, and at other times guanine. It promotes transition mutations.

Nitrous acid (HNO_2) also readily produces transitions by replacing amino groups on nucleotides with keto groups ($-\text{NH}_2$ to $=\text{O}$). The result is that cytosine is converted to uracil, adenine to hypoxanthine, and guanine to xanthine. As figure 12.22 shows, transition

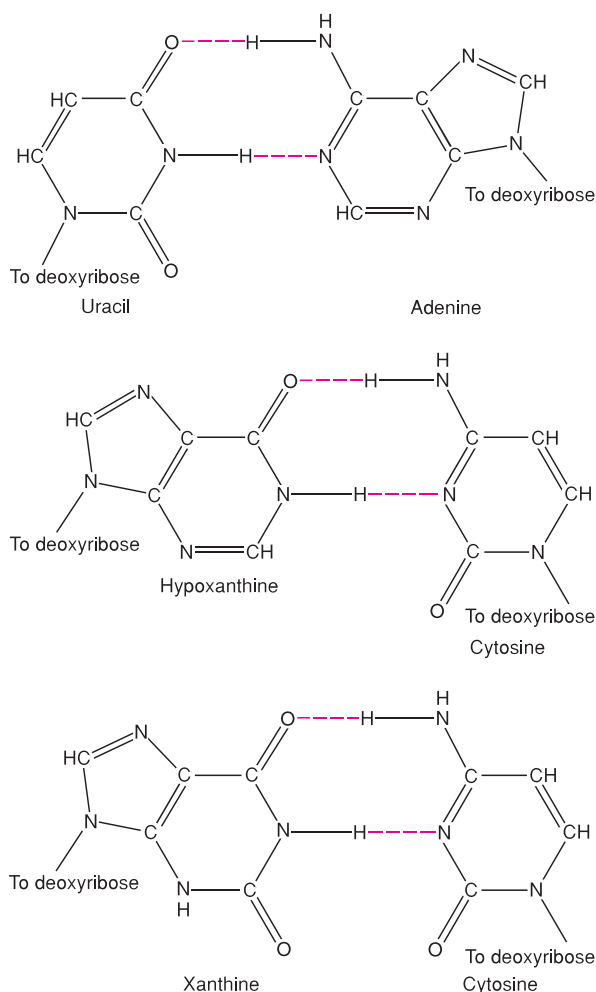


Figure 12.22 Nitrous acid converts cytosine to uracil, adenine to hypoxanthine, and guanine to xanthine. Uracil pairs with adenine, whereas its progenitor, cytosine, normally pairs with thymine; hypoxanthine pairs with cytosine, whereas its progenitor, adenine, normally pairs with thymine; and xanthine pairs with cytosine—the same base that guanine, its progenitor, pairs with. Thus, only the first two pairings result in transition mutations.

BOX 12.1

Which chemicals cause cancer in human beings? It is difficult to determine whether any substance is a carcinogen. Tests for carcinogenicity usually involve administering the substance in question to laboratory rats or mice to determine whether the substance actually causes cancer in these animals. Even these tests, however, are not absolute predictors of cancer in people. Tests of this nature are very expensive (\$1 million to \$2 million each) and time-consuming (three to four years). Since more than fifty thousand different chemical compounds are used in industry, with thousands more added each year, the challenge of making the working environment as well as the general environment safe seems overwhelming. We can, however, make a preliminary determination about the cancer-causing properties of any substance very quickly because of the relationship between mutagenicity and carcinogenicity. Many substances in the environment that can cause cancer also cause mutations. Both kinds of effects are related to DNA damage.

Bruce Ames, at the University of California at Berkeley, developed a routine screening test for mutagenicity. Substances that prove positive in this test are suspected of being carcinogens and would have to be tested further to determine their potential to cause cancer in mammals.

Ames worked with a strain of *Salmonella typhimurium* that requires histidine to grow. This strain will not grow on minimal medium.

Biomedical Applications

The Ames Test for Carcinogens

However, the strain will grow if a mutagen is added to the medium, causing the defective gene in the histidine pathway to revert to the wild-type. (Mutagens inducing gross chromosomal damage, such as deletions or inversions, will not be detected.) Under normal circumstances, there is a background mutation rate; a certain number of *Salmonella* cells revert spontaneously, and therefore a certain number of colonies will grow on the minimal medium. A mutagen, however, increases the number of colonies that can grow on minimal medium. This procedure is, therefore, a rapid, inexpensive, and easy test for mutagenicity.

To improve this test's ability to detect carcinogens, Ames added a supplement of rat liver extract to the medium. It is known that, although many substances are themselves not carcinogens, the breakdown of these substances in the liver creates substances that are carcinogenic. Rat liver enzymes act on a substance the same way human livers do, converting a noncarcinogenic primary substance into a possible carcinogen. The liver enzymes can also make a mutagen nonmutagenic.

Other short-term tests are in use that effectively duplicate the Ames test. These include tests for mutagenicity in mouse lymphoma cells and two tests in Chinese hamster ovary cells: a test for chromosomal aberrations and a test for sister-chromatid exchanges. None of these tests surpasses the Ames test, which has scored better than 90% correct when tested with hundreds of known carcinogens. Thousands of other substances have been subjected to this test; many have proven to be mutagenic. These substances are usually withdrawn from the workplace or home environment. From time to time, we read that a certain substance is believed to be carcinogenic and is being removed from grocery store shelves. Examples have included hair dyes, food preservatives, food-coloring agents, and artificial sweeteners. Many of these first were suspected after they failed the Ames test.



Bruce Ames (1928–).
(Courtesy Dr. Bruce Ames.)

BOX 12.2

Experimental
Methods*In Vitro Site-Directed
Mutagenesis*

One way of studying the way that proteins work is to change the sequence of amino acids in the protein. For example, if a scientist were working on the active site of a particular enzyme, he or she could learn how the enzyme modifies its substrates by changing one or a few amino acids. Changes could be made, for example, in order to study the role of shape or charge on the functioning of the enzyme. Advances in recombinant DNA techniques have made it possible for a research scientist to create exactly the changes he or she wants in a protein.

To begin with, the gene for the protein or enzyme must be cloned so that it can be manipulated (see chapter 13). Once cloned, deletions are easy to create with restriction en-

donucleases (described in chapter 13). If a particular endonuclease cuts the gene in two places, the intervening segment can be spliced out (fig. 1a; see chapter 13). If the endonuclease cuts only once, exonucleases can digest the ends of the cut, extending the deletion away from the cut in both directions (fig. 1b). Insertions can be created by either cutting the gene and repairing the single-stranded ends (fig. 1c) or by creating

an oligonucleotide (a linker) with the desired sequence and inserting the linker at the site of an endonuclease cut (fig. 1d).

Far more impressive, however, is the ability to change a single specific codon in order to replace any amino acid in the protein with any other amino acid. The process involves directed mutagenesis using artificially created oligonucleotides.

Basically, a short sequence of DNA (an oligonucleotide) is synthesized complementary to a region of the cloned gene, but with a change in one or more bases of a codon to specify a different amino acid. That oligonucleotide is then hybridized with the single-stranded form of the clone (fig. 2). Although one or more

continued

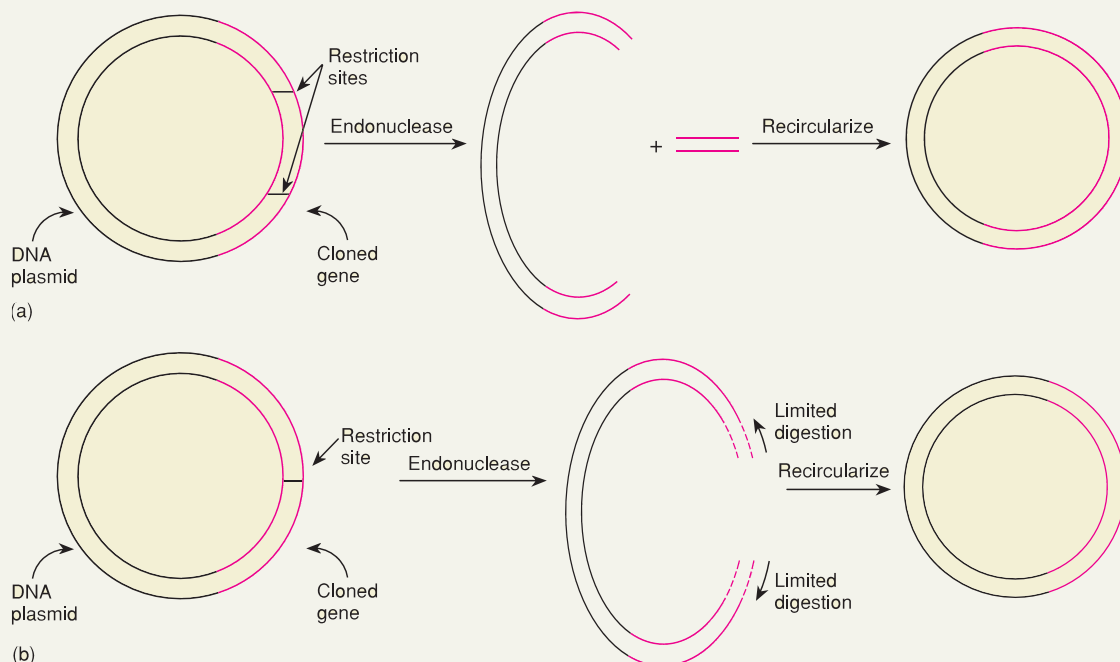


Figure 1 A cloned gene can be mutated in several ways. (a) If a restriction endonuclease has two sites in the gene, the intermediate piece can be spliced out. (b) If the endonuclease has only one site, the gene can be opened at that site, and limited digestion by exonucleases will delete part of the gene.

continued

BOX 12.2 CONTINUED

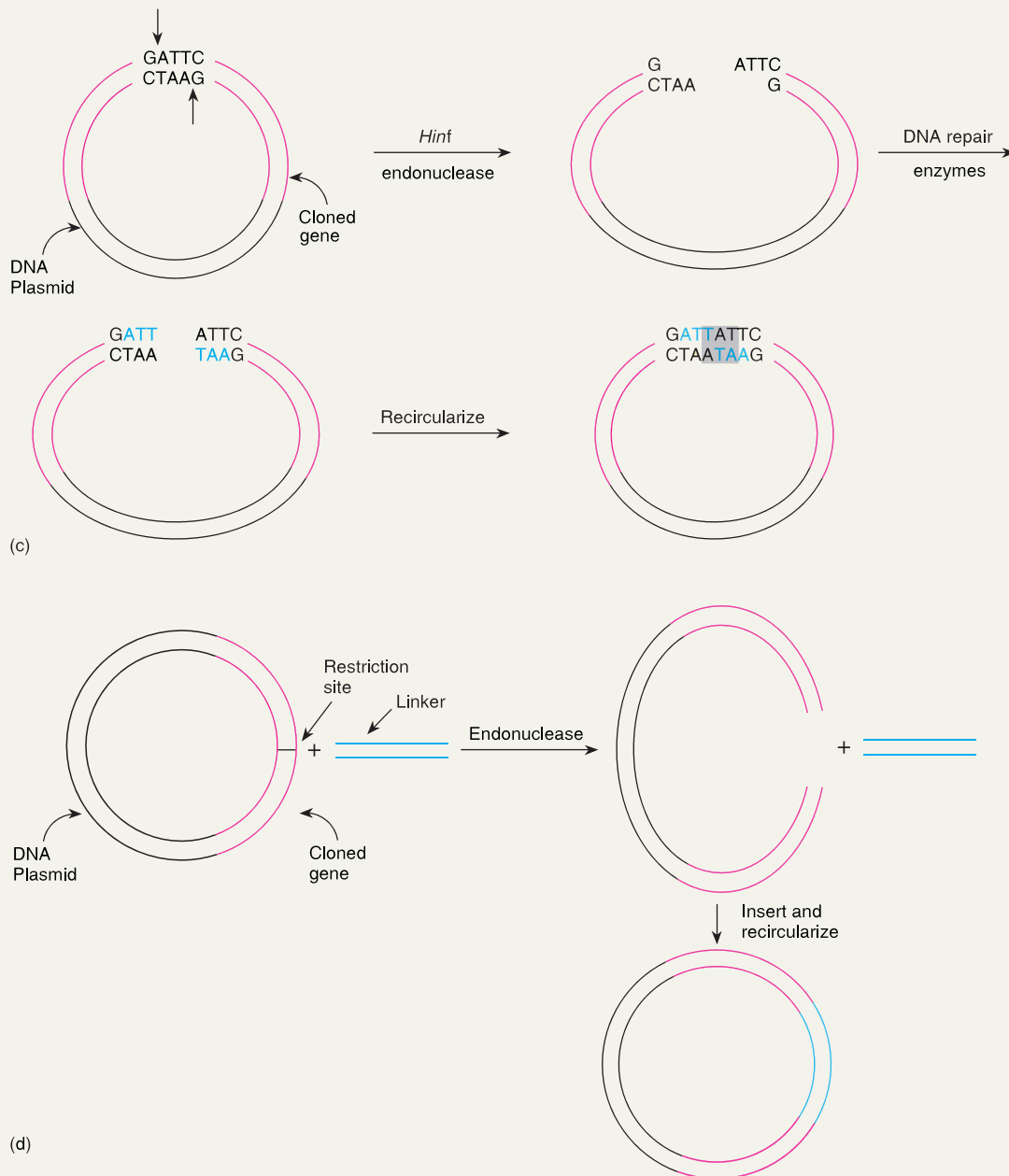


Figure 1—continued (c) If an endonuclease has an offset region between its splice points of three or six nucleotides (one or two codons), that length can be inserted by repairing the single-stranded ends after cutting by the endonuclease. The resulting blunt ends can be spliced together. (Note that actually an ATT region has been converted to an ATTATT region. If reading codons along the DNA, the actual insertion is of a TAT codon.) (d) A linker of any length (usually the length of a specific number of codons) can be inserted at a restriction site.

bases will not match, hybridization can usually be facilitated by adjusting the pH or ionic strength of the solution. The hybridized oligonucleotide is then used as a primer for DNA replication; the whole plasmid is replicated, resulting in hybrid DNA. In subsequent DNA replications of the hybrid, both the original gene and the mutated DNA will be produced. The latter can be isolated by appropriate selection methods; it is a plasmid with a cloned gene that has the exact mutation the researcher wanted. Using techniques of this type, geneticists have made many advancements in understanding exactly how various components of an enzyme contribute to its function.

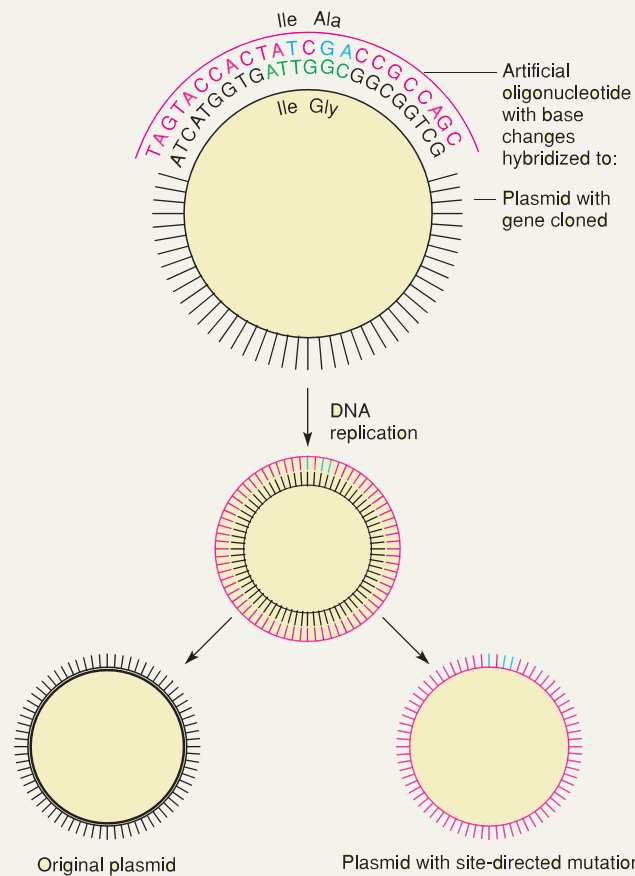


Figure 2 Site-directed mutagenesis can involve any nucleotide(s). In this case, an inserted gene with an Ile-Gly sequence is converted, at the direction of the investigator, to an Ile-Ala sequence. A single-stranded form of the plasmid is isolated. A synthetically prepared oligonucleotide (twenty-three bases in this example) is added. It can be made to hybridize at the complementary site despite differing by three bases. Then DNA replication is carried out using the oligonucleotide configuration as a primer. After the strands of the duplex are separated, the investigators can isolate the original plasmid as well as the mutated plasmid. (Note that the investigators changed two codons, although they changed only one amino acid, because they also wanted to introduce an Alu site at that point for future studies.) (From J. E. Villafranca, et al., "Directed mutagenesis of dihydrofolate reductase," *Science* 222:782–88. Copyright 1983 by the AAAS.)

mutation results from two of the changes. Uracil pairs with adenine instead of guanine, thus leading to a UA base pair in place of a CG base pair; hypoxanthine (H) pairs with cytosine instead of thymine, the original base paired with adenine. Thus, in this case, an HC base pair replaces an AT base pair. Both of these base pairs (UA and HC) are transition mutations. Xanthine, however, pairs with cytosine just as guanine does. Thus, the replacement of guanine with xanthine does not cause changes in base pairing.

Like nitrous acid, heat can also deaminate cytosine to form uracil and thus bring about transitions (CG to TA). Apparently, heat can also bring about transversions by an unknown mechanism.

Transversions

Ethyl methane sulfonate ($\text{CH}_3\text{SO}_3\text{CH}_2\text{CH}_3$) and ethyl ethane sulfonate ($\text{CH}_3\text{CH}_2\text{SO}_3\text{CH}_2\text{CH}_3$) are agents that cause the removal of purine rings from DNA. The multistep process begins with the ethylation of a purine ring and ends with the hydrolysis of the glycosidic (purine-deoxyribose) bond, causing the loss of the base.

These sites where this happens are referred to as AP (apurinic-apyrimidinic) sites. If the AP site is not repaired, any of the four DNA bases could be inserted into the new strand opposite the gap (fig. 12.23). If thymine is placed in the newly formed strand, then the original base pair is restored; insertion of cytosine results in a transition mutation; insertion of either adenine or guanine results in a transversion mutation. Of course, the gap is still there, and it continues to generate new mutations each generation until it is repaired. During DNA replication in *E. coli*, the polymerase tends to place adenine opposite the gap more frequently than it places other bases.

Insertions and Deletions

The molecules of the acridine dyes, such as proflavin and acridine orange (fig. 12.24), are flat. Presumably, they initiate mutation by inserting into the DNA double helix, causing the helix to buckle in the region of insertion, possibly leading to base additions and deletions during DNA replication. Crick and Brenner used acridine-induced mutations to demonstrate both that the genetic

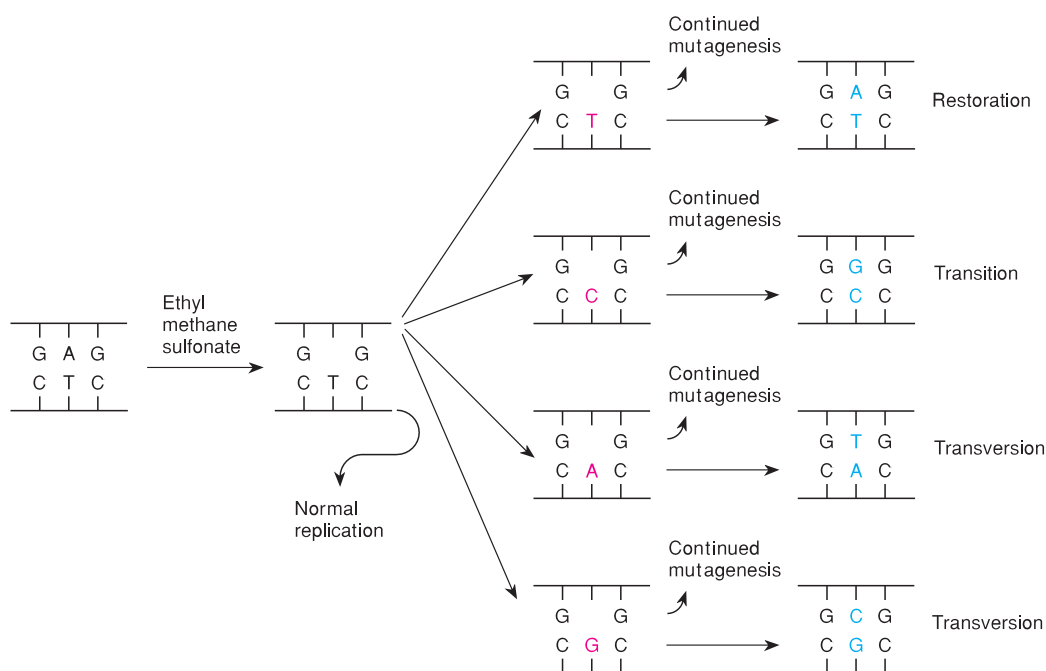


Figure 12.23 Four possible outcomes after treatment of DNA with an alkylating agent, which removes the purine—adenine in this example. The bases shown in red are the four bases that DNA polymerase may insert opposite the gap. After another round of DNA replication, the gap remains to generate further mutations. The inserted base forms a base pair (blue), which can be a restoration or a transition or transversion mutation.

code was read from a fixed point and that it was triplet (chapter 11).

Misalignment Mutagenesis

Additions and deletions in DNA can also come about by misalignment of a template strand and the newly formed (progeny) strand in a region containing a repeated se-

quence. For example, in figure 12.25 we expect the progeny strand to contain six adjacent adenines because the template strand contains six adjacent thymines. Misalignment of the progeny strand results in seven consecutive adenines: six thymines replicated, plus one already replicated but misaligned. Misalignment of the template strand results in five consecutive adenines because one thymine is not available in the template. Regions with long runs of a particular base may be very mutation prone. They may explain the “hot spots” observed by Benzer (see fig. 12.8) and others.

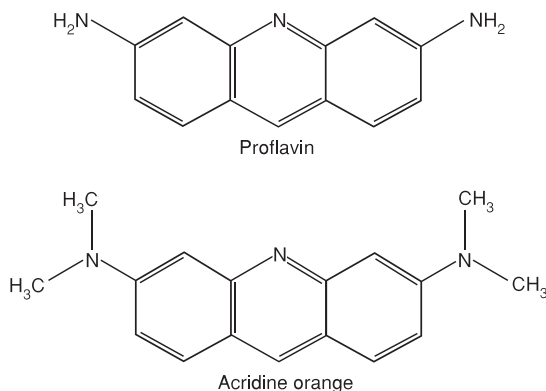


Figure 12.24 Structure of two acridine dyes: proflavin and acridine orange.

Intergenic Suppression

When a critical mutation occurs in a codon, several routes can still lead to survival of the individual; simple reversion and intragenic suppression are two that we have already considered. A third route is through **intergenic suppression**—restoration of the function of a mutated gene by changes in a different gene, called a **suppressor gene**. Suppressor genes are usually transfer RNA genes. When mutated, intergenic suppressors change the way in which a codon is read.

Suppressor genes can restore proper reading to nonsense, missense, and frameshift mutations. **Nonsense mutations** convert a codon that originally specified an amino acid into one of the three nonsense

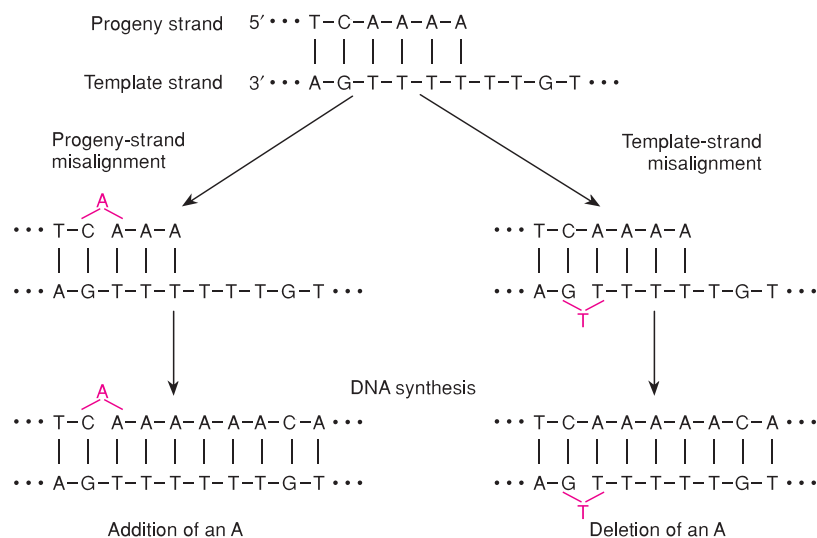


Figure 12.25 Misalignment of a template or progeny strand during DNA synthesis. If the progeny strand is misaligned after DNA replication has begun, the resulting progeny strand will have an additional base. If the template strand is misaligned during DNA replication, the resulting progeny strand will have a deleted base. These changes will show up after another round of DNA replication. (From J.W. Drake, B.W. Glickman, and L.S. Ripley, “Updating the Theory of Mutation,” *American Scientist*, 71:621–630, 1983. Reprinted with permission of American Scientist, magazine of Sigma Xi, The Scientific Research Society.)

codons. **Missense mutations** change a codon so that it specifies a different amino acid. Frameshift mutations, by additions or deletions of nucleotides, cause an alteration in the reading frame of codons. A frameshift mutation, caused by the insertion of a single base, can be suppressed by a transfer RNA that has an added base in its anticodon (fig. 12.26a). It reads four bases as a codon and thus restores the original reading frame.

The transfer RNA produced by a nonsense suppressor gene reads the nonsense codon as if it were a codon for an amino acid; an amino acid is placed into

the protein, and reading of the messenger RNA continues. At least three suppressors of the mutant amber codon (UAG) are known in *E. coli*. One suppressor puts tyrosine, one puts glutamine, and one puts serine into the protein chain at the point of an amber codon. Normally, tyrosine transfer RNA has the anticodon 3'-AUG-5'. The suppressor transfer RNA that reads amber as a tyrosine codon has the anticodon 3'-AUC-5', which is complementary to amber. Hence, a mutated tyrosine transfer RNA reads amber as a tyrosine codon (fig. 12.26b).

If the amber nonsense codon is no longer read as a stop signal, then won't all the genes terminating in the amber codon continue to be translated beyond their ends, causing the cell to die? In the tyrosine case, two genes for tyrosine transfer RNA were found; one contributes the major fraction of the transfer RNAs, and the other, the minor fraction. It is the minor-fraction gene that mutates to act as the suppressor. Thus, most messenger RNAs are translated normally, and most amber mutations result in premature termination, although a sufficient number are translated (suppressed) to ensure the viability of the mutant cell. In general, intergenic suppressor mutants would be eliminated quickly in nature because they are inefficient—the cells are not healthy. In the laboratory, we can provide special conditions that allow them to be grown and studied.

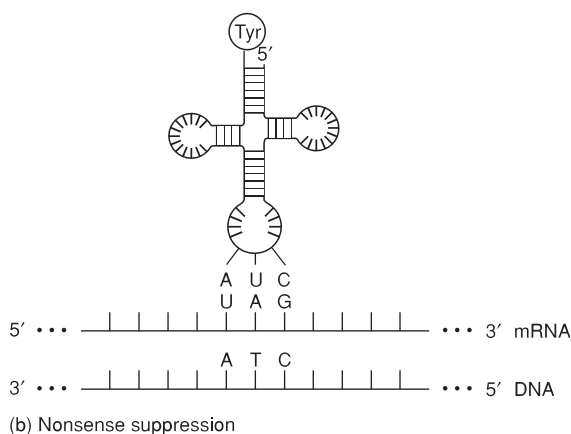
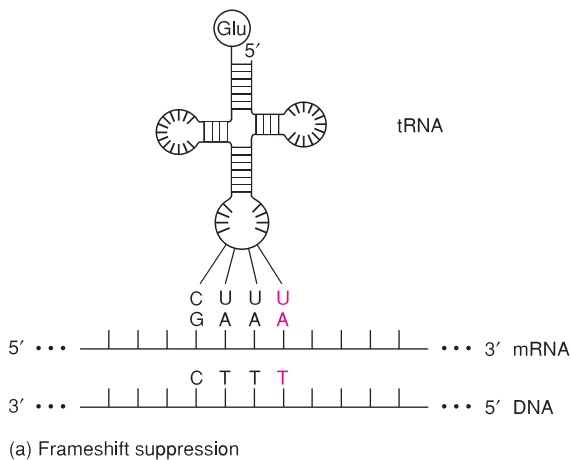


Figure 12.26 Frameshift and nonsense suppression by mutant transfer RNAs. In (a), a thymine has been inserted into DNA, resulting in a frameshift. However, a transfer RNA with four bases in the codon region reads the inserted base as part of the previous codon in the messenger RNA. The frameshift thus does not occur. In (b), an *amber* mutation (UAG), which normally results in chain termination, is read as tyrosine by a mutant tyrosine transfer RNA that has the anticodon sequence complementary to the *amber* codon.

Mutator and Antimutator Mutations

Whereas intergenic suppressors represent mutations that “restore” the normal phenotype, mostly through mutation of transfer RNA loci, **mutator** and **antimutator mutations** cause an increase or decrease in the overall mutation rate of the cell. They are frequently mutations of DNA polymerase, which, as you remember, not only polymerizes DNA nucleotides 5' → 3' complementary to the template strand, but also checks to be sure that the correct base was put in (they proofread). If, in the proofreading process, the polymerase discovers an error, it can correct this error with its 3' → 5' exonuclease activity. Mutator and antimutator mutations sometimes involve changes in the polymerase's proofreading ability (exonuclease activity).

Phage T4 has its own DNA polymerase with known mutator and antimutator mutants. Mutator mutants are very poor proofreaders (they have low exonuclease-to-polymerase ratios), and thus they introduce mutations throughout the phage genome. Antimutator mutants, however, have exceptionally efficient proofreading ability (high exonuclease-to-polymerase ratios) and, therefore, result in a very low mutation rate for the whole genome (box 12.3).

BOX 12.3

Although we discuss evolution in detail at the end of the book, note here that we view mutation as a random process, not one that occurs because a cell “needs” a particular mutation. For example, Luria and Delbrück’s work showed that the mutation in *E. coli* for resistance to phage T1 occurred randomly before exposure to the phage, not because the cells would benefit from the mutation.

Because of the entrenched dogma that mutations occur through random processes, the scientific community was startled when, in 1988, John Cairns—a highly respected senior scientist—and colleagues reported a new observation—**adaptive**, or **directed**, **mutation**, occurring when the cell needed it. Their system was the *lacZ* gene in *E. coli*. Cells that could not use lactose as an energy source (*lacZ*[−]) were plated on a medium in which lactose was the sole energy source. Some mutants already there of course produced colonies (*lacZ*⁺). The expectation was that no new mutations would occur over time because the *lacZ*[−] cells would have either died or stopped metabolizing. Unexpectedly, Cairns and colleagues found that more and more colonies appeared over time, coming from cells that had mutated to *lacZ*⁺. As a control, they looked for revertants of other genes not involved with lactose metabolism. These mutations did not occur in a directed manner.

Scientists were extremely skeptical of this work for two reasons. First,

Experimental Methods

Adaptive Mutation

it seemed to fly in the face of our common understanding of the mutational process. Second, there were no obvious explanations for how this could occur. Numerous articles were published refuting the notion of adaptive mutations and suggesting other explanations for the results. These explanations included artifacts of miscounting cells to mutants that were extremely slow growing, but there all the time.

In the past several years, other scientists have found at least a half dozen similar results in other organisms and other genes. The debate is ongoing, but work published in mid-1994 seems to have recast it into the realm of methods of mutagenesis rather than non-Darwinian processes. Several scientists found that the “directed” mutations seem to be of a certain type, mainly single nucleotide deletions within runs of the same nucleotide: for example, a deletion of a C in a CCCC sequence. This type of error happens during DNA replication and could be the result of a repair deficiency. That is, under extreme duress, the cells may be going into a “hypermutational” mode, or

selection may favor hypermutable genotypes in which repair mechanisms are shut down in order to intentionally create lots of errors in the DNA. Any errors that do not alleviate the problem result in cell death, a death that was inevitable anyway; however, some errors will correct the problem (*lacZ*[−] to *lacZ*⁺). Those cells will survive. One recent study indicated that a subpopulation of about 0.06% of the population was hypermutable, with a mutation rate about 200 times that of the normal cells. That group of cells could account for the adaptive mutations. Other scientists found that when the locus of importance was on a plasmid, the adaptive mutation could occur by increased replication of the plasmid. (Currently, the favored term is *adaptive mutation*, rather than *directed mutation*, indicating that the mutations are useful to the cell, not that some unknown process directs them.)

This controversy, although not necessarily resolved, has actually brought out the best in the scientific method: A skeptical scientific community tried its best to refute an unreasonable observation. Other scientists then repeated the observation and extended it, making it more worthy of further study. Finally, further work has given us reasonable mechanisms that not only require no rejection of the original concept of random mutation, but actually give us hypotheses to test further.

DNA REPAIR



Radiation, chemical mutagens, heat, enzymatic errors, and spontaneous decay constantly damage DNA. For example, it is estimated that several thousand DNA bases are lost each day in every mammalian cell due to spontaneous decay. Some types of DNA damage interfere with

DNA replication and transcription. In the long evolutionary challenge to minimize mutation, cells have evolved numerous mechanisms to repair damaged or incorrectly replicated DNA. Many enzymes, acting alone or in concert with other enzymes, repair DNA. Repair systems are generally placed in four broad categories: damage reversal, excision repair, double-strand break repair, and

postreplicative repair. Enzymes that process repair steps have been conserved during evolution. That is, enzymes found in *E. coli* have homologues in yeast, fruit flies, and human beings. However, eukaryotic systems are almost always more complex.

Damage Reversal

Ultraviolet (UV) light causes linkage, or **dimerization**, of adjacent pyrimidines in DNA (fig. 12.27). Although cytosine-cytosine and cytosine-thymine dimers are occasionally produced, the principal products of UV irradiation are thymine-thymine dimers. These can be repaired in several different ways. The simplest is to reverse the dimerization process and restore the original unlinked thymines.

In *E. coli*, an enzyme called DNA photolyase, the product of the *pbr* gene (for **photoreactivation**), binds to dimerized thymines. When light shines on the cell,

the enzyme breaks the dimer bonds with light energy. The enzyme then falls free of the DNA. This enzyme thus reverses the UV-induced dimerization. Another example of an enzyme that performs direct DNA repair is O^6 -mGua DNA methyltransferase, which removes the methyl groups from O^6 -methylguanine, the major product of DNA-methylating agents (fig. 12.28). Although other repair mechanisms seem to be present in all organisms, photoreactivation is not; it is apparently absent in human beings.

Excision Repair

Excision repair refers to the general mechanism of DNA repair that works by removing the damaged portion of a DNA molecule. Various enzymes can sense damage or distortion in the DNA double helix. During excision repair, bases and nucleotides are removed from the damaged strand. The gap is then patched using complementarity with the remaining strand. We can broadly categorize these systems as base excision repair and nucleotide excision repair, which includes mismatch repair. We will discuss only the major repair pathways; others exist. Presumably, redundancy in repair has been selected for because of the critical need to keep DNA intact and relatively mutation free.

Base Excision Repair

A base can be removed from a nucleotide within DNA in several ways: by direct action of an agent such as radiation, by spontaneous hydrolysis, by an attack of oxygen free radicals, or by **DNA glycosylases**, enzymes that sense damaged bases and remove them. Currently, at least five DNA glycosylases are known. For example, uracil-DNA glycosylase, the product of the *ung* gene in

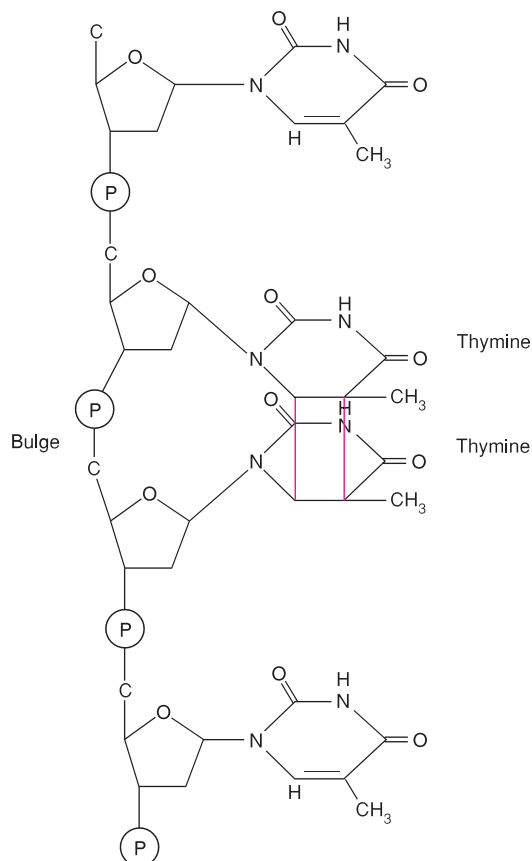


Figure 12.27 UV-induced dimerization of adjacent thymines in DNA. The red lines represent the dimer bonds in the adjacent thymines.

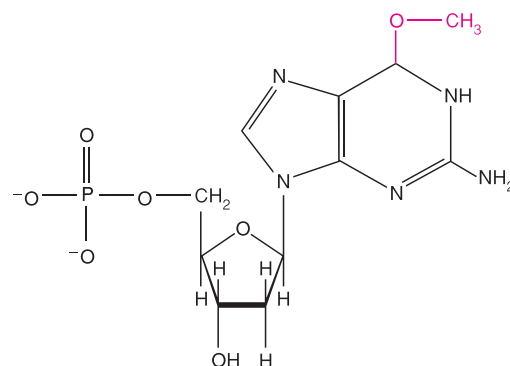


Figure 12.28 The structure of O^6 -methylguanine. The red color shows the modification of guanine, in which the normal configuration is a double-bonded oxygen (keto form).